

19. EXDMA Controller (EXDMACa)

This MCU incorporates a 2-channel direct memory access controller (EXDMAC) designed exclusively for external bus transfer. The EXDMAC is a module to transfer data without the CPU. When a DMA transfer request is generated, the EXDMAC transfers data stored at the transfer source address to the transfer destination address.

19.1 Overview

Table 19.1 lists the specifications of the EXDMAC, and Figure 19.1 shows a block diagram of the EXDMAC.

Table 19.1 Specifications of EXDMAC (1/2)

Item		Description
Number of channels		2 (EXDMAC0 and EXDMAC1)
Transfer space		512 Mbytes (External areas at addresses 0000 0000h to 0FFF FFFFh and F000 0000h to FFFF FFFFh excluding reserved areas)
Maximum transfer volume		1 M data (Maximum number of transfer operations in block transfer mode: 1,024 data × 1,024 blocks)
DMA request source		- Request source selectable from the following three sources for each channel - Software trigger - External DMA transfer request input - DMA transfer request from peripheral modules (TPU1.TRGA or MTU1.TRGA) (Channel 0: a software configurable interrupt B request from TPU1.TRGA selected in ICU.SLIBR144 or a software configurable interrupt A request from MTU1.TRGA selected in ICU.SLIAR208; Channel 1: a software configurable interrupt B request from TPU1.TRGA selected in ICU.SLIBR145 or a software configurable interrupt A request from MTU1.TRGA selected in ICU.SLIAR209)
Channel priority		Channel 0 > Channel 1 (Channel 0: Highest)
Transfer data	Single data	Bit length: 8, 16, 32 bits
	Block size	Number of data: 1 to 1,024 data
	Cluster size	Number of data: 1 to 8 data
Transfer mode	Normal transfer mode	- One data transfer by one DMA transfer request - Free running mode (setting in which total number of data transfer operations is not specified) settable
	Repeat transfer mode	- One data transfer by one DMA transfer request - Program returns to the transfer start address on completion of the repeat size of data transfer specified for the transfer source or destination. - Maximum settable repeat size: 1,024 data
	Block transfer mode	- One block data transfer by one DMA transfer request - Maximum settable block size: 1,024 data
	Cluster transfer mode	- One cluster data transfer by one DMA transfer request - Maximum settable cluster size: 8 data (32 bytes)
Address mode	Single address mode	- Transfers data by accessing the transfer source or destination peripheral device with the EDACKn signal (n = 0, 1) and specifying the address of the other peripheral device. - Available in normal transfer mode, repeat transfer mode, and block transfer mode.
	Dual address mode	- Transfers data by specifying the addresses of transfer source and destination. - Available in normal transfer mode, repeat transfer mode, block transfer mode, and cluster transfer mode.
Selective functions	Extended repeat area function	- Function in which data can be transferred by repeating the address values in the specified range with the upper bit values in the transfer address register fixed - Area of 2 bytes to 128 Mbytes separately settable as extended repeat area for transfer source and destination

Table 19.1 Specifications of EXDMAC (2/2)

Item	Description
Interrupt request	Transfer end interrupt Generated when the specified number of transfers is completed in normal transfer mode Generated when the specified repeat count of transfers is completed in repeat transfer mode Generated when the specified block count of transfers is completed in block transfer mode Generated when the specified cluster count of transfers is completed in cluster transfer mode
	Transfer escape end interrupt Generated when the repeat size of data transfer is completed or the extended repeat area overflows.
Low-power consumption function	The module-stop state can be set.

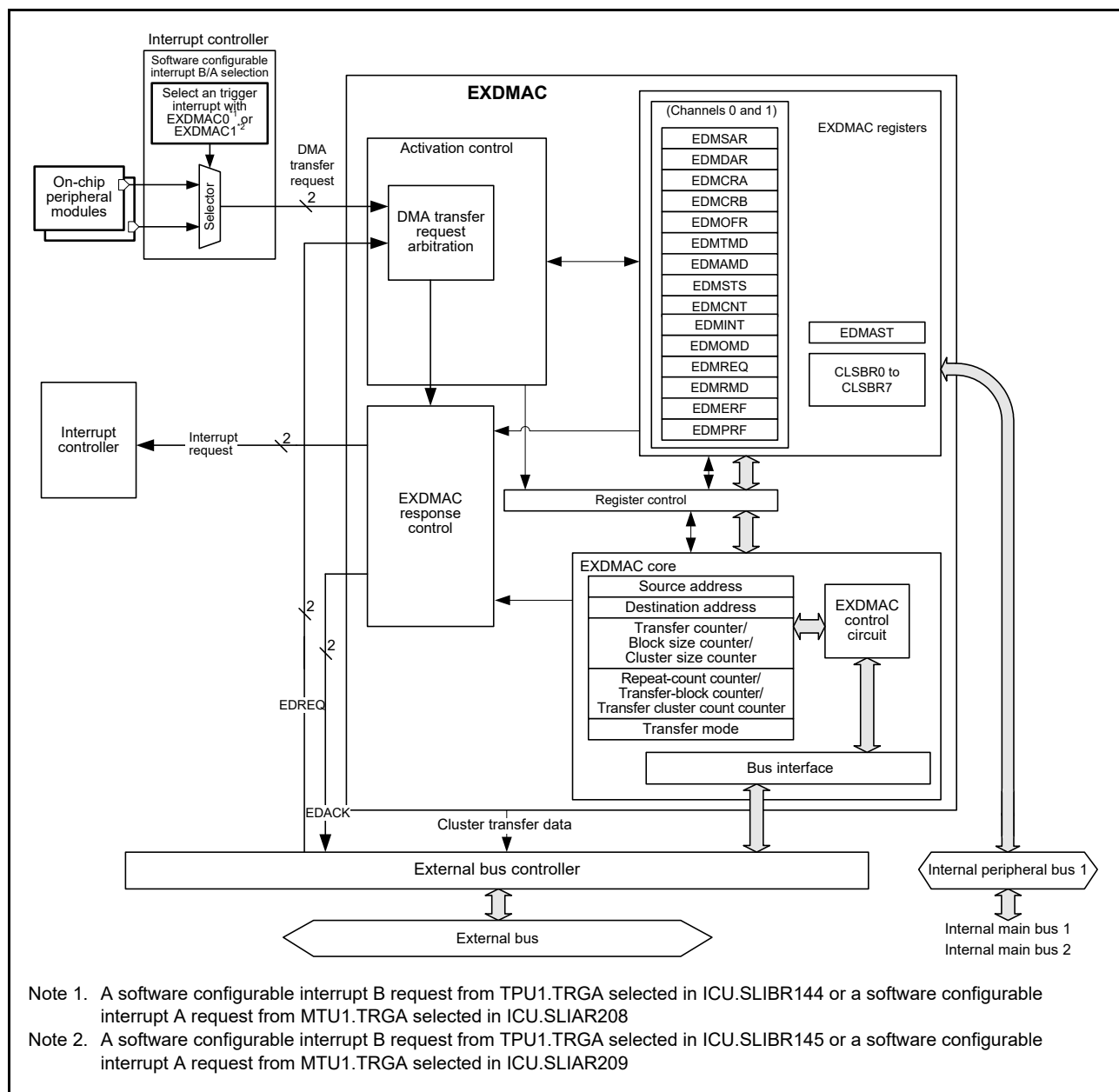
**Figure 19.1 Block Diagram of EXDMAC**

Table 19.2 lists the input/output pins of the EXDMAC.

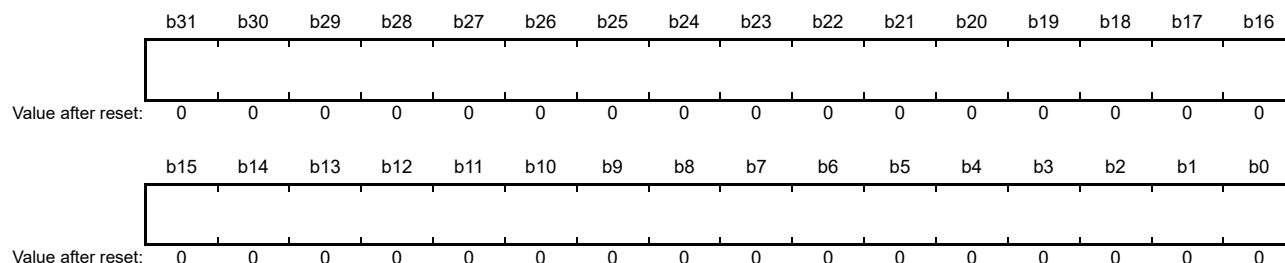
Table 19.2 Pin Configuration of EXDMAC

Channel	Pin Name	I/O	Description
EXDMAC0	EDREQ0	Input	EXDMAC0 external DMA transfer request
	EDACK0	Output	EXDMAC0 single address transfer acknowledge
EXDMAC1	EDREQ1	Input	EXDMAC1 external DMA transfer request
	EDACK1	Output	EXDMAC1 single address transfer acknowledge

19.2 Register Descriptions

19.2.1 EXDMA Source Address Register (EDMSAR)

Address(es): EXDMAC0.EDMSAR 0008 2800h, EXDMAC1.EDMSAR 0008 2840h



Bit	Description	Setting Range	R/W
b31 to b0	Specifies the transfer source start address.	0000 0000h to 0FFF FFFFh (256 Mbytes) F000 0000h to FFFF FFFFh (256 Mbytes)	R/W

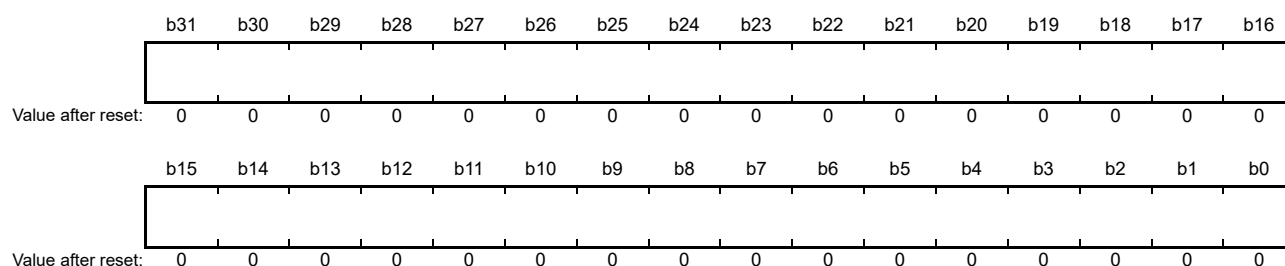
This register is used to set the start address of the transfer source.

Set the EDMSAR register while EXDMAC stops (EDMAST.DMST bit = 0) or DMA transfer is disabled (EDMCNT.DTE bit = 0).

Setting bits 31 to 29 is invalid; a value of bit 28 is extended to bits 31 to 29. Reading the EDMSAR register returns the extended value.

19.2.2 EXDMA Destination Address Register (EDMDAR)

Address(es): EXDMAC0.EDMDAR 0008 2804h, EXDMAC1.EDMDAR 0008 2844h



Bit	Description	Setting Range	R/W
b31 to b0	Specifies the transfer destination start address.	0000 0000h to 0FFF FFFFh (256 Mbytes) F000 0000h to FFFF FFFFh (256 Mbytes)	R/W

This register is used to set the start address of the transfer destination.

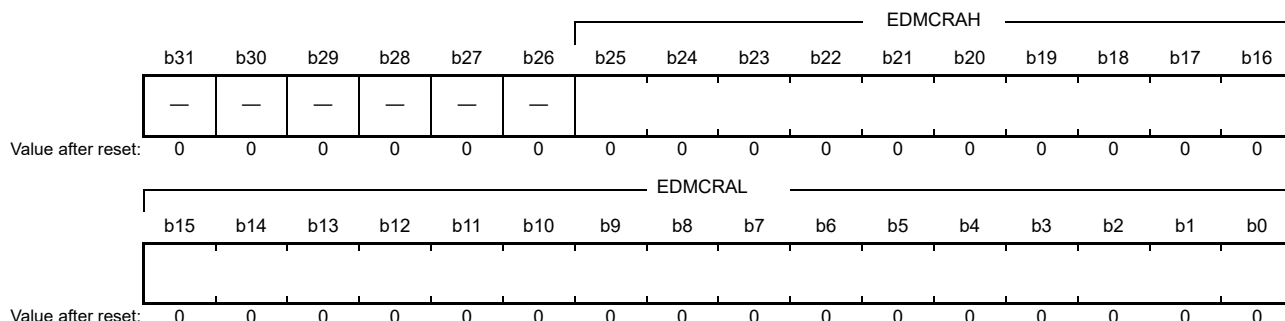
Set the EDMDAR register while EXDMAC stops (EDMAST.DMST bit = 0) or DMA transfer is disabled (EDMCNT.DTE bit = 0).

Setting bits 31 to 29 is invalid; a value of bit 28 is extended to bits 31 to 29. Reading the EDMDAR register returns the extended value.

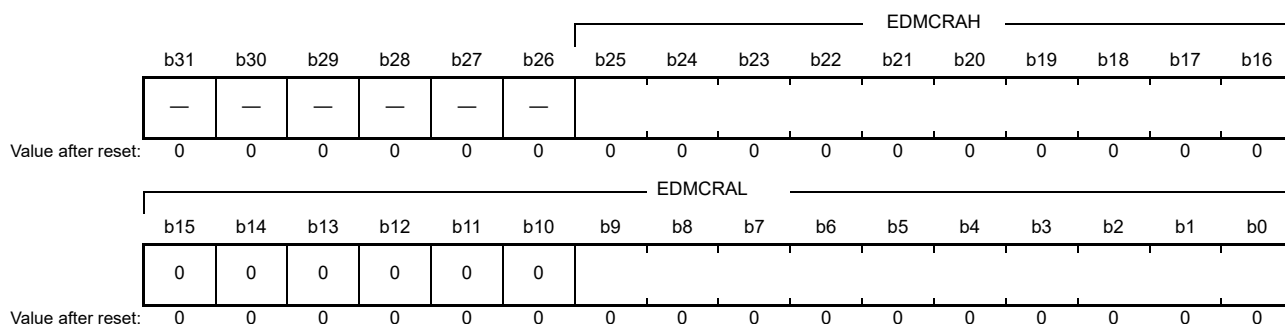
19.2.3 EXDMA Transfer Count Register (EDMCRA)

Address(es): EXDMAC0.EDMCRA 0008 2808h, EXDMAC1.EDMCRA 0008 2848h

· Normal transfer mode



· Repeat transfer mode, block transfer mode



Note: The function differs depending on the transfer mode.

· Cluster transfer mode



Symbol	Bit Name	Description	R/W
EDMCRAL	Lower bits of transfer count	Specifies the number of transfer operations.	R/W
EDMCRAH	Upper bits of transfer count		R/W

Note: Set the same value for EDMCRAH and EDMCRL in repeat transfer mode, block transfer mode, and cluster transfer mode.

This register is used to specify the transfer count of the DMA. The function of this register depends on transfer mode.

(1) Normal transfer mode (EXDMACn.EDMTMD.MD[1:0] bits = 00b)

EDMCRAL functions as a 16-bit transfer counter.

The number of transfer operations is one when the setting is 0001h, and 65535 when it is FFFFh. The value is decremented by one each time data is transferred.

When the setting is 0000h, no specific number of transfer operations is set; data transfer is performed with the transfer counter stopped (free running mode).

EDMCRAH is not used in normal transfer mode. Write 0000h to EDMCRAH.

(2) Repeat transfer mode (EXDMACn.EDMTMD.MD[1:0] bits = 01b)

EDMCRAH specifies the repeat size and EDMCRAL functions as a 10-bit transfer counter.

The number of transfer operations is one when the setting is 001h, 1023 when it is 3FFh, and 1024 when it is 000h. In repeat transfer mode, a value in the range of 000h to 3FFh (the number of transfer operations: 1 to 1024) can be set for EDMCRAH and EDMCRAL.

Setting bits 15 to 10 in EDMCRAL is invalid. Write 0 to these bits.

The value in EDMCRAL is decremented by one each time data is transferred until it reaches 000h, at which the value in EDMCRAH is loaded into EDMCRAL.

(3) Block transfer mode (EXDMACn.EDMTMD.MD[1:0] bits = 10b)

EDMCRAH specifies the block size and EDMCRAL functions as a 10-bit block size counter.

The block size is 1 when the setting is 001h, 1023 when it is 3FFh, and 1024 when it is 000h. In block transfer mode, a value in the range of 000h to 3FFh (the number of transfer operations: 1 to 1024) can be set for EDMCRAH and EDMCRAL.

Setting bits 15 to 10 in EDMCRAL is invalid. Write 0 to these bits.

The value in EDMCRAL is decremented by one each time data is transferred until it reaches 000h, at which the value in EDMCRAH is loaded into EDMCRAL.

(4) Cluster transfer mode (EXDMACn.EDMTMD.MD[1:0] bits = 11b)

EDMCRAH specifies the cluster size and EDMCRAL functions as a 3-bit cluster size counter.

The cluster size is 1 when the setting is 001h, 7 when it is 007h, and 8 when it is 000h. In cluster transfer mode, a value in the range of 000h to 007h (the number of transfer operations: 1 to 8) can be set for EDMCRAH and EDMCRAL.

Setting bits 15 to 3 in EDMCRAL is invalid. Write 0 to these bits.

The value in EDMCRAL is decremented by one each time data is transferred until it reaches 000h, at which the value in EDMCRAH is loaded into EDMCRAL.

19.2.4 EXDMA Block Transfer Count Register (EDMCRB)

Address(es): EXDMAC0.EDMCRB 0008 280Ch, EXDMAC1.EDMCRB 0008 284Ch

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—										
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Description	Setting Range	R/W
b9 to b0	Specifies the block count of transfers in block transfer mode, the repeat count in repeat transfer mode, or the transfer cluster count in cluster transfer mode.	001h to 3FFh (1 to 1023) 000h (1024)	R/W
b15 to b10	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register is used to specify the transfer block count in block transfer mode, the repeat count in repeat transfer mode, or the transfer cluster count in cluster transfer mode.

Decrement (by 1) when the last data of 1 repeat size is transferred in repeat transfer mode.

Decrement (by 1) when the last data of 1 block size is transferred in block transfer mode.

Decrement (by 1) when the last data of 1 cluster size is transferred in cluster transfer mode.

In normal transfer mode, EDMCRB is not used and setting this register is invalid.

19.2.5 EXDMA Transfer Mode Register (EDMTMD)

Address(es): EXDMAC0.EDMTMD 0008 2810h, EXDMAC1.EDMTMD 0008 2850h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
MD[1:0]		DTS[1:0]		—		SZ[1:0]		—		—		—		DCTG[1:0]	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	DCTG[1:0]	Transfer Request Source Select	b1 b0 0 0: Software 0 1: Setting prohibited 1 0: External DMA transfer request pin (EDREQn) 1 1: DMA transfer request from peripheral modules (TPU1.TRGA or MTU1.TRGA) *1	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b9, b8	SZ[1:0]	Transfer Data Size Select	b9 b8 0 0: 8 bits 0 1: 16 bits 1 0: 32 bits 1 1: Setting prohibited	R/W
b11, b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13, b12	DTS[1:0]	Repeat Area Select	b13 b12 0 0: The destination is specified as the repeat area or block area. 0 1: The source is specified as the repeat area or block area. 1 0: The repeat area or block area is not specified. 1 1: Setting prohibited	R/W
b15, b14	MD[1:0]	Transfer Mode Select	b15 b14 0 0: Normal transfer 0 1: Repeat transfer 1 0: Block transfer 1 1: Cluster transfer	R/W

Note 1. EXDMAC0: A software configurable interrupt B request from TPU1.TRGA selected in ICU.SLIBR144 or a software configurable interrupt A request from MTU1.TRGA selected in ICU.SLIAR208
 EXDMAC1: A software configurable interrupt B request from TPU1.TRGA selected in ICU.SLIBR145 or a software configurable interrupt A request from MTU1.TRGA selected in ICU.SLIAR209
 For the setting procedure, refer to section 19.5.1, Request Sources, (3).

This register is used to set DMA transfer mode.

DCTG[1:0] Bits (Transfer Request Source Select)

These bits select the startup source of the EXDMAC from software, the external DMA transfer request pin, or a DMA transfer request generated from a peripheral module.

SZ[1:0] Bits (Transfer Data Size Select)

These bits select data size for a single data transfer from 8, 16, and 32 bits.

DTS[1:0] Bits (Repeat Area Select)

DTS[1:0] select either the source or destination as the repeat area in repeat, block, or cluster transfer mode. In normal transfer mode, setting these bits is invalid.

MD[1:0] Bits (Transfer Mode Select)

These bits specify DMA transfer mode from normal, repeat, block, and cluster.

19.2.6 EXDMA Output Setting Register (EDMOMD)

Address(es): EXDMAC0.EDMOMD 0008 2812h, EXDMAC1.EDMOMD 0008 2852h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	DACKS	DACE	DACKW	DACKSEL
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	DACKSEL	EDACKn Pin Toggling Select	0: EDACKn pin toggle is disabled. 1: EDACKn pin toggle is enabled.	R/W
b1	DACKW	EDACKn Pin Negate Wait	0: The EDACKn pin is negated at the same time as the RD# or WRn# pin is negated. 1: The EDACKn is negated one BCLK cycle before the RD# pin is negated or one BCLK cycle after the WRn# pin is negated.	R/W
b2	DACE	EDACKn Pin Output Enable	0: EDACKn output is disabled. 1: EDACKn output is enabled.	R/W
b3	DACKS	EDACKn Pin Output Polarity Select	0: EDACKn pin polarity is active low. 1: EDACKn pin polarity is active high.	R/W
b7 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register is used to set the output signal of the EXDMAC.

DACKSEL Bit (EDACKn Pin Toggling Select)

This bit is used to disable or enable toggled outputs on the EDACKn pin during normal, repeated, or block transfer to and from the SDRAM area in single-address mode.

If the DACKSEL bit is 0 during normal, repeated, or block transfer to and from the SDRAM area in single-address mode (i.e. while the EDMAMD.AMS bit is 1), the signal on the EDACKn pin is asserted throughout the interval where the data is valid (data-valid interval). If the DACKSEL bit is 1, the signal on the EDACKn pin is only asserted for one-half of the SDCLK period in the latter half of the data-valid interval.

The value of the DACKSEL pin has no effect in the case of CS areas.

The value of the DACKSEL pin also has no effect for transfer in dual-address mode and in cluster mode.

In these cases, there is no output on the EDACKn pin.

DACKW Bit (EDACKn Pin Negate Wait)

DACKW selects the EDACKn pin negation timing in single address mode during normal, repeat, or block transfer to/from the CS area.

During the above transfer in single address mode (EDMAMD.AMS bit = 1), the EDACKn pin is negated at the same time as the RD# or WRn# pin is negated if this bit is 0; and the EDACKn pin is negated one BCLK cycle before the RD# pin is negated, or one BCLK cycle after the WRn# pin is negated if this bit is 1. In the SDRAM area, setting this bit is invalid. EDACKn pin negation timing cannot be changed. Setting this bit is also invalid in dual address mode and during cluster transfer. In these cases, the EDACKn pin does not provide output.

DACE Bit (EDACKn Pin Output Enable)

DACE enables or disables EDACKn pin output. Setting this bit is invalid in dual address mode and during cluster transfer (EDACKn pin output not provided).

19.2.7 EXDMA Interrupt Setting Register (EDMINT)

Address(es): EXDMAC0.EDMINT 0008 2813h, EXDMAC1.EDMINT 0008 2853h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	DTIE	ESIE	RPTIE	SARIE	DARIE
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	DARIE	Destination Address Extended Repeat Area Overflow Interrupt Enable	0: Disables an interrupt request for an extended repeat area overflow on the destination address 1: Enables an interrupt request for an extended repeat area overflow on the destination address	R/W
b1	SARIE	Source Address Extended Repeat Area Overflow Interrupt Enable	0: Disables an interrupt request for an extended repeat area overflow on the source address 1: Enables an interrupt request for an extended repeat area overflow on the source address	R/W
b2	RPTIE	Repeat Size End Interrupt Enable	0: Disables the repeat size end interrupt request. 1: Enables the repeat size end interrupt request.	R/W
b3	ESIE	Transfer Escape End Interrupt Enable	0: Disables the transfer escape end interrupt request. 1: Enables the transfer escape end interrupt request.	R/W
b4	DTIE	Transfer End Interrupt Enable	0: Disables the transfer end interrupt request. 1: Enables the transfer end interrupt request.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register is used to set the interrupt request output of the EXDMAC.

DARIE Bit (Destination Address Extended Repeat Area Overflow Interrupt Enable)

When an extended repeat area overflow on the destination address occurs while this bit is set to 1, the EDMCNT.DTE bit is set to 0 (DMA transfer is disabled). At the same time, the EDMSTS.ESIF flag is set to 1 to indicate that an interrupt by an extended repeat area overflow on the destination address is requested.

When block transfer mode is used with the extended repeat area function, an interrupt is requested after completion of a 1-block size transfer. When setting 1 (DMA transfer is enabled) in the EDMCNT.DTE bit of the channel for which a transfer has been stopped, the transfer is resumed from the state when the transfer is stopped.

When the extended repeat area is not specified for the destination address, this bit is ignored.

SARIE Bit (Source Address Extended Repeat Area Overflow Interrupt Enable)

When an extended repeat area overflow on the source address occurs while this bit is set to 1, the DTE bit in EDMCNT is set to 0 (DMA transfer is disabled). At the same time, the EDMSTS.ESIF flag is set to 1 to indicate that an interrupt by an extended repeat area overflow on the source address is requested.

When block transfer mode is used with the extended repeat area function, an interrupt is requested after completion of a 1-block size transfer. When setting 1 (DMA transfer is enabled) in the EDMCNT.DTE bit of the channel for which a transfer has been stopped, the transfer is resumed from the state when the transfer is stopped.

When the extended repeat area is not specified for the source address, this bit is ignored.

RPTIE Bit (Repeat Size End Interrupt Enable)

When this bit is set to 1 in repeat transfer mode, the EDMCNT.DTE bit is set to 0 (DMA transfer is disabled) after completion of a 1-repeat size data transfer. At the same time, the EDMSTS.ESIF flag is set to 1 to indicate that the repeat size end interrupt request has been generated. The repeat size end interrupt request can be generated even when the EDMTMD.DTS[1:0] bits are 10b (repeat area or block area is not specified).

When this bit is set to 1 in block transfer mode, the EDMCNT.DTE bit is set to 0 after completion of a 1-block data transfer in the same way as repeat transfer mode. At the same time, the EDMSTS.ESIF flag is set to 1 to indicate that the repeat size end interrupt request has been generated. The repeat size end interrupt request can be generated even when the EDMTMD.DTS[1:0] bits are 10b.

When this bit is set to 1 in cluster transfer mode, the EDMCNT.DTE bit is set to 0 after completion of a 1-cluster data transfer in the same way as repeat transfer mode. At the same time, the EDMSTS.ESIF flag is set to 1 to indicate that the repeat size end interrupt request has been generated. The repeat size end interrupt request can be generated even when the EDMTMD.DTS[1:0] bits are 10b.

ESIE Bit (Transfer Escape End Interrupt Enable)

This bit enables or disables the transfer escape end interrupt requests (repeat size end interrupt request and extended repeat area overflow interrupt request) that are generated during DMA transfer.

The transfer escape end interrupt is generated when the ESIF flag in EXDMACn.EDMSTS is set to 1 with this bit set to 1. The transfer escape end interrupt is cleared by setting this bit or the EDMSTS.ESIF flag to 0.

DTIE Bit (Transfer End Interrupt Enable)

This bit enables or disables the transfer end interrupt request to be generated on completion of a specified number of data transfers.

The transfer end interrupt is generated when the EDMSTS.DTIF flag is set to 1 with this bit set to 1. The transfer end interrupt is cleared by setting this bit or the EDMSTS.DTIF flag to 0.

19.2.8 EXDMA Address Mode Register (EDMAMD)

Address(es): EXDMAC0.EDMAMD 0008 2814h, EXDMAC1.EDMAMD 0008 2854h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	AMS	DIR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	SM[1:0]	—				SARA[4:0]				DM[1:0]	—				DARA[4:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	DARA[4:0]	Destination Address Extended Repeat Area Set	Specifies the extended repeat area on the destination address. For details on the settings, refer to Table 19.3.	R/W
b5	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7, b6	DM[1:0]	Destination Address Update Mode Set	b7 b6 0 0: Destination address is fixed. 0 1: Offset addition*1 1 0: Destination address is incremented. 1 1: Destination address is decremented.	R/W
b12 to b8	SARA[4:0]	Source Address Extended Repeat Area Set	Specifies the extended repeat area on the source address. For details on the settings, refer to Table 19.3.	R/W
b13	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b15, b14	SM[1:0]	Source Address Update Mode Set	b15 b14 0 0: Destination address is fixed. 0 1: Offset addition*1 1 0: Destination address is incremented. 1 1: Destination address is decremented.	R/W
b16	DIR	Single Address Direction Select	0: Data is transferred in single address mode using the EDMSAR register value as the transfer source address. EDACKn is output to the transfer destination. 1: Data is transferred in single address mode using the EDMDAR register value as the transfer destination address. EDACKn is output to the transfer destination.	R/W
b17	AMS	Address Mode Select	0: Dual address mode 1: Single address mode	R/W
b31 to b18	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Offset addition can be specified only for EXDMAC0.

This register is used to set address mode of the EXDMACn.

DARA[4:0] Bits (Destination Address Extended Repeat Area Set)

These bits specify the extended repeat area on the destination address. The extended repeat area function is realized by updating the specified lower address bits with the remaining upper address bits fixed. The size of the extended repeat area can be any power of two between 21 (2 bytes) and 217 (128 Mbytes).

When the lower address overflows the extended repeat area by address increment, the top address of the extended repeat area is set. Similarly, when the lower address underflows the extended repeat area by address decrement, the bottom address of the extended repeat area is set.

When the repeat area or block area is specified as a transfer destination, do not specify the extended repeat area on the destination address. When repeat transfer, block transfer, or cluster transfer is selected, or when

EXDMACn.EDMTMD.DTS[1:0] = 00b (the transfer destination is specified as the repeat area or block area), write 00000b in the DARA[4:0] bits.

An interrupt can be requested when an overflow occurs in the extended repeat area with the DARIE bit in EDMINT set to 1. Table 19.3 lists the settings and the corresponding extended repeat areas.

DM[1:0] Bits (Destination Address Update Mode Set)

These bits select the mode of updating the destination address.

When increment is selected and the SZ[1:0] bits in EDMTMD are set to 00b, 01b, and 10b, the destination address is incremented by 1, 2, and 4, respectively.

When decrement is selected and the SZ[1:0] bits in EDMTMD are set to 00b, 01b, and 10b, the destination address is decremented by 1, 2, and 4, respectively.

When offset addition is selected, the offset specified by the EXDMAC0.EDMOFR register is added to the address.

Offset addition can be specified only for EXDMAC0.

SARA[4:0] Bits (Source Address Extended Repeat Area Set)

These bits specify the extended repeat area on the source address. The extended repeat area function is realized by updating the specified lower address bits with the remaining upper address bits fixed. The size of the extended repeat area can be any power of two between 21 (2 bytes) and 217 (128 Mbytes).

When the lower address overflows the extended repeat area by address increment, the top address of the extended repeat area is set. Similarly, when the lower address underflows the extended repeat area by address decrement, the bottom address of the extended repeat area is set.

When the repeat area or block area is specified as a transfer destination, do not specify the extended repeat area on the destination address. When repeat transfer, block transfer, or cluster transfer is selected, or when

EXDMACn.EDMTMD.DTS[1:0] = 01b (the transfer destination is specified as the repeat area or block area), write 00000b in the SARA[4:0] bits.

An interrupt can be requested when an overflow occurs in the extended repeat area with the SARIE bit in EDMINT set to 1. Table 19.3 lists the settings and the corresponding extended repeat areas.

SM[1:0] Bits (Source Address Update Mode Set)

These bits select the mode of updating the source address.

When increment is selected and the SZ[1:0] bits in EDMTMD are set to 00b, 01b, and 10b, the destination address is incremented by 1, 2, and 4, respectively.

When decrement is selected and the SZ[1:0] bits in EDMTMD are set to 00b, 01b, and 10b, the destination address is decremented by 1, 2, and 4, respectively.

When offset addition is selected, the offset specified by the EXDMAC0.EDMOFR register is added to the address.

Offset addition can be specified only for EXDMAC0.

DIR Bit (Single Address Direction Select)

This bit selects the transfer destination or source, to which the addresses should be output in single address mode.

- Normal/Repeat/Block Transfer

When this bit is set to 0, data is transferred in single address mode using the EDMSAR register value as the transfer source address. Here, EDACKn can be output to the transfer destination device by setting the DACK bit in EDMOMD to 1 (EDACKn output is enabled).

When this bit is set to 1, data is transferred in single address mode using the EDMDAR register value as the transfer destination address. Here, EDACKn can be output to the transfer source device by setting the DACK bit in EDMOMD to 1 (EDACKn output is enabled). Setting the DIR bit is valid when the AMS bit in EDMAMD is 1 (single address mode); setting the DIR bit is invalid when the AMS bit is 0 (dual address mode).

- Cluster Transfer

When this bit is set to 0, data is transferred in cluster transfer read address mode using the EDMSAR register value as the transfer source address. Here, data can be transferred to the cluster buffers from the external device.

When this bit is set to 1, data is transferred in cluster transfer write address mode using the EDMDAR register value as

the transfer destination address. Here, data can be transferred to the external device from the cluster buffers. Setting the DIR bit is valid when the AMS bit in EDMAMD is 1 (single address mode); setting the DIR bit is invalid when the AMS bit is 0 (dual address mode).

AMS Bit (Address Mode Select)

This bit selects the address mode.

- Normal/Repeat/Block Transfer

When this bit is set to 0, dual address mode is selected and when set to 1, single address mode is selected.

When using single address mode, select the transfer source or destination device to which the addresses should be output using the DIR bit in EDMAMD.

- Cluster Transfer

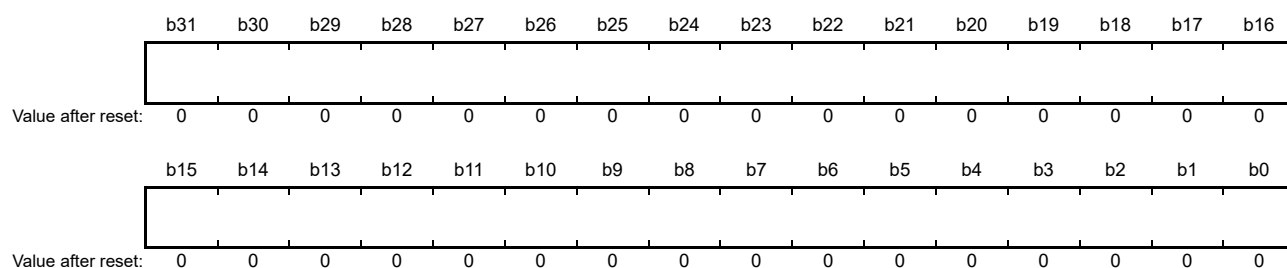
When this bit is set to 0, dual address mode is selected and when set to 1, read or write address mode is selected. Select read or write address mode using the DIR bit in EDMAMD.

Table 19.3 Settings and Range of Extended Repeat Areas

SARA[4:0] or DARA[4:0]	Extended Repeat Area
00000b	Not specified
00001b	2 bytes specified as extended repeat area by the lower 1 bit of the address
00010b	4 bytes specified as extended repeat area by the lower 2 bits of the address
00011b	8 bytes specified as extended repeat area by the lower 3 bits of the address
00100b	16 bytes specified as extended repeat area by the lower 4 bits of the address
00101b	32 bytes specified as extended repeat area by the lower 5 bits of the address
00110b	64 bytes specified as extended repeat area by the lower 6 bits of the address
00111b	128 bytes specified as extended repeat area by the lower 7 bits of the address
01000b	256 bytes specified as extended repeat area by the lower 8 bits of the address
01001b	512 bytes specified as extended repeat area by the lower 9 bits of the address
01010b	1 Kbyte specified as extended repeat area by the lower 10 bits of the address
01011b	2 Kbytes specified as extended repeat area by the lower 11 bits of the address
01100b	4 Kbytes specified as extended repeat area by the lower 12 bits of the address
01101b	8 Kbytes specified as extended repeat area by the lower 13 bits of the address
01110b	16 Kbytes specified as extended repeat area by the lower 14 bits of the address
01111b	32 Kbytes specified as extended repeat area by the lower 15 bits of the address
10000b	64 Kbytes specified as extended repeat area by the lower 16 bits of the address
10001b	128 Kbytes specified as extended repeat area by the lower 17 bits of the address
10010b	256 Kbytes specified as extended repeat area by the lower 18 bits of the address
10011b	512 Kbytes specified as extended repeat area by the lower 19 bits of the address
10100b	1 Mbyte specified as extended repeat area by the lower 20 bits of the address
10101b	2 Mbytes specified as extended repeat area by the lower 21 bits of the address
10110b	4 Mbytes specified as extended repeat area by the lower 22 bits of the address
10111b	8 Mbytes specified as extended repeat area by the lower 23 bits of the address
11000b	16 Mbytes specified as extended repeat area by the lower 24 bits of the address
11001b	32 Mbytes specified as extended repeat area by the lower 25 bits of the address
11010b	64 Mbytes specified as extended repeat area by the lower 26 bits of the address
11011b	128 Mbytes specified as extended repeat area by the lower 27 bits of the address
11100b to 11111b	Setting prohibited

19.2.9 EXDMA Offset Register (EDMOFR)

Address(es): EXDMAC0.EDMOFR 0008 2818h



Bit	Description	Setting Range	R/W
b31 to b0	Specifies the offset when offset addition is selected as the address update mode for transfer source or destination.	0000 0000h to 00FF FFFFh (0 bytes to (16M – 1) bytes) FF00 0000h to FFFF FFFFh (–16M bytes to –1 byte)	R/W

This register is used to set the address offset value.

Write to this register while the EXDMAC operation is stopped or DMA transfer is disabled (not during data transfer). Setting bits 31 to 25 is invalid; a value of bit 24 is extended to bits 31 to 25. Reading EDMOFR returns the extended value.

19.2.10 EXDMA Transfer Enable Register (EDMCNT)

Address(es): EXDMAC0.EDMCNT 0008 281Ch, EXDMAC1.EDMCNT 0008 285Ch

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	DTE
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	DTE	DMA Transfer Enable	0: Disables DMA transfer. 1: Enables DMA transfer.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register enables or disables the DMA transfer to the corresponding channel.

DTE Bit (DMA Transfer Enable)

When the EDMAST.DMST bit is set to 1 (EXDMAC module start) and this bit is set to 1 (DMA transfer is enabled), DMA transfer can be started for the corresponding channel.

When the DTE bit is set to 0 during DMA transfer, DMA transfer is suspended after completion of the current data transfer corresponding to a single transfer request. DMA transfer is resumed by setting the DTE bit to 1 again.

While the DTE bit is 1, writing to registers other than the DTE bit of the EXDMAC channel is prohibited.

[Setting condition]

- When 1 is written to this bit.

[Clearing conditions]

- When 0 is written to this bit.
- When the specified total volume of data transfer is completed.
- When DMA transfer is stopped by the repeat size end interrupt.
- When DMA transfer is stopped by the extended repeat area overflow interrupt.

19.2.11 EXDMA Software Start Register (EDMREQ)

Address(es): EXDMAC0.EDMREQ 0008 281Dh, EXDMAC1.EDMREQ 0008 285Dh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	CLRS	—	—	—	SWREQ
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	SWREQ	DMA Software Start	0: DMA transfer is not requested. 1: DMA transfer is requested.	R/W
b3 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	CLRS	DMA Software Start Bit Auto Clear Select	0: SWREQ bit is cleared after DMA transfer is started by software. 1: SWREQ bit is not cleared after DMA transfer is started by software.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register is used to execute the DMA by software.

SWREQ Bit (DMA Software Start)

When 1 is written to this bit, a DMA transfer request is generated. After DMA transfer is started in response to the request, this bit is set to 0 if the CLRS bit is set to 0. This bit is not set to 0 while the CLRS bit is set to 1. In this case, a DMA transfer request can be issued again after completion of a transfer.

Note that, however, setting this bit is valid and DMA transfer by software is enabled only when the DCTG[1:0] bits in EDMTMD are set to 00b (DMA request source is software).

Setting this bit is invalid when the DCTG[1:0] bits in EDMTMD are set to a value other than 00b.

To start DMA transfer by software with CLRS set to 0, check that the SWREQ bit is 0 and then write 1 to SWREQ.

[Setting condition]

- When 1 is written to this bit.

[Clearing conditions]

- When a DMA transfer request by software is accepted and DMA transfer is started while the CLRS bit is set to 0 (the SWREQ bit is cleared after DMA transfer is started by software).
- When 0 is written to this bit.

CLRS Bit (DMA Software Start Bit Auto Clear Select)

This bit specifies whether to clear the SWREQ bit to 0 after DMA transfer is started in response to the DMA transfer request generated by setting the SWREQ bit to 1. With this bit set to 0, the SWREQ bit is set to 0 after DMA transfer is started. With this bit set to 1, the SWREQ bit is not set to 0. In this case, a DMA transfer request can be issued again after completion of a transfer.

19.2.12 EXDMA Status Register (EDMSTS)

Address(es): EXDMAC0.EDMSTS 0008 281Eh, EXDMAC1.EDMSTS 0008 285Eh

b7	b6	b5	b4	b3	b2	b1	b0
ACT	—	—	DTIF	—	—	—	ESIF
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ESIF	Transfer Escape End Interrupt Flag	0: A transfer escape end interrupt has not been generated. 1: A transfer escape end interrupt has been generated.	R/W
b3 to b1	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b4	DTIF	Transfer End Interrupt Flag	0: A transfer end interrupt has not been generated. 1: A transfer end interrupt has been generated.	R/W
b6, b5	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b7	ACT	EXDMA Active Flag	0: EXDMAC operation is suspended. 1: EXDMAC is operating.	R

This register indicates the state of the DMA.

ESIF Flag (Transfer Escape End Interrupt Flag)

This flag indicates that the transfer escape end interrupt has been generated.

[Setting conditions]

- When 1-repeat size data transfer is completed in repeat transfer mode with the EDMINT.RPTIE bit set to 1.
- When 1-block data transfer is completed in block transfer mode with the EDMINT.RPTIE bit set to 1.
- When 1-cluster data transfer is completed in cluster transfer mode with the EDMINT.RPTIE bit set to 1.
- When an extended repeat area overflow on the source address occurs while the EDMINT.SARIE bit is set to 1 and the EDMAMD.SARA[4:0] bits are set to a value other than 00000b (extended repeat area is specified on the transfer source address)
- When an extended repeat area overflow on the destination address occurs while the EDMINT.DARIE bit is set to 1 and the EDMAMD.DADR[4:0] bits are set to a value other than 00000b (extended repeat area is specified on the transfer destination address)

[Clearing conditions]

- When 0 is written to this bit.
- When 1 is written to the EDMCNT.DTE bit.

DTIF Flag (Transfer End Interrupt Flag)

This flag indicates that the transfer end interrupt has been generated.

[Setting conditions]

- When the specified number of unit-transfers are completed in normal transfer mode (the value of EDMCRAL becoming 0 on completion of transfer)
- When the specified number of repeat transfer operations are completed in repeat transfer mode (the value of EDMCRB becoming 0 on completion of transfer)
- When the specified number of blocks have been transferred in block transfer mode (the value of EDMCRB becoming 0 on completion of transfer)
- When the specified number of clusters have been transferred in cluster transfer mode (the value of EDMCRB becoming 0 on completion of transfer)

[Clearing conditions]

- When 0 is written to this bit
- When 1 is written to the DTE bit in EDMCNT

ACT Flag (EXDMA Active Flag)

- This flag indicates whether the EXDMACn is in the idle or active state.

[Setting condition]

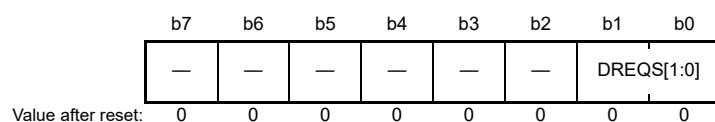
- When the EXDMACn starts data transfer operation

[Clearing condition]

- When data transfer in response to one transfer request is completed

19.2.13 EXDMA External Request Sense Mode Register (EDMRMD)

Address(es): EXDMAC0.EDMRMD 0008 2820h, EXDMAC1.EDMRMD 0008 2860h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	DREQS[1:0]	Request Input Sense Mode Set	b1 b0 0 0: Rising edge 0 1: Falling edge 1 0: Low level 1 1: (Setting prohibited)	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register is used to set sense mode of the EDREQn pin.

DREQS[1:0] Bits (Request Input Sense Mode Set)

These bits specify the sense mode for the external DMA transfer request signal (EDREQn pin).

19.2.14 EXDMA External Request Flag Register (EDMERF)

Address(es): EXDMAC0.EDMERF 0008 2821h, EXDMAC1.EDMERF 0008 2861h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	EREQ
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	EREQ	External Request Flag	0: No request 1: Requested	R/(W) *1
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Writing 0 has no effect.

This register detects a request generated by the EDREQn pin.

EREQ Flag (External Request Flag)

This flag indicates the DMA transfer request from the external DMA transfer request signal (the EDREQn pin).

[Setting conditions]

- When the level on the EDREQn pin changes from 0 to 1 while EXDMACn.EDMRMD.DREQS[1:0] = 00b
- When the level on the EDREQn pin changes from 1 to 0 while EXDMACn.EDMRMD.DREQS[1:0] = 01b
- When the level on the EDREQn pin is 0 while EXDMACn.EDMRMD.DREQS[1:0] = 10b (low level)

[Clearing conditions]

- When the DMA transfer is started while EXDMACn.EDMRMD.DREQS[1:0] = 00b (rising edge) or 01b (falling edge) and then the DMA transfer is started
- When 1 is written to this flag while EXDMACn.EDMRMD.DREQS[1:0] = 00b (rising edge) or 01b (falling edge)
- When the EDREQn pin is set to 1 while EXDMACn.EDMRMD.DREQS[1:0] = 10b (low level)

19.2.15 EXDMA Peripheral Request Flag Register (EDMPRF)

Address(es): EXDMAC0.EDMPRF 0008 2822h, EXDMAC1.EDMPRF 0008 2862h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	PREQ
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	PREQ	Peripheral Module Request Flag	0: No request 1: Requested	R/(W) *1
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Writing 0 has no effect.

This register detects a DMA transfer request generated by a peripheral module.

PREQ Flag (Peripheral Module Request Flag)

This flag detects the DMA transfer request from the peripheral modules.

[Setting condition]

- When the DMA transfer request is generated from the peripheral modules

[Clearing conditions]

- When the DMA transfer request is generated from the peripheral modules and the DMA transfer is started
- When 1 is written to this flag

19.2.16 EXDMAC Module Start Register (EDMAST)

Address(es): 0008 2A00h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	DMST
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	DMST	EXDMAC Module Start	0: EXDMAC module stop 1: EXDMAC module start	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register enables or disables the startup of all channels of the EXDMAC.

DMST Bit (EXDMAC Module Start)

When this bit is set to 1, EXDMAC is ready to accept transfer requests for all the channels.

When 1 is written to the DTE bit in EDMCNT (DMA transfer is enabled) of all the EXDMACn channels and then this bit is set to 1, all the channels can be placed in the transfer request acceptable state at the same time.

When the DMST bit is set to 0 during DMA transfer, DMA transfer for all channels is suspended after completion of the current data transfer corresponding to a single transfer request. DMA transfer is resumed by setting the DMST bit to 1 again.

[Setting condition]

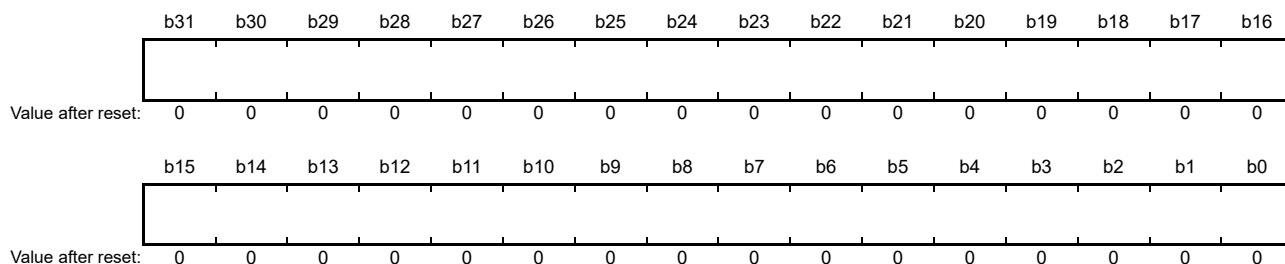
- When 1 is written to this bit

[Clearing condition]

- When 0 is written to this bit

19.2.17 Cluster Buffer Register y (CLSB_{Ry}) (y = 0 to 7)

Address(es): CLSBR0 0008 2BE0h, CLSBR1 0008 2BE4h, CLSBR2 0008 2BE8h, CLSBR3 0008 2BECh,
CLSBR4 0008 2BF0h, CLSBR5 0008 2BF4h, CLSBR6 0008 2BF8h, CLSBR7 0008 2BFCh



Bit	Description	R/W
b31 to b0	Buffer area for cluster transfer.	R/W

CLSB_{Ry} are buffer registers for cluster transfer.

During cluster transfer, transferred data is sequentially stored in CLSB_{Ry} starting from CLSBR0. The cluster-transferred data or data written by the CPU is retained until another cluster transfer or data write by the CPU. When reading the cluster-transferred data with the CPU, confirm that cluster transfer has been completed and only refer to the data of the specified size for cluster; the other data is invalid.

During cluster transfer, the same CLSB_{Ry} is used for all the channels. If a conflict occurs between the write to CLSB_{Ry} by the CPU and cluster transfer, transferred data is not guaranteed. If a channel is set to cluster transfer in read or write address mode and another channel is set to cluster transfer, data to be transferred may be erroneously modified.

Data is stored in cluster buffers in the different manner depending on the transfer size setting (EDMTMD.SZ[1:0] bits).

(1) Transfer Size is 8 Bits (EXDMACn.EDMTMD.SZ[1:0] = 00b)

Data is stored in the lower 8 bits in the cluster buffers. Here, the upper 24 bits are invalid. When the maximum cluster size is set to 8, 8-byte data is one cluster.

Data is stored in CLSBR in the order of CLSBR0 to CLSBR_j (j = cluster size value – 1).

(2) Transfer Size is 16 Bits (EXDMACn.EDMTMD.SZ[1:0] = 01b)

Data is stored in the lower 16 bits in the cluster buffers. Here, the upper 16 bits are invalid. When the maximum cluster size is set to 8, 16-byte data is one cluster.

Data is stored in CLSBR in the order of CLSBR0 to CLSBR_j (j = cluster size value – 1).

(3) Transfer Size is 32 Bits (EXDMACn.EDMTMD.SZ[1:0] = 10b)

Data is stored in all the 32 bits in the cluster buffers. When the maximum cluster size is set to 8, 32-byte data is one cluster.

Data is stored in CLSBR in the order of CLSBR0 to CLSBR_j (j = cluster size value – 1).

19.3 Operation

19.3.1 Transfer Mode

(1) Normal Transfer Mode

In normal transfer mode, one data is transferred by one transfer request. A maximum of 65535 can be set as the number of transfer operations using EDMCRA of EXDMACn. When the EXDMACn.EDMCRAL bits are set to 0000h, no specific number of transfer operations is set; data transfer is performed with the transfer counter stopped (free running mode). Setting EDMCRB of EXDMACn is invalid in normal transfer mode. Except in free running mode, a transfer end interrupt request can be generated after completion of the specified number of transfer operations.

Table 19.4 summarizes the register update operation in normal transfer mode, and Figure 19.2 shows the operation in normal transfer mode.

Table 19.4 Register Update Operation in Normal Transfer Mode

Register	Function	Update Operation after Completion of a Transfer by One Transfer Request
EXDMACn.EDMSAR	Transfer source address	Increment/decrement/fixed/offset addition*1
EXDMACn.EDMDAR	Transfer destination address	Increment/decrement/fixed/offset addition*1
EXDMACn.EDMCRAL	Transfer counter	Decrement by one/not updated (in free running mode)
EXDMACn.EDMCRAH	—	Not updated (Not used in normal transfer mode)
EXDMACn.EDMCRB	—	Not updated (Not used in normal transfer mode)

Note 1. Offset addition can be specified only for EXDMAC0.

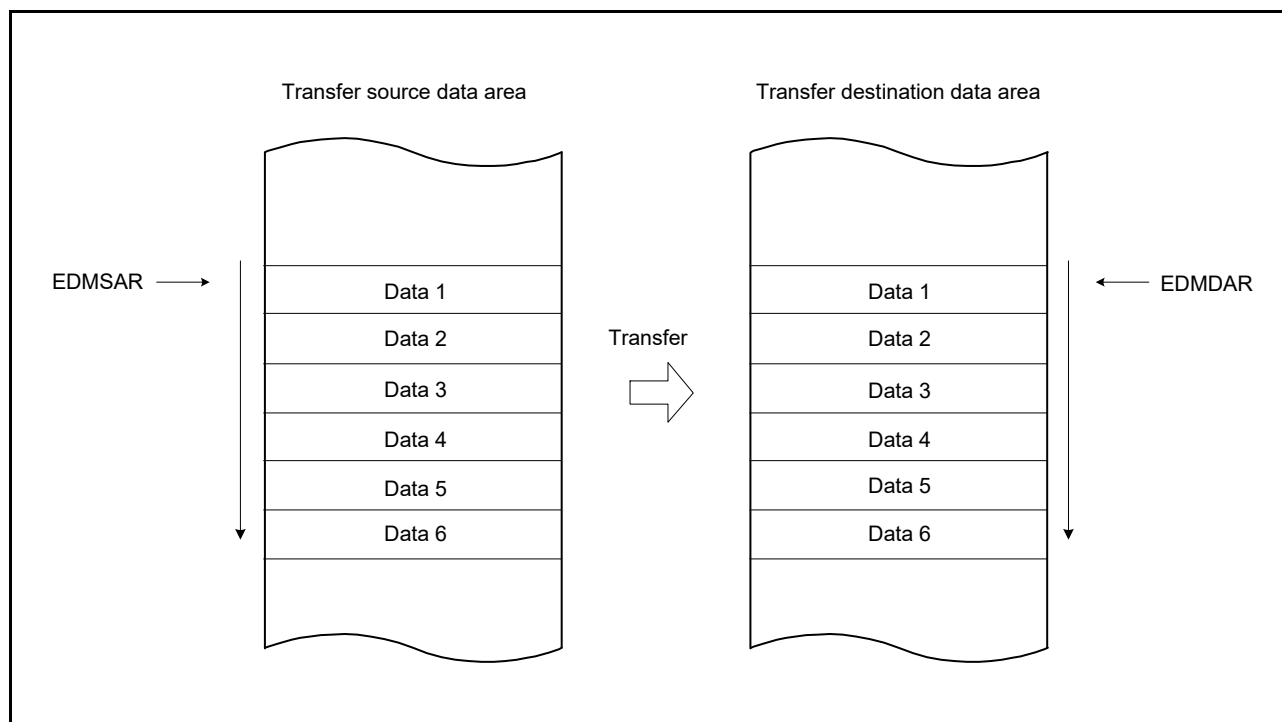


Figure 19.2 Operation in Normal Transfer Mode

(2) Repeat Transfer Mode

In repeat transfer mode, one data is transferred by one transfer request.

A maximum of 1K data can be set as a total repeat transfer size using EDMCRA of the EXDMACn.

A maximum of 1K can be set as the number of repeat transfer operations using EDMCRB of the EXDMACn; therefore, a maximum of 1K data $\times$ 1K = 1M can be set as a total data transfer size.

Either the transfer source or transfer destination can be specified as a repeat area. When transfer of the repeat size data is completed, the address of the specified repeat area (EDMSAR or EDMDAR of the EXDMACn) returns to the transfer start address. When data of the specified repeat size has all been transferred in repeat transfer mode, DMA transfer can be stopped and the repeat size end interrupt can be requested. DMA transfer can be resumed by writing 1 to the DTE bit in EDMCNT of EXDMACn in the repeat size end interrupt handling.

A transfer end interrupt request can be generated after completion of the specified number of repeat transfer operations. Table 19.5 summarizes the register update operation in repeat transfer mode, and Figure 19.3 shows the operation in repeat transfer mode.

Table 19.5 Register Update Operation in Repeat Transfer Mode

Register	Function	Update Operation after Completion of a Transfer by One Transfer Request	
		When EXDMACn.EDMCRAL is not 1	When EXDMACn.EDMCRAL is 1 (Transfer of the Last Data in Repeat Size)
EXDMACn.EDMSAR	Transfer source address	Increment/decrement/fixed/offset addition*1	- EXDMACn.EDMTMD.DTS[1:0] = 00b Increment/decrement/fixed/offset addition*1 - EXDMACn.EDMTMD.DTS[1:0] = 01b Initial value of EXDMACn.EDMSAR - EXDMACn.EDMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition*1
EXDMACn.EDMDAR	Transfer destination address	Increment/decrement/fixed/offset addition*1	- EXDMACn.EDMTMD.DTS[1:0] = 00b Initial value of EXDMACn.EDMDAR - EXDMACn.EDMTMD.DTS[1:0] = 01b Increment/decrement/fixed/offset addition*1 - EXDMACn.EDMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition*1
EXDMACn.EDMCRAH	Repeat size	Not updated	Not updated
EXDMACn.EDMCRAL	Transfer counter	Decrement by one	EXDMACn.EDMCRAH
EXDMACn.EDMCRB	Repeat-count counter	Not updated	Decrement by one

Note 1. Offset addition can be specified only for EXDMAC0.

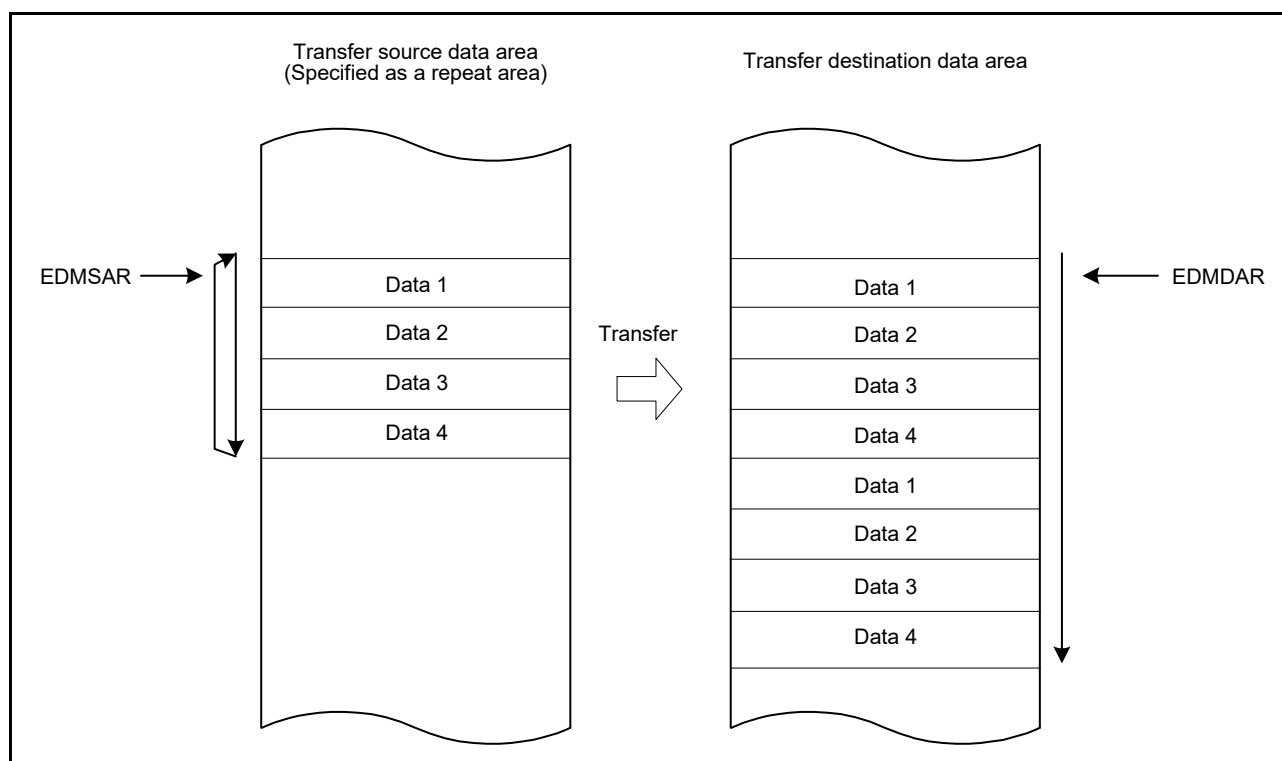


Figure 19.3 Operation in Repeat Transfer Mode

(3) Block Transfer Mode

In block transfer mode, a single block data is transferred by one transfer request.

A maximum of 1K data can be set as a total block transfer size using EDMCRA of the EXDMACn.

A maximum of 1K can be set as the number of block transfer operations using EDMCRB of the EXDMACn; therefore, a maximum of 1K data × 1K blocks = 1M can be set as a total data transfer size.

Either the transfer source or transfer destination can be specified as a block area. When transfer of a single block data is completed, the address of the specified block area (EDMSAR or EDMDAR of the EXDMACn) returns to the transfer start address. When a single block data has all been transferred in block transfer mode, DMA transfer can be stopped and the repeat size end interrupt can be requested. DMA transfer can be resumed by writing 1 to the DTE bit in EDMCNT of EXDMACn in the repeat size end interrupt handling.

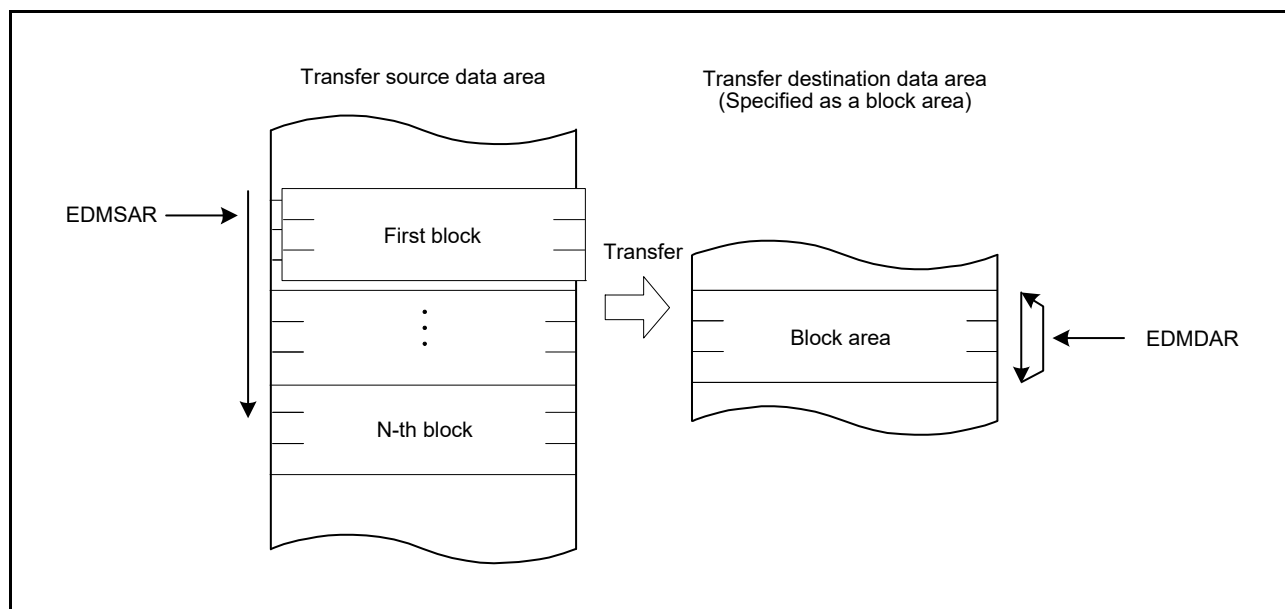
Transfer end interrupt request can be generated after completion of the specified number of block transfer operations.

Table 19.6 summarizes the register update operation in block transfer mode, and Figure 19.4 shows the operation in block transfer mode.

Table 19.6 Register Update Operation in Block Transfer Mode

Register	Function	Update Operation after Completion of Single-Block Transfer by One Transfer Request
EXDMACn.EDMSAR	Transfer source address	- EXDMACn.EDMTMD.DTS[1:0] = 00b Increment/decrement/fixed/offset addition*1 - EXDMACn.EDMTMD.DTS[1:0] = 01b Initial value of EXDMACn.EDMSAR - EXDMACn.EDMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition*1
EXDMACn.EDMDAR	Transfer destination address	- EXDMACn.EDMTMD.DTS[1:0] = 00b Initial value of EXDMACn.EDMDAR - EXDMACn.EDMTMD.DTS[1:0] = 01b Increment/decrement/fixed/offset addition*1 - EXDMACn.EDMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition*1
EXDMACn.EDMCRAH	Block size	Not updated
EXDMACn.EDMCRAL	Block size counter	EXDMACn.EDMCRAH
EXDMACn.EDMCRB	Transfer-block counter	Decrement by one

Note 1. Offset addition can be specified only for EXDMAC0.

**Figure 19.4 Operation in Block Transfer Mode**

(4) Cluster Transfer Mode

In cluster transfer mode, a single cluster data is transferred by one transfer request. A maximum of 8 data can be set as a total cluster transfer size using EDMCRA of the EXDMACn.

A maximum of 1K can be set as the number of cluster transfer operations using EDMCRB of the EXDMACn; therefore, a maximum of 8 data × 1K = 8K can be set as a total data transfer size.

The cluster transfer mode can be selected from among cluster transfer dual address mode, cluster transfer read address mode, and cluster transfer write address mode.

- Cluster transfer dual address mode

(EXDMACn.EDMTMD.MD[1:0] = 11b, EXDMACn.EDMAMD.AMS = 0)

A single cluster data is transferred by one transfer request from the transfer source address to the cluster buffers. A single cluster data is then transferred from the cluster buffers to the transfer destination address.

- Cluster transfer read address mode

(EXDMACn.EDMTMD.MD[1:0] = 11b, EXDMACn.EDMAMD.AMS = 1, EXDMACn.EDMAMD.DIR = 0)

A single cluster data is transferred by one transfer request from the transfer source address to the cluster buffers.

- Cluster transfer write address mode

(EXDMACn.EDMTMD.MD[1:0] = 11b, EXDMACn.EDMAMD.AMS = 1, EXDMACn.EDMAMD.DIR = 1)

A single cluster data is transferred by one transfer request from the cluster buffers to the transfer destination address.

In cluster-transfer mode, DMA transfer stops on completion of the transfer of each cluster of data, and a repeat-size-completed interrupt request can be generated. DMA transfer can be restarted by writing 1 to the EXDMACn.EDMCNT.DTE bit during processing of the repeat-size-completed interrupt.

A repeat-size-completed interrupt request can also be generated on completion of transfer of clusters the specified number of times.

Table 19.7 summarizes the register update operation in cluster transfer mode, and Figure 19.5 shows the operation in cluster transfer mode.

Table 19.7 Register Update Operation in Cluster Transfer Mode (Dual Address Mode)

Register	Function	Update Operation after Completion of Single-Cluster Transfer by One Transfer Request
EXDMACn.EDMSAR	Transfer source address	- EXDMACn.EDMTMD.DTS[1:0] = 00b Increment/decrement/fixed/offset addition*1 - EXDMACn.EDMTMD.DTS[1:0] = 01b Initial value of EXDMACn.EDMSAR - EXDMACn.EDMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition*1
EXDMACn.EDMDAR	Transfer destination address	- EXDMACn.EDMTMD.DTS[1:0] = 00b Initial value of EXDMACn.EDMDAR - EXDMACn.EDMTMD.DTS[1:0] = 01b Increment/decrement/fixed/offset addition*1 - EXDMACn.EDMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition*1
EXDMACn.EDMCRAH	Cluster size	Not updated
EXDMACn.EDMCRAL	Cluster size counter	EXDMACn.EDMCRAH
EXDMACn.EDMCRB	Transfer cluster count counter	Decrement by one

Note 1. Offset addition can be specified only for EXDMAC0.

In read address mode, the transfer destination address EXDMACn.EDMDAR is fixed (invalid).

In write address mode, the transfer source address EXDMACn.EDMSAR is fixed (invalid).

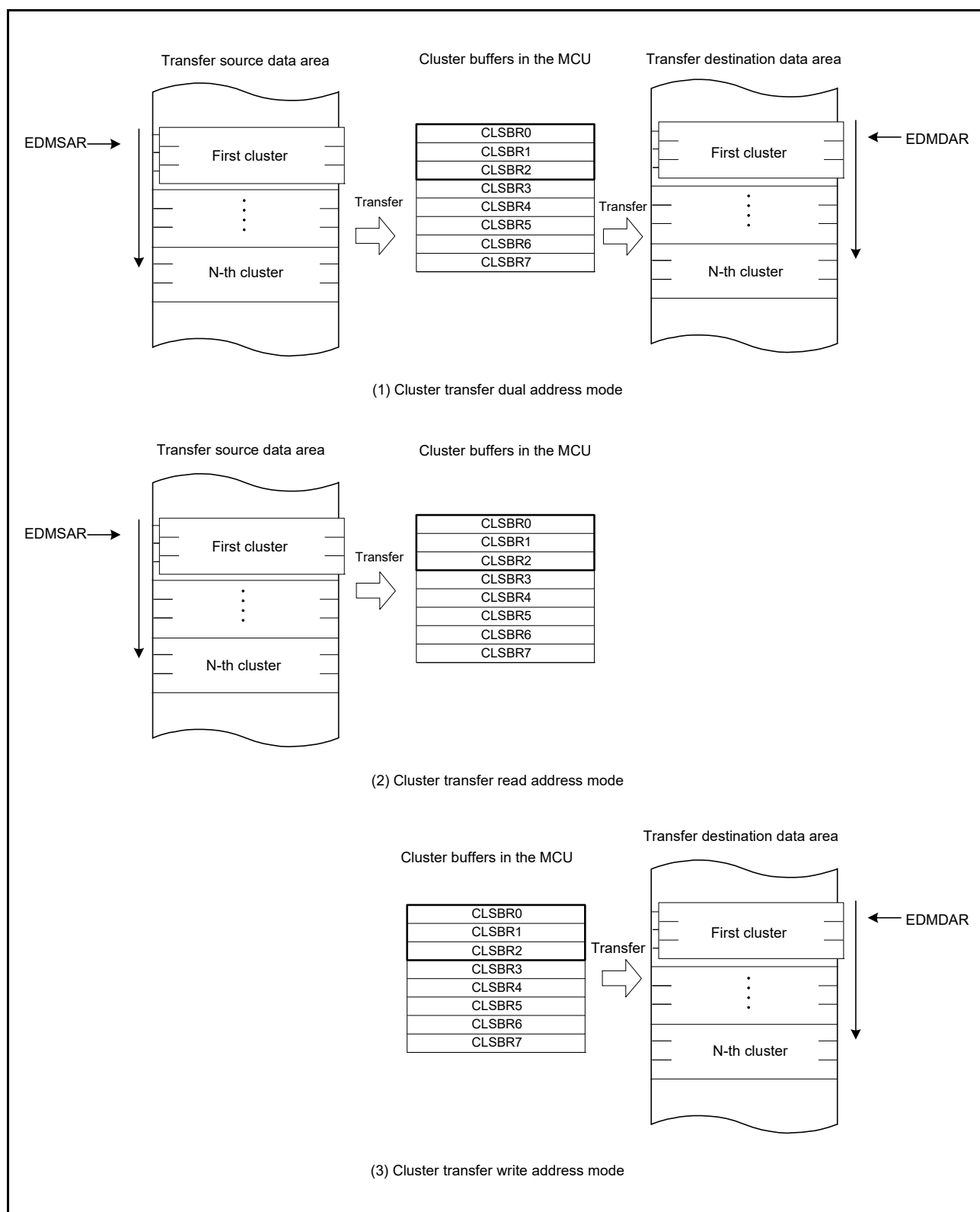


Figure 19.5 Operation in Cluster Transfer Mode

19.3.2 Extended Repeat Area Function

The EXDMAC supports a function to specify the extended repeat areas on the transfer source and destination addresses. With the extended repeat areas set, the address registers repeatedly indicate the addresses of the specified extended repeat areas.

The extended repeat areas can be specified separately to the transfer source address register (EDMSAR) and transfer destination address register (EDMDAR) of EXDMACn.

The extended repeat area on the source address is specified by the SARA[4:0] bits in EDMAMD of EXDMACn. The extended repeat area on the destination address is specified by the DARA[4:0] bits in EDMAMD of EXDMACn. The size can be specified separately for the source and destination sides. However, do not specify a repeat area or block area that is also an extended repeat area.

When the address register value reaches the end address of the extended repeat area and the extended repeat area overflows, DMA transfer is stopped and an interrupt by an extended repeat area overflow can be requested. When an overflow occurs in the extended repeat area on the transfer source while the SARIE bit in EDMINT of EXDMACn is set to 1, the ESIF flag in EDMSTS of EXDMACn is set to 1 and the DTE bit in EDMINT of EXDMACn is set to 0 to stop DMA transfer. At this time, if the ESIE bit in EDMINT of EXDMACn is set to 1, an interrupt by an extended repeat area overflow is requested.

When the DARIE bit in EDMINT of EXDMACn is set to 1, an overflow on the extended repeat area set in EDMDAR occurs, meaning that the destination side is a target. DMA transfer can be resumed by writing 1 to the DTE bit in EDMCINT of EXDMACn in the extended repeat area overflow interrupt handling.

Figure 19.6 shows an example of the extended repeat area operation.

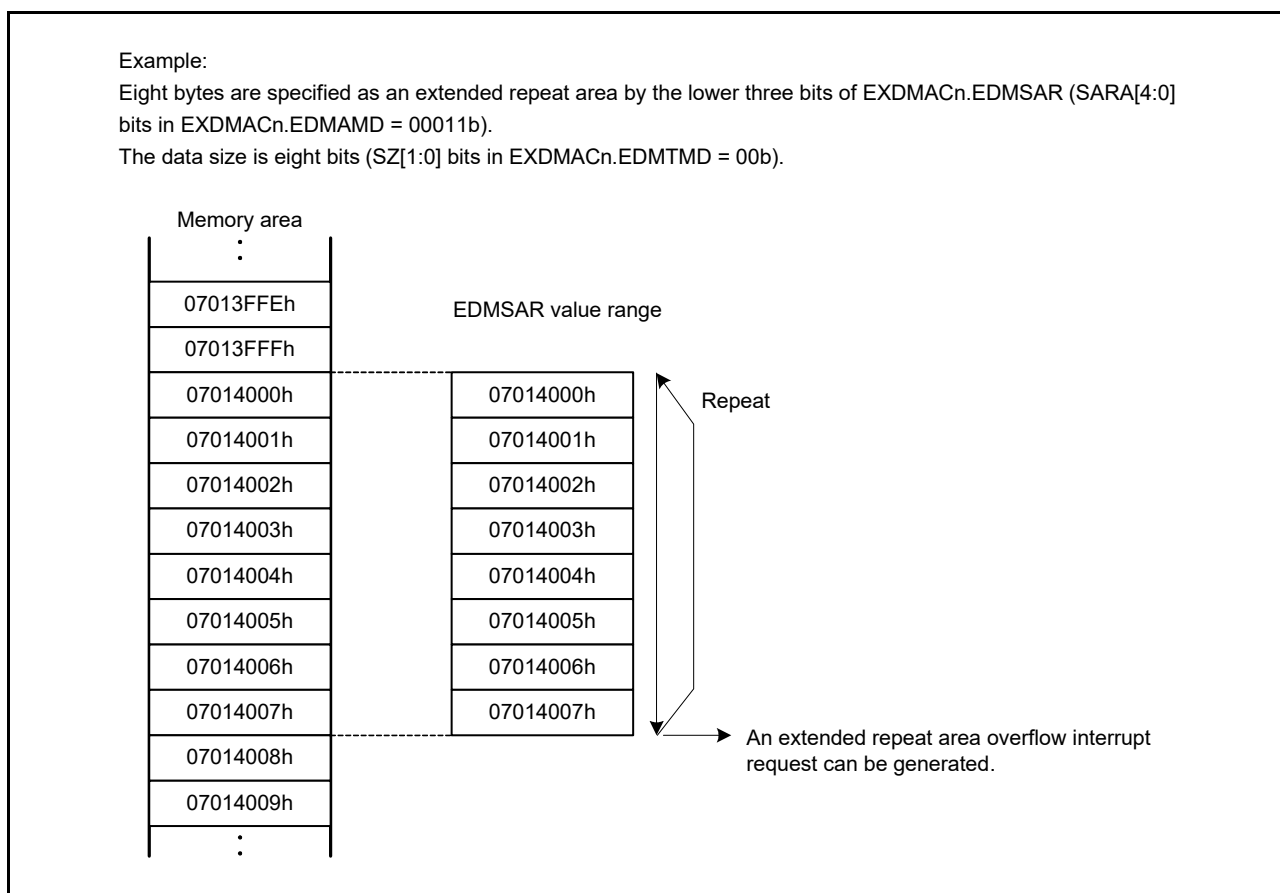


Figure 19.6 Example of Extended Repeat Area Operation

When an interrupt by an extended repeat area overflow is used in block transfer mode or cluster transfer mode, the following should be taken into consideration.

When a transfer is stopped by an interrupt by an extended repeat area overflow, the address register must be set so that the block size (or cluster size) is a power of 2 or the block size (or cluster size) boundary is aligned with the extended repeat area boundary. When an overflow on the extended repeat area occurs during a transfer of one block (or one cluster), the interrupt by the overflow is suspended until transfer of the block (or the cluster) is completed, and the transfer overruns.

Figure 19.7 shows an example when the extended repeat area function is used in block transfer mode.

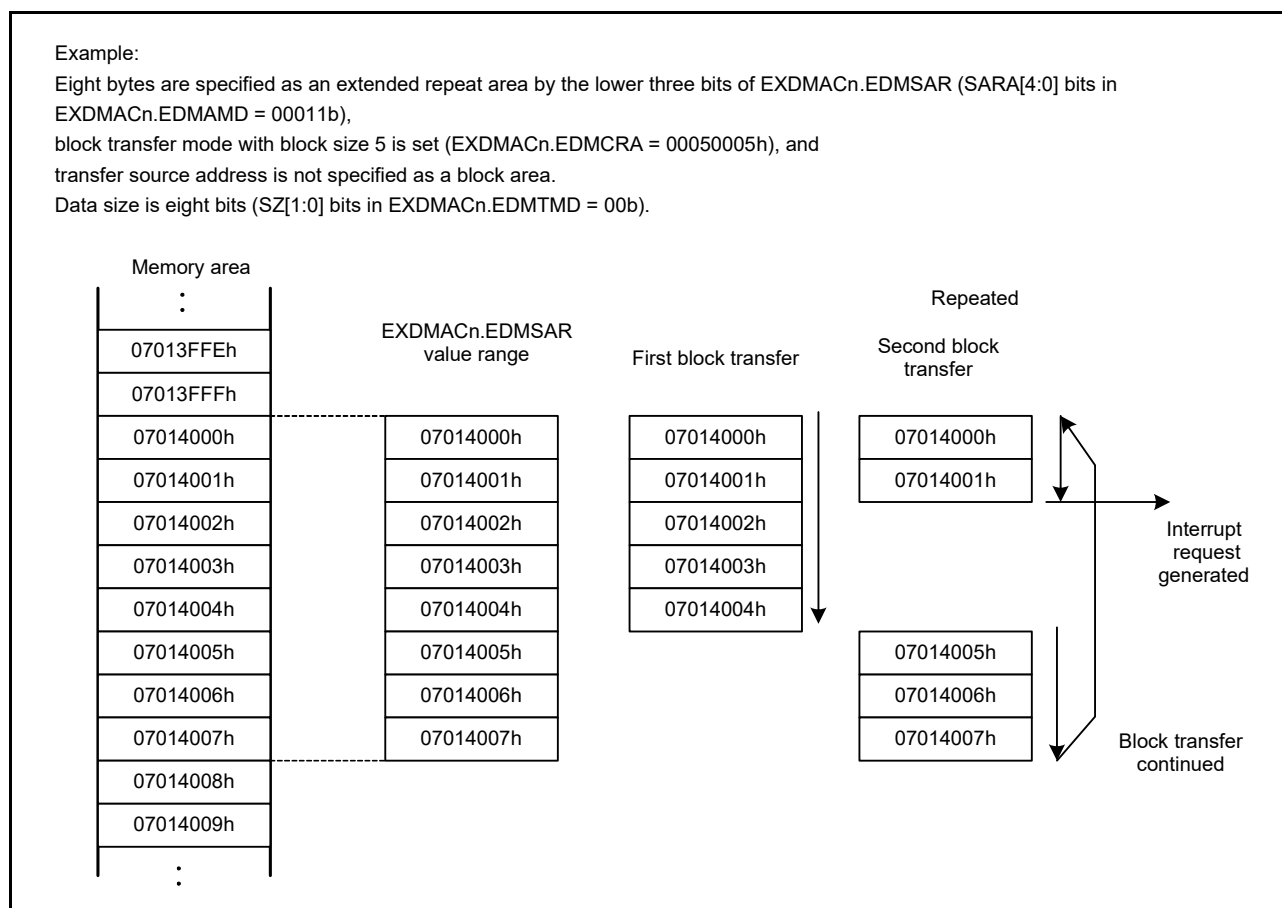


Figure 19.7 Example of Extended Repeat Area Function in Block Transfer Mode

19.3.3 Address Update Function using Offset

The source and destination addresses can be updated by fixing, increment, decrement, or offset addition. When the offset addition is selected, the offset specified by the EXDMA offset register (EDMOFR of EXDMAC0) is added to the address every time the EXDMAC performs one data transfer. This function realizes a data transfer where addresses are allocated to separated areas.

Offset subtraction can also be realized by setting a negative value in EDMOFR of EXDMAC0. In this case, the negative value must be 2's complement.

Address update function using offset can be specified only for the EXDMAC0 channel.

Table 19.8 lists the address update method in each address update mode.

Table 19.8 Address Update Method in Each Address Update Mode

Address Update Mode	Settings of EXDMACn.EDMAMD.SM[1:0] and EXDMACn.EDMAMD.DM[1:0] for Address Update Modes	Address Update Method (for Different SZ[1:0] Settings in EDTMD of EXDMACn)		
		SZ[1:0] = 00b	SZ[1:0] = 01b	SZ[1:0] = 10b
Address fixed	00b	Fixed		
Offset addition	01b	+EXDMAC0.EDMOFR*1		
Increment	10b	+1	+2	+4
Decrement	11b	-1	-2	-4

Note 1. When setting a negative value in the EXDMA offset register, the value must be 2's complement. The 2's complement is obtained by the following formula.

2's complement of a negative offset value = $\sim(\text{offset}) + 1$ ($\sim$: bit inversion)

(1) Basic Transfer Using Offset Addition

Figure 19.8 shows an example of address updating using offset addition.

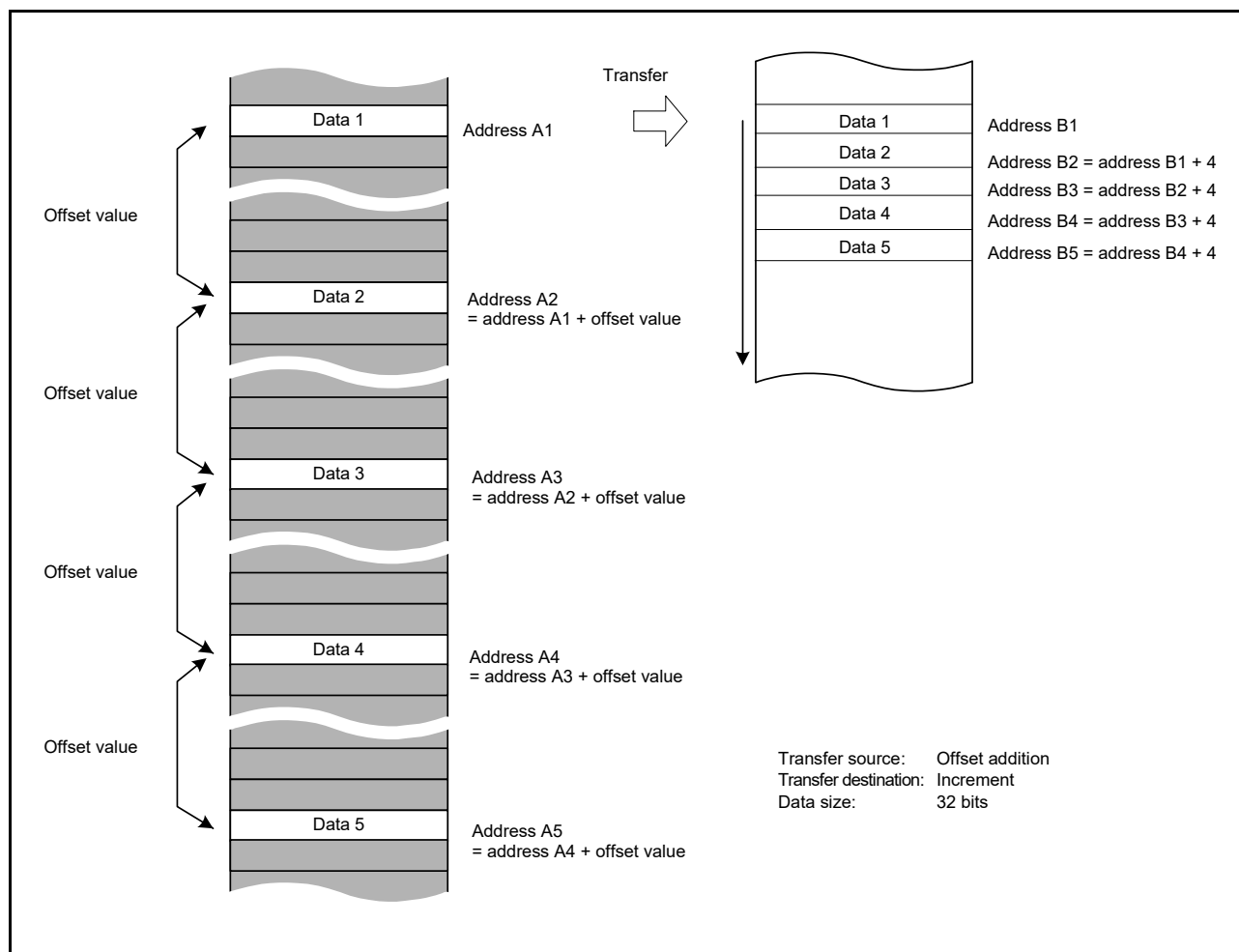


Figure 19.8 Example of Address Updating by Offset Addition

In Figure 19.8, the transfer data is 32 bits long, and offset addition and increment are set as the transfer source address update mode and transfer destination address update mode, respectively. The second and subsequent data is each read from the transfer source address obtained by adding the offset value to the previous address. The data read from the addresses at the specified intervals is written to the continuous locations on the destination.

(2) Example of XY Conversion Using Offset Addition

Figure 19.9 shows the XY conversion using offset addition in repeat transfer mode.

The settings are as follows.

- EXDMAC0.EDMAMD register: Source address update mode (offset addition)
- EXDMAC0.EDMAMD register: Source address update mode (incremented)
- EXDMAC0.EDMTMD register: Transfer data size select (32-bit transfer)
- EXDMAC0.EDMTMD register: Transfer mode select (repeat transfer)
- EXDMAC0.EDMTMD register: Repeat area select (the source is specified as the repeat area)
- EXDMAC0.EDMOFR register: Address offset setting (10h)
- EXDMAC0.EDMCRA register: The number of repeat transfer setting (4h)
- EXDMAC0.EDMINT register: Repeat size end interrupt enable

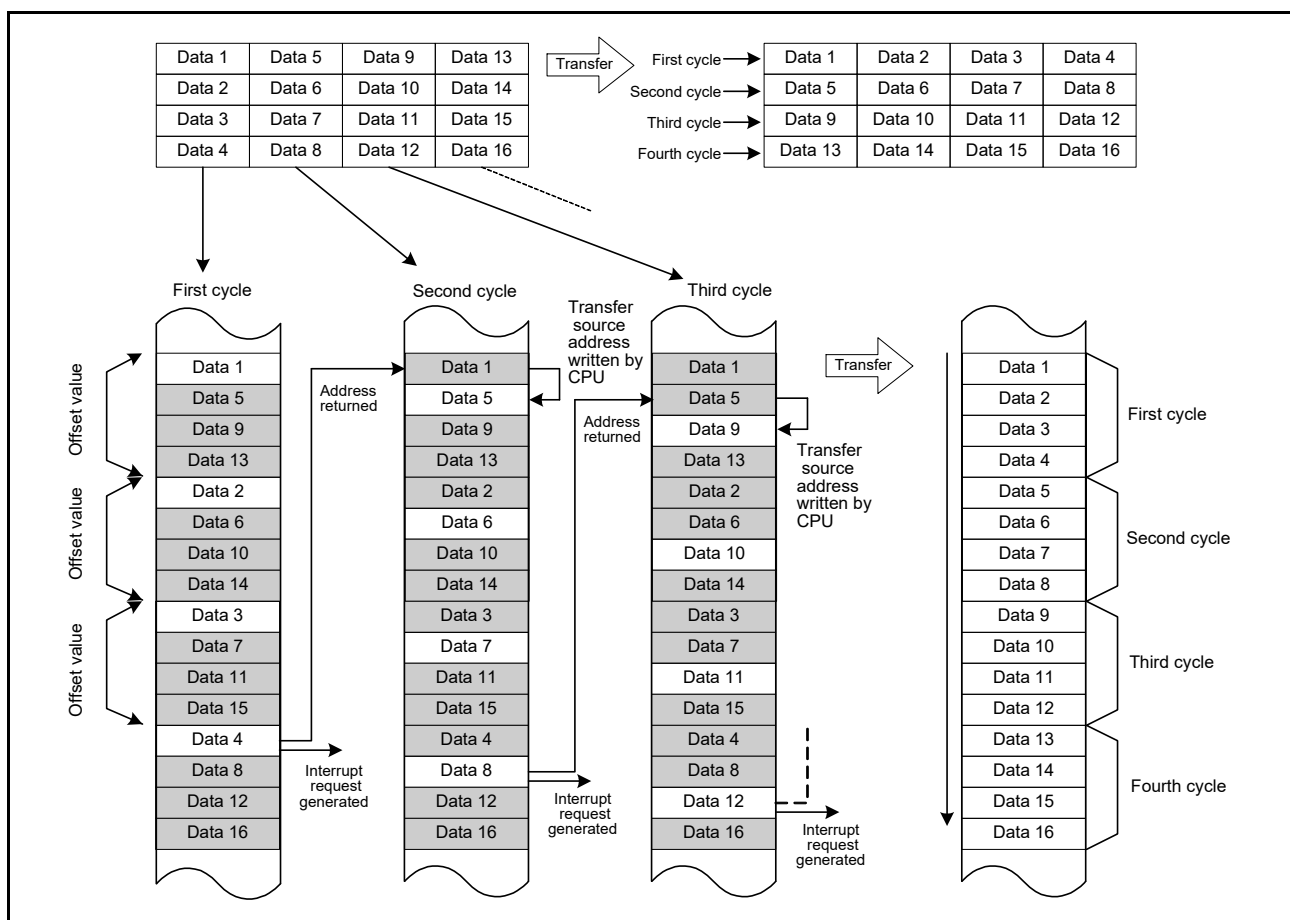


Figure 19.9 XY Conversion Operation Using Offset Addition in Repeat Transfer Mode

When a transfer starts, the offset value is added to the transfer source address every time data is transferred. The transfer data is written to the destination continuous addresses. When data 4 is transferred, which means that the repeat size of transfers is completed, EXDMAC returns the transfer source address to the transfer start address (address of data 1 on the transfer source) and a repeat size end interrupt is requested. While this interrupt stops the transfer temporarily, perform the following.

- EXDMAC0.EDMSAR: Rewrite the DMA transfer source address to the address of data 5 (with the above example, the data 1 address + 4).
- EXDMAC0.EDMCNT: Set the DTE bit to 1.

The DMA transfer is resumed from the state when the DMA transfer is stopped. After that, the operations described above are repeated until the transfer source data is transposed to the destination area (XY conversion)

Figure 19.10 shows a flowchart of the XY conversion.

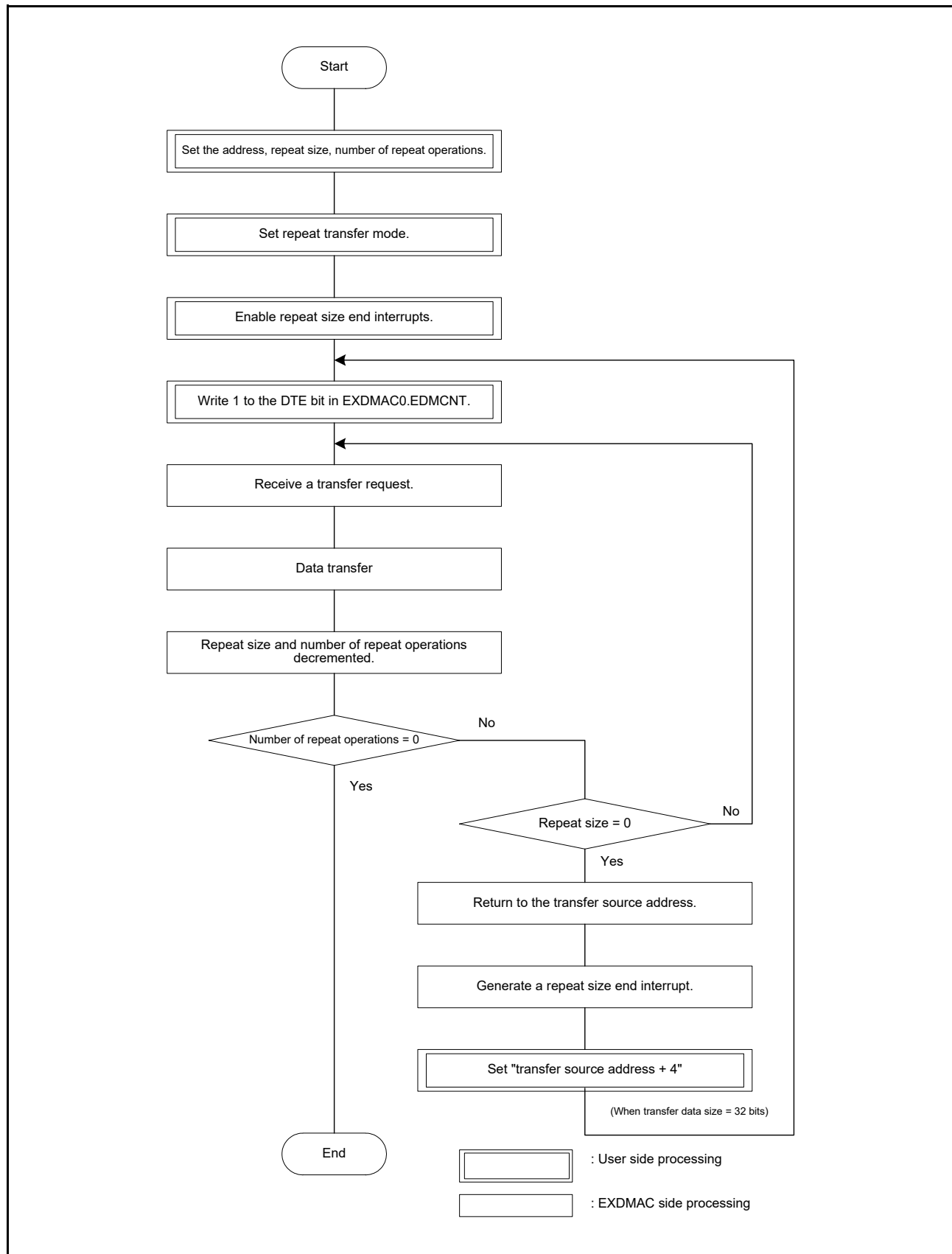


Figure 19.10 XY Conversion Flowchart Using Offset Addition in Repeat Transfer Mode

19.3.4 Address Modes

The EXDMAC provides dual and single address modes (dual, read, and write address modes in cluster transfer), either of which can be selectable. Table 19.9 lists the relationship between transfer modes and address modes.

Table 19.9 Relationship between Transfer Modes and Address Modes

Transfer Mode	Address Mode	Single Address Direction	EXDMAC Operation
Normal transfer mode (EDMTMD.MD[1:0] = 00b)	Dual address mode (EDMAMD.AMS = 0)	—	Reads and then writes data.
	Single address mode (EDMAMD.AMS = 1)	Transfer source (EDMAMD.DIR = 0)	Only reads data; outputs EDACKn to the device to be written to. This MCU receives no read data.
		Transfer destination (EDMAMD.DIR = 1)	Only writes data; outputs EDACKn to the device to be read from. This MCU outputs no write data.
Repeat transfer mode (EDMTMD.MD[1:0] = 01b)	Dual address mode (EDMAMD.AMS = 0)	—	Reads and then writes data.
	Single address mode (EDMAMD.AMS = 1)	Transfer source (EDMAMD.DIR = 0)	Only reads data; outputs EDACKn to the device to be written to. This MCU receives no read data.
		Transfer destination (EDMAMD.DIR = 1)	Only writes data; outputs EDACKn to the device to be read from. This MCU outputs no write data.
Block transfer mode (EDMTMD.MD[1:0] = 10b)	Dual address mode (EDMAMD.AMS = 0)	—	Reads and writes data alternately for every data transfer specified by EDMTMD.SZ[1:0] (transfer data size)
	Single address mode (EDMAMD.AMS = 1)	Transfer source (EDMAMD.DIR = 0)	Only reads data; outputs EDACKn to the device to be written to. This MCU receives no read data.
		Transfer destination (EDMAMD.DIR = 1)	Only writes data; outputs EDACKn to the device to be read from. This MCU outputs no write data.
Cluster transfer mode (EDMTMD.MD[1:0] = 11b)	Dual address mode (EDMAMD.AMS = 0)	—	Reads data of cluster size and then writes data of cluster size.
	Read address mode (EDMAMD.AMS = 1)	Transfer source (EDMAMD.DIR = 0)	Only reads data of cluster size. Transfers data to the cluster buffers.
	Write address mode (EDMAMD.AMS = 1)	Transfer destination (EDMAMD.DIR = 1)	Only writes data of cluster size. Transfers data from the cluster buffers.

19.4 Transfer Operation

Descriptions of examples of operations in transfer by the EXDMAC are given in the following passages.

Operations of the EXDMAC are synchronized by the external bus clock (BCLK). The examples that follow are for cases where the external bus clock (BCLK) and the signal output on the BCLK pin are at the same frequency unless there is a particular reason for doing otherwise.

Note: An idle cycle is inserted between two adjacent accesses depending on the external bus recovery cycle setting. One idle cycle, however, is inserted between a read access and a write access even if the recovery cycle is set to 0. refer to section 16, Buses, for details on the recovery cycle.

19.4.1 Normal/Repeat Transfer Operation

(1) Dual Address Mode

Figure 19.11 shows the bus cycle example in normal-transfer dual address mode. In the example, a 16-bit data (SZ[1:0] = 01b in EDMTMD of EXDMACn) is transferred from a device with 16-bit 2-cycle access to another device with 16-bit 2-cycle access, which is started at the falling edge of EDREQn.

The bus cycles in repeat-transfer dual address mode are the same as those in normal-transfer dual address mode.

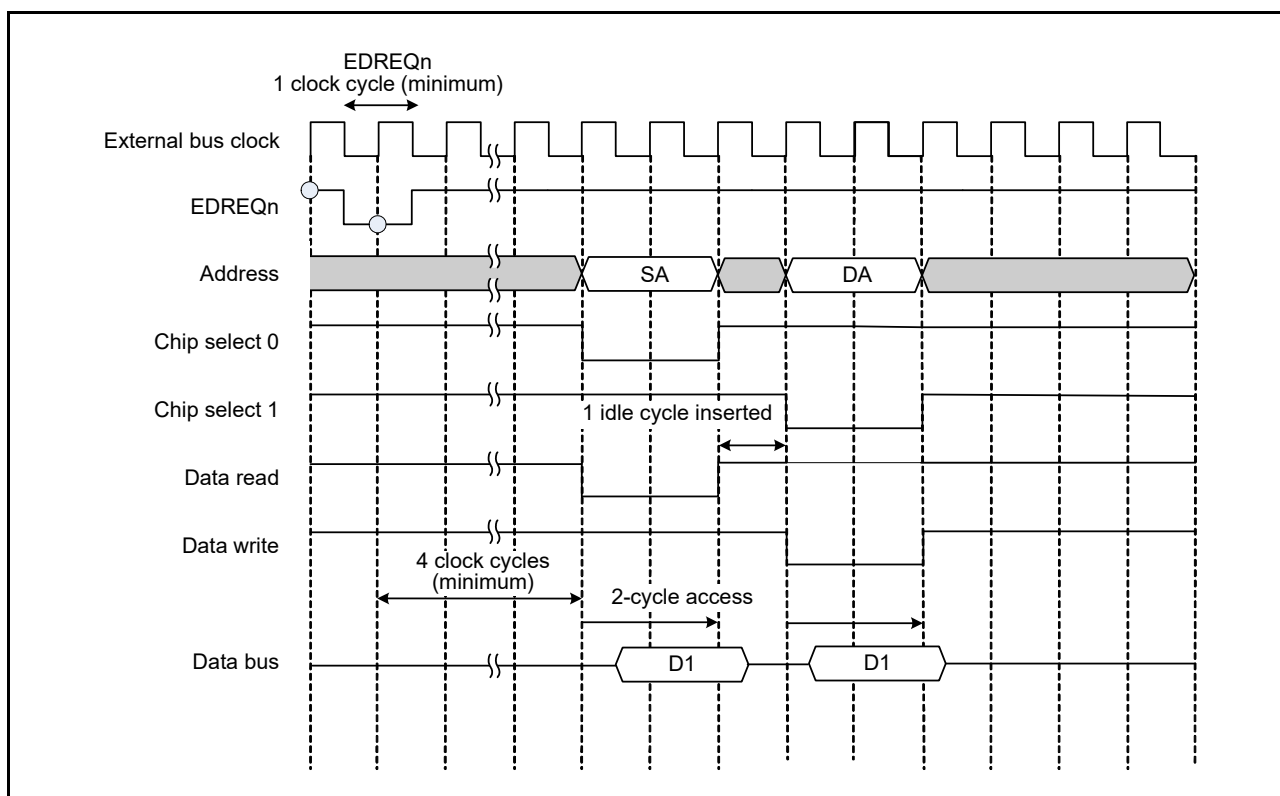


Figure 19.11 Bus Cycle Example in Normal-Transfer Dual Address Mode

(2) Single Address Mode

In single address mode, data is read from the transfer source address and directly transferred to the transfer destination device without being taken in the MCU. Here, EDACKn is output to one of the external transfer-destination and transfer-source devices, and the address is simultaneously output to the other transfer device for access.

With the DIR bit in EDMAMD of EXDMACn set to 0, the transfer source address is output to the external bus and EDACKn is output to the transfer destination. With the DIR bit in EDMAMD of EXDMACn set to 1, the transfer destination address is output to the external bus and the EDACKn is output to the transfer source. Figure 19.12 shows the data flow in single address mode.

Figure 19.13 shows the bus cycle example in normal-transfer single address mode. In the example, one data is transferred in 2-cycle access when the DIR bit in EDMAMD of EXDMACn is set to 1 (transfer destination address is output) and when set to 0 (transfer source address is output).

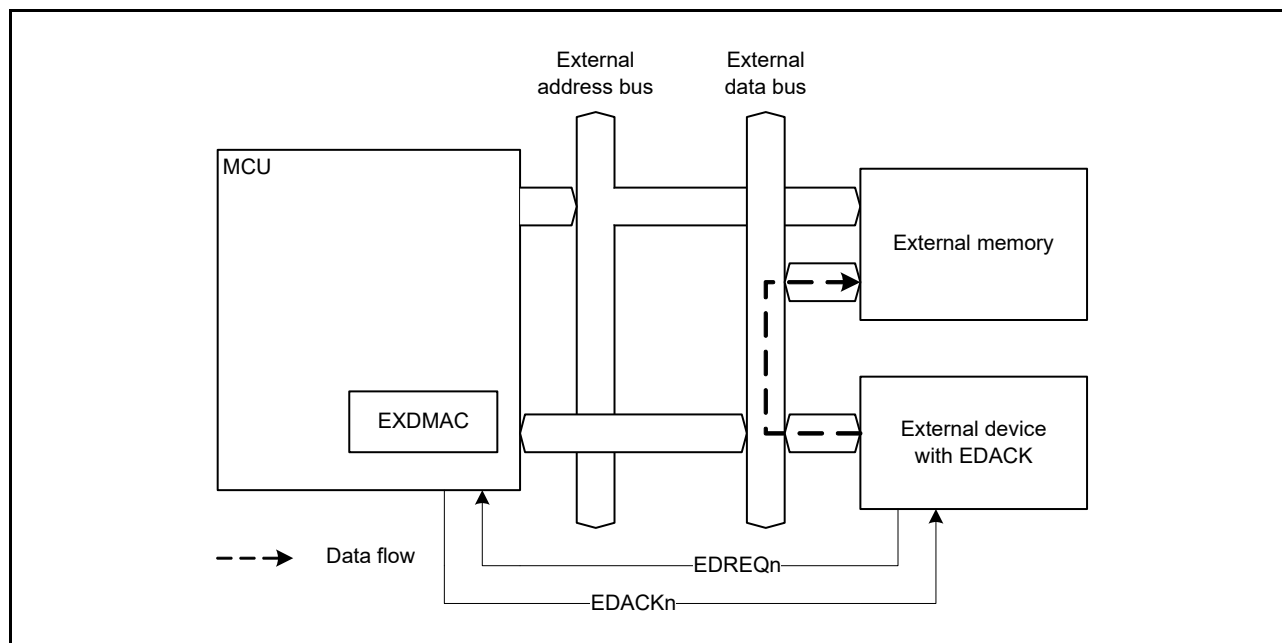


Figure 19.12 Data Flow in Single Address Mode (when EDMAMD.DIR = 1)

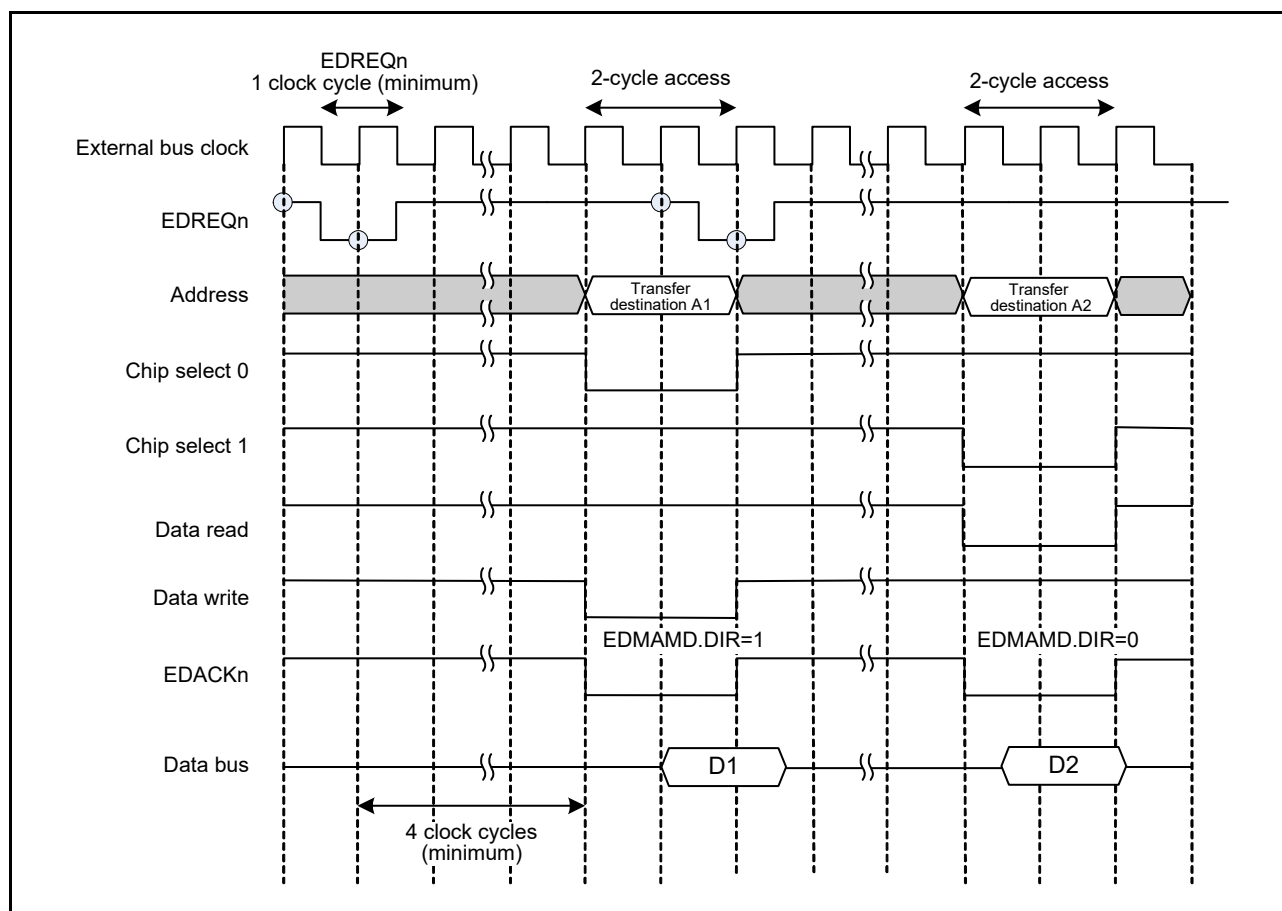


Figure 19.13 Bus Cycle Example in Normal-Transfer Single Address Mode

19.4.2 Block Transfer Operation

(1) Dual Address Mode

Figure 19.14 shows the bus cycle example in block-transfer dual address mode. In the example, a 16-bit data (SZ[1:0] = 01 in EDMTMD of EXDMACn) is transferred from a device with 16-bit 2-cycle access to another device with 16-bit 2-cycle access when the block size is 3.

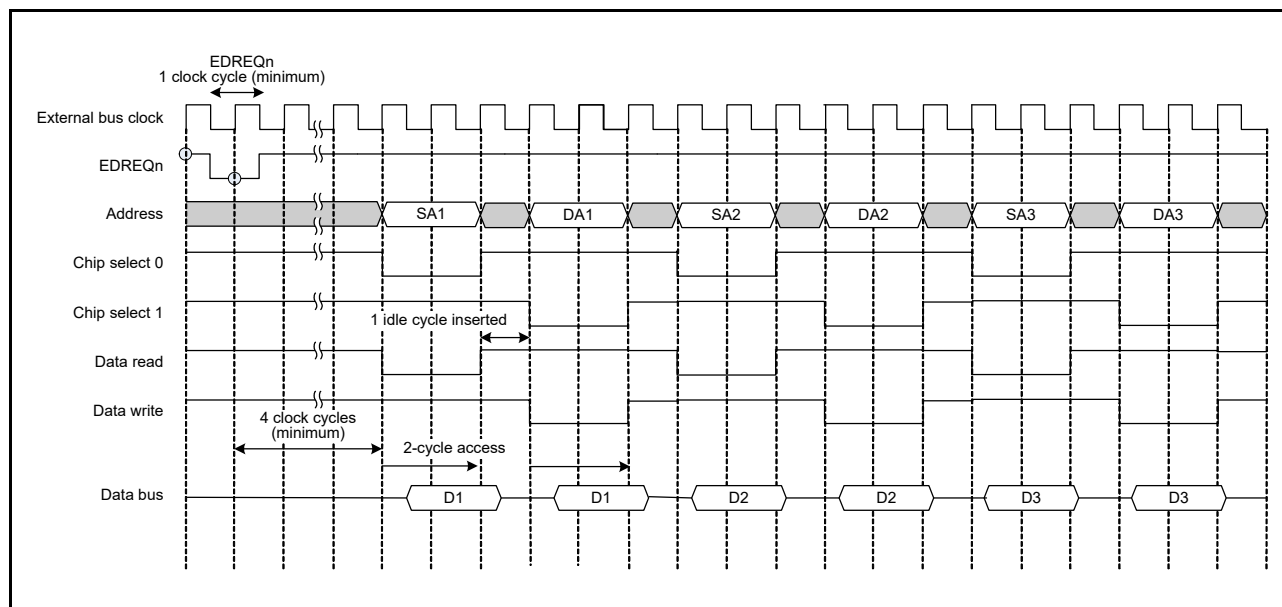


Figure 19.14 Bus Cycle Example in Block-Transfer Dual Address Mode

(2) Single Address Mode

Figure 19.15 shows the bus cycle example in block-transfer single address mode. In the example, a 16-bit data (SZ[1:0] = 01 in EDMTMD of EXDMACn) is transferred in three bus clock cycles from a device with EDACKn with 16-bit access to another device with 16-bit access when the block size is 3.

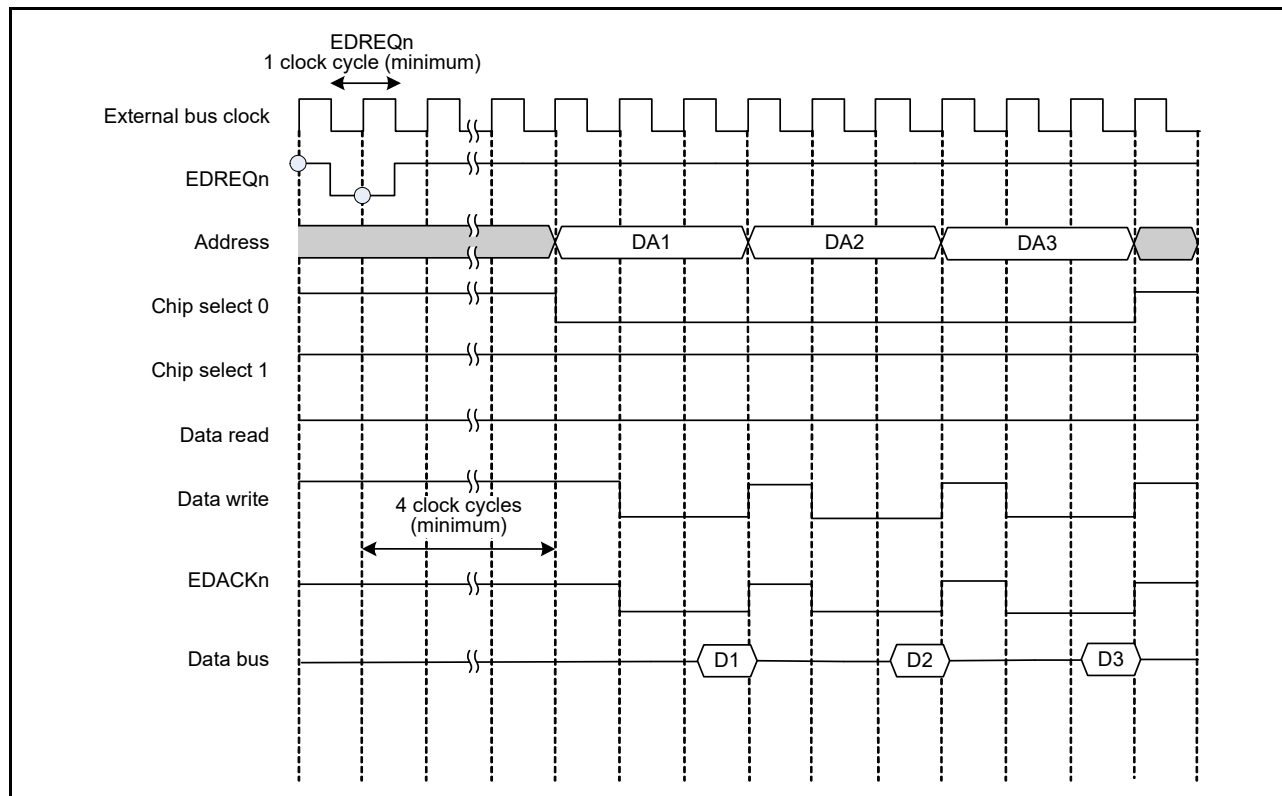


Figure 19.15 Data Flow in Block Transfer Single Address Mode

19.4.3 Cluster Transfer Operation

(1) Dual Address Mode

In cluster-transfer dual address mode, cluster-size data is transferred from the external transfer source device to the external transfer destination device via the cluster buffers. Figure 19.16 shows the data flow in cluster-transfer dual address mode and Figure 19.17 shows the bus cycle example, in which one cluster is transferred in two clock cycles when the cluster size is three.

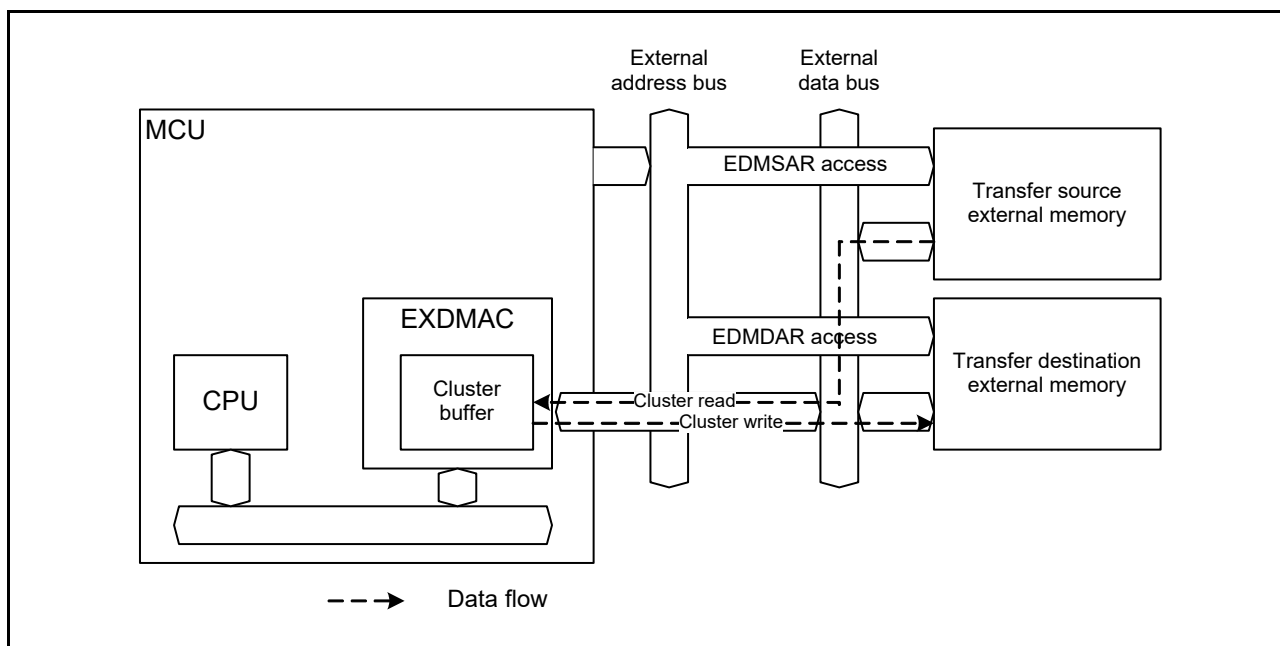


Figure 19.16 Data Flow in Cluster-Transfer Dual Address Mode

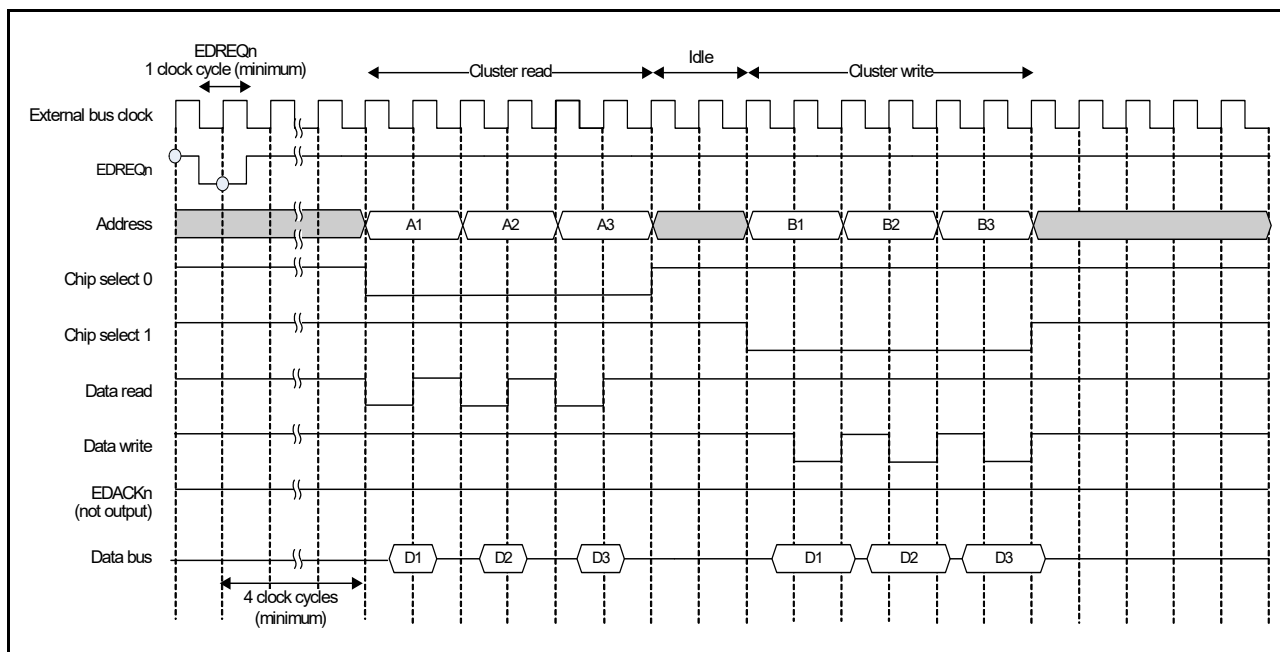


Figure 19.17 Bus Cycle Example in Cluster-Transfer Dual Address Mode

(2) Read Address Mode

In cluster-transfer read address mode, cluster-size data is transferred from the external transfer source device to the cluster buffers. The data transferred in the cluster buffers can be read by the CPU. Figure 19.18 shows the data flow in cluster-transfer read address mode and Figure 19.19 shows the bus cycle example, in which one cluster is transferred in two clock cycles when the cluster size is six.

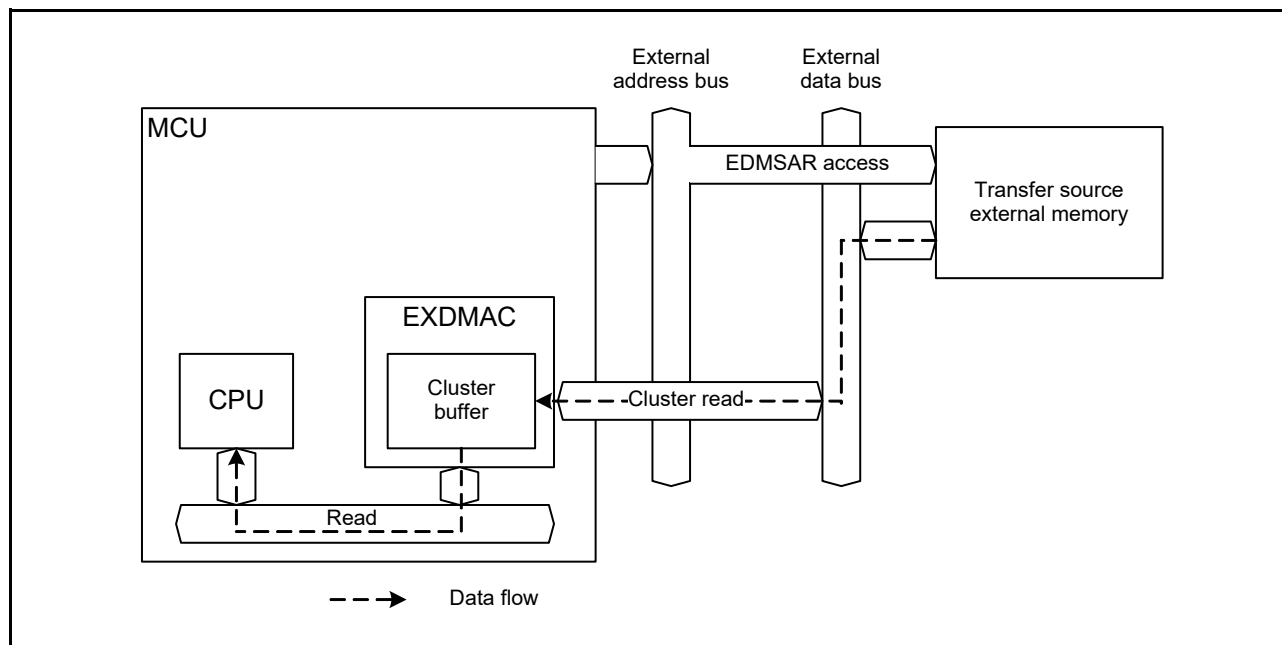


Figure 19.18 Data Flow in Cluster-Transfer Read Address Mode

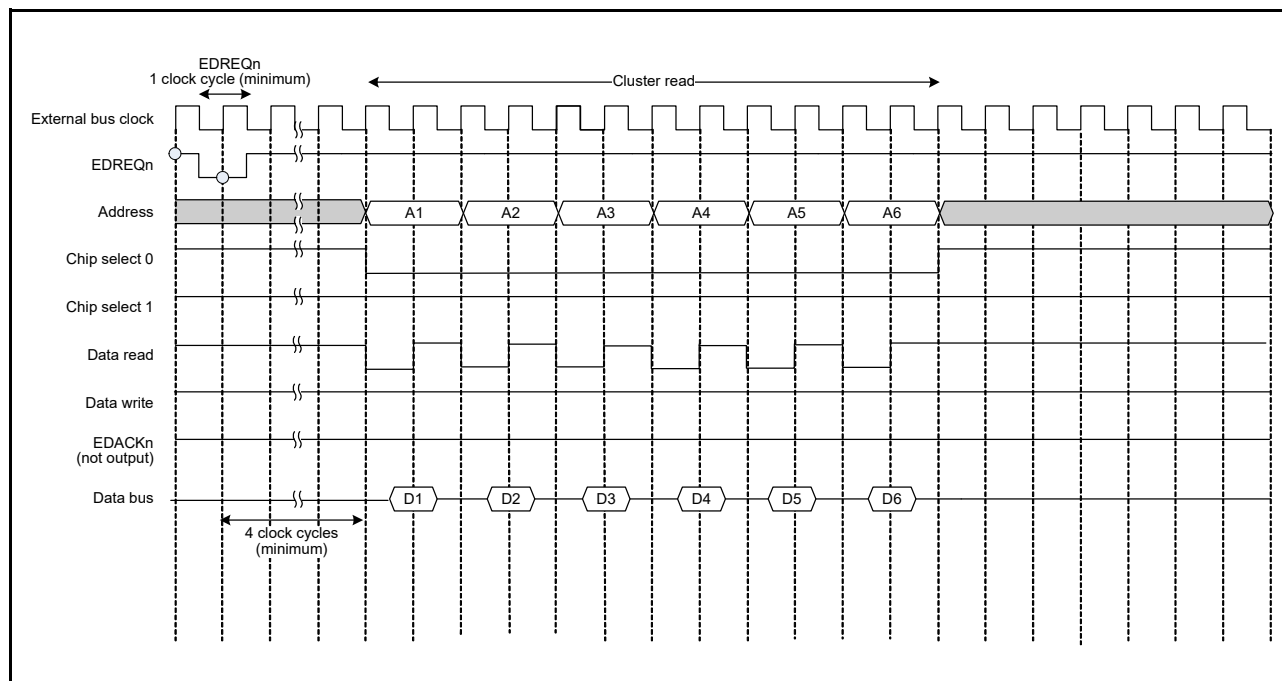


Figure 19.19 Bus Cycle Example in Cluster-Transfer Read Address Mode

(3) Write Address Mode

In cluster-transfer write address mode, the data is written to the cluster buffers by the internal bus master such as the CPU, DMACA, and DTC and then transferred to the external transfer destination device. Figure 19.20 shows the data flow in cluster-transfer write address mode and Figure 19.21 shows the bus cycle example, in which one cluster is transferred in two clock cycles when the cluster size is six.

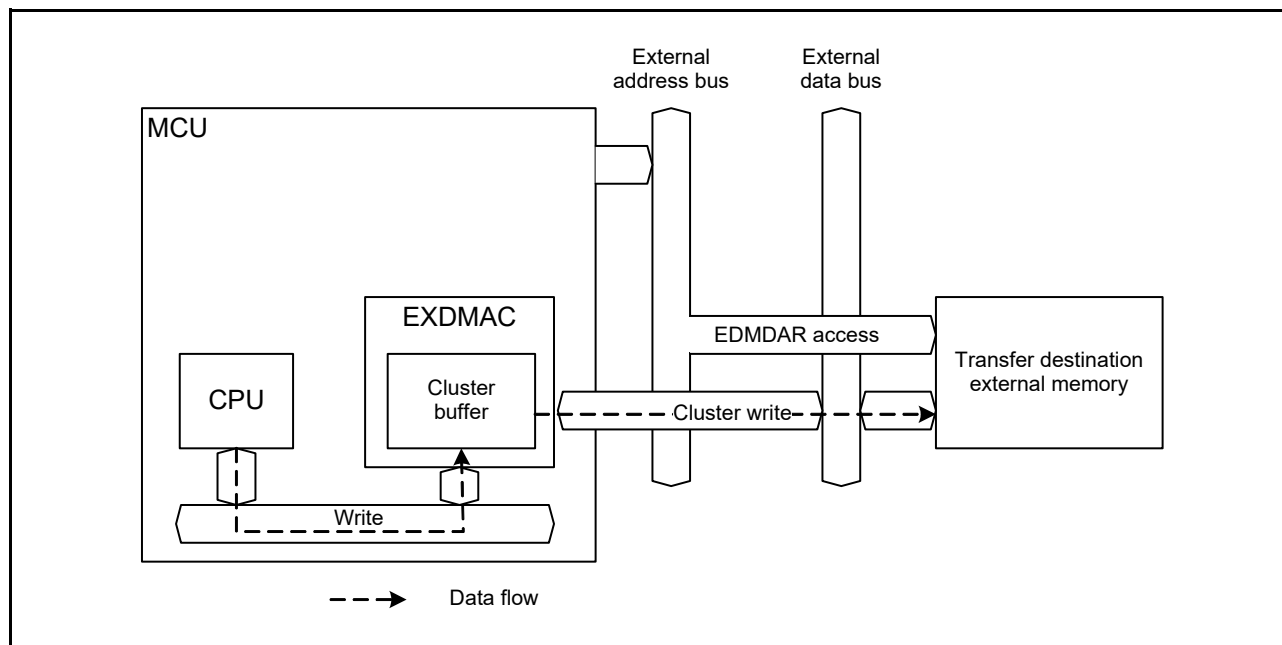


Figure 19.20 Data Flow in Cluster-Transfer Write Address Mode

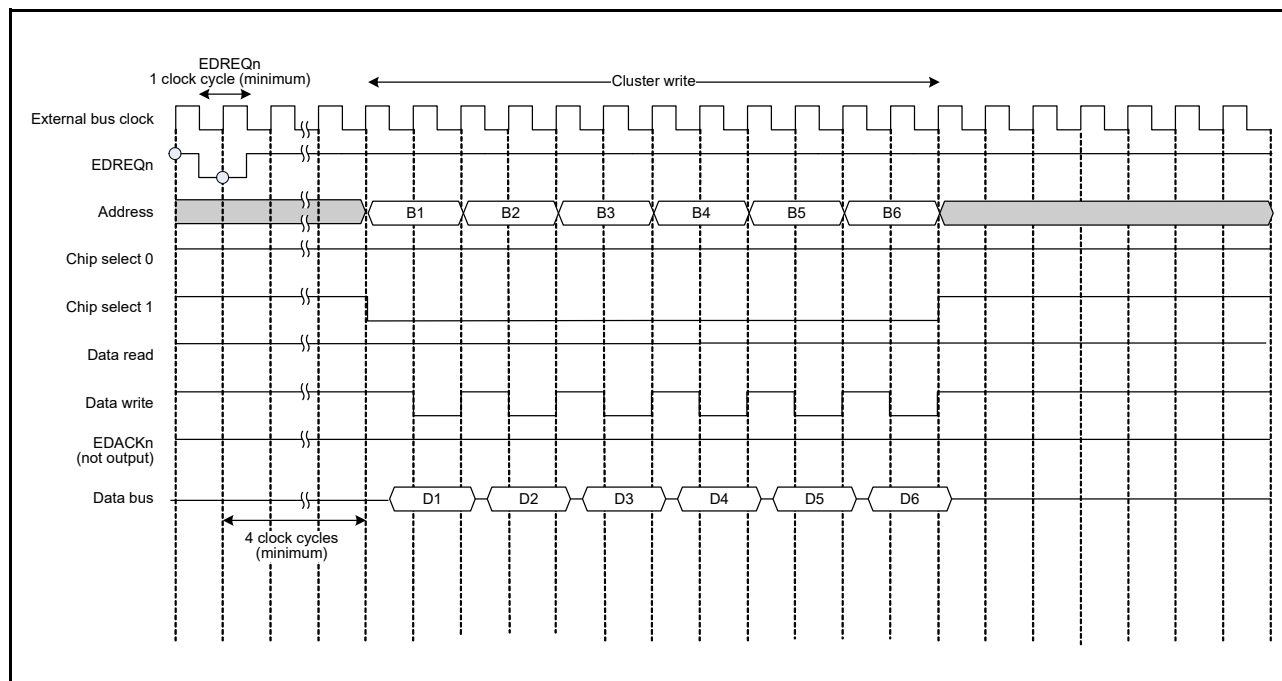


Figure 19.21 Bus Cycle Example in Cluster-Transfer Write Address Mode

19.5 Request Sources and Procedures

19.5.1 Request Sources

The EXDMAC can be triggered by software, by an external DMA transfer request pin (EDREQn pins), or by the DMA transfer requests from the peripheral modules (the software configurable interrupt B source select register 144 (SLIBR144) or the software configurable interrupt A source select register 208 (SLIAR208) should be set for channel 0, and the software configurable interrupt B source select register 145 (SLIBR145) or the software configurable interrupt A source select register 209 (SLIAR209) should be set for channel 1). Setting the DCTG[1:0] bits in EDMTMD of EXDMACn selects the request source.

(1) Trigger by Software

Setting the EXDMACn.EDMREQ.DCTG[1:0] bits to 00b enables the trigger by software.

To start DMA transfer by software, follow the procedure below.

1. Check that the EXDMACn.EDMREQ.SWREQ bit is 0 (DMA transfer is not requested).
2. Set the EXDMACn.EDMTMD.DCTG[1:0] bits to 00b (activation by software).
3. Set the EXDMACn.EDMCNT.DTE bit to 1 (enables DMA transfer).
4. Set the EXDMACn.EDMREQ.CLRS bit (DMA software start bit auto clear select) and also set the EXDMACn.EDMREQ.SWREQ bit to 1 (DMA transfer is requested).

When the EXDMACn.DMREQ.CLRS bit is 0 (SWREQ bit is cleared after DMA transfer is started by software), the EXDMACn.EDMREQ.SWREQ bit is set to 0 after data transfer is started in response to a DMA transfer request. When the CLRS bit is 1 (SWREQ bit is not cleared after DMA transfer is started by software), the SWREQ bit is not set to 0. In this case, a DMA transfer request is issued again after completion of a transfer.

(2) Trigger by external DMA transfer request Pin (EDREQn)

Setting the EXDMACn.EDMTMD.DCTG[1:0] bits to 10b enables the trigger by the external DMA transfer request pins.

To set the trigger by the external DMA transfer request pin, follow the procedure below.

1. Set the detection mode by the EXDMACn.EDMRMD.DREQS[1:0] bits.
2. Set the EXDMACn.EDMTMD.DCTG[1:0] bits to 10b (external DMA transfer request pins).
3. Set the EXDMACn.EDMERF.EREQ flag to 1 to clear the EREQ flag.
4. Set the EXDMACn.EDMCNT.DTE bit to 1 (enables DMA transfer).

When the falling edge or rising edge is selected by using the EXDMACn.EDMRMD.DREQS[1:0] bits, if the external DMA transfer request pin detects an edge, the EXDMACn.EDMERF.EREQ flag is set to 1. The EXDMACn.EDMERF.EREQ flag is set to 0 when the DMA transfer is started by the external request. Moreover, this flag is set to 0 by writing 1 to it.

When the low level detection is set by the EXDMACn.EDMRMD.DREQS[1:0] bits, if the external DMA transfer request pin is low level, the EXDMACn.EDMERF.EREQ flag is set to 1 (DMA transfer is requested). If the external DMA transfer request pin is high level, the EXDMACn.EDMERF.EREQ flag is 0 (DMA transfer is not requested). In case of the low level detection, when the DMA transfer is started by the external request or 1 is written to the flag, the EXDMACn.EDMERF.EREQ flag is not set to 0.

When the EDMAST.DMST bit and the EXDMACn.EDMCNT.DTE bit are set to 1 (enables DMA transfer) while the EXDMACn.EDMERF.EREQ flag is 1 (DMA transfer is requested), the DMA transfer is started.

The value of the EXDMACn.EDMERF.EREQ flag is retained regardless of the settings in the EDMAST.DMST and EXDMACn.EDMCNT.DTE bits.

Figure 19.22 and Figure 19.23 show the external DMA transfer request timings in the falling edge detection mode and low level detection mode, respectively.

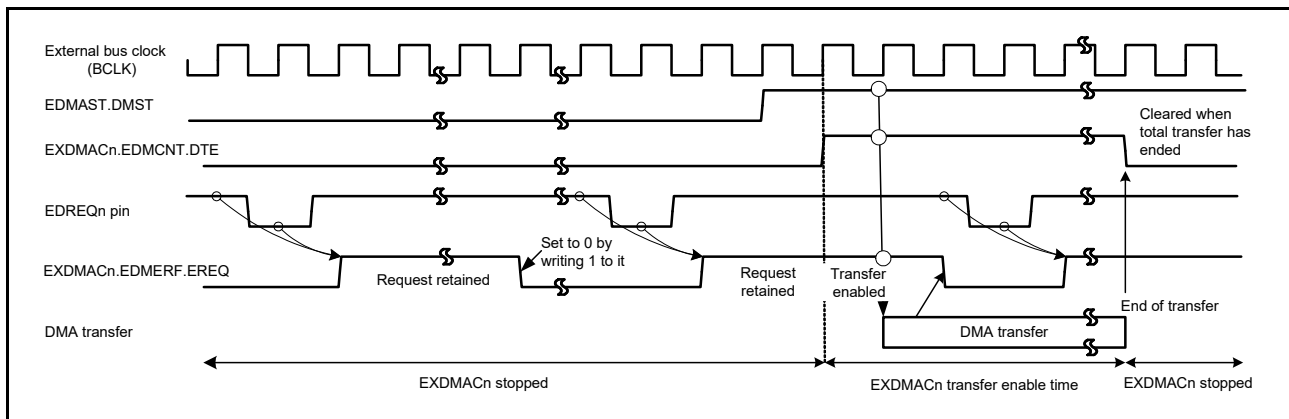


Figure 19.22 External DMA Transfer Request Timing in Falling-Edge Detection Mode

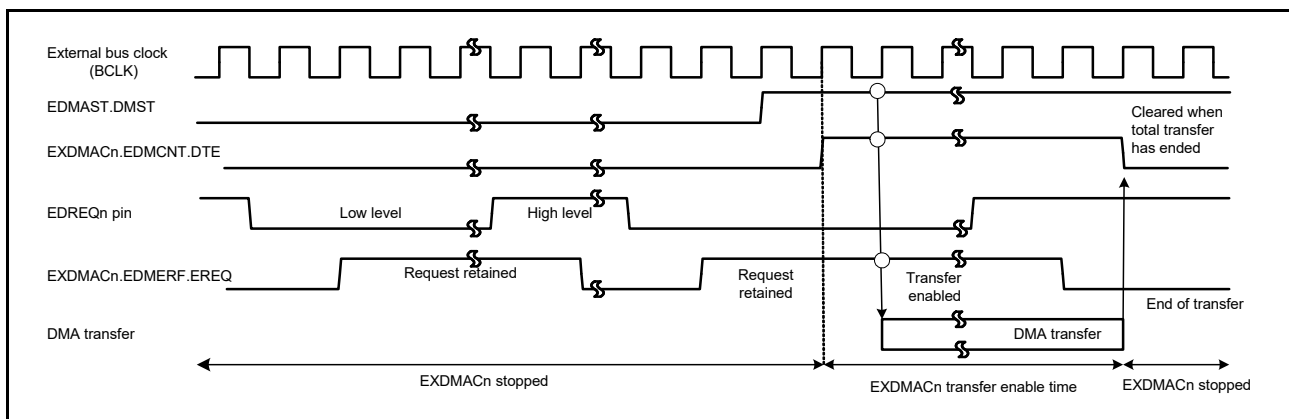


Figure 19.23 External DMA Transfer Request Timing in Low-Level Detection Mode

(3) Trigger by DMA Transfer Requests from Peripheral Modules

Setting the DCTG[1:0] bits in EDMTMD of EXDMACn to 11b enables the trigger by DMA transfer requests from the peripheral modules (an interrupt specified by the software configurable interrupt B source select register (ICU.SLIBR144 or ICU.SLIBR145) or the software configurable interrupt A source select register (ICU.SLIAR208 or ICU.SLIAR209)).

To start DMA transfer by DMA transfer requests from the peripheral modules follow the procedure below.

1. Set the software configurable interrupt B source select register 144 (SLIBR144) or the software configurable interrupt A source select register 208 (SLIAR208) for channel 0, and set the software configurable interrupt B source select register 145 (SLIBR145) or the software configurable interrupt A source select register 209 (SLIAR209) for channel 1 (For the setting procedure, refer to section 15.4.5, Software Configurable Interrupts, in section 15, Interrupt Controller (ICUD)).
2. Set the EXDMACn.EDMTMD.DCTG[1:0] bits to 11b (DMA transfer requests from the peripheral modules).
3. Set the EXDMACn.EDMPRF.PREQ flag to 1, and then clear it to 0.
4. Set the EXDMACn.EDMCNT.DTE bit to 1 (DMA transfer is enabled).

None of the values of the interrupt request enable bits (IERm.IENj) affect EXDMACn requests initiated by peripheral modules.

When a DMA transfer request is input from the peripheral modules, the EXDMACn.EDMPRF.PREQ flag is set to 1 (DMA transfer is requested). The EXDMACn.EDMPRF.PREQ flag is set to 0 (DMA transfer is not requested) when the DMA transfer is started by the peripheral module request.

This flag is set to 0 by writing 1 to it.

When the EDMAST.DMST bit is set to 1 (EXDMAC module start) and the EXDMACn.EDMCNT.DTE bit is set to 1 (DMA transfer is enabled) while the EXDMACn.EDMPRF.PREQ flag is 1, the DMA transfer is started.

The value of the EXDMACn.EDMPRF.PREQ flag is retained regardless of the settings of the EDMAST.DMST and EXDMACn.EDMCNT.DTE bits.

The EXDMACn.EDMPRF.PREQ flag is set to 1 regardless of the state of the module stop bit when the corresponding EXDMACn is started by an internal peripheral module.

19.5.2 Activating the EXDMAC

Figure 19.24 shows the register setting procedure.

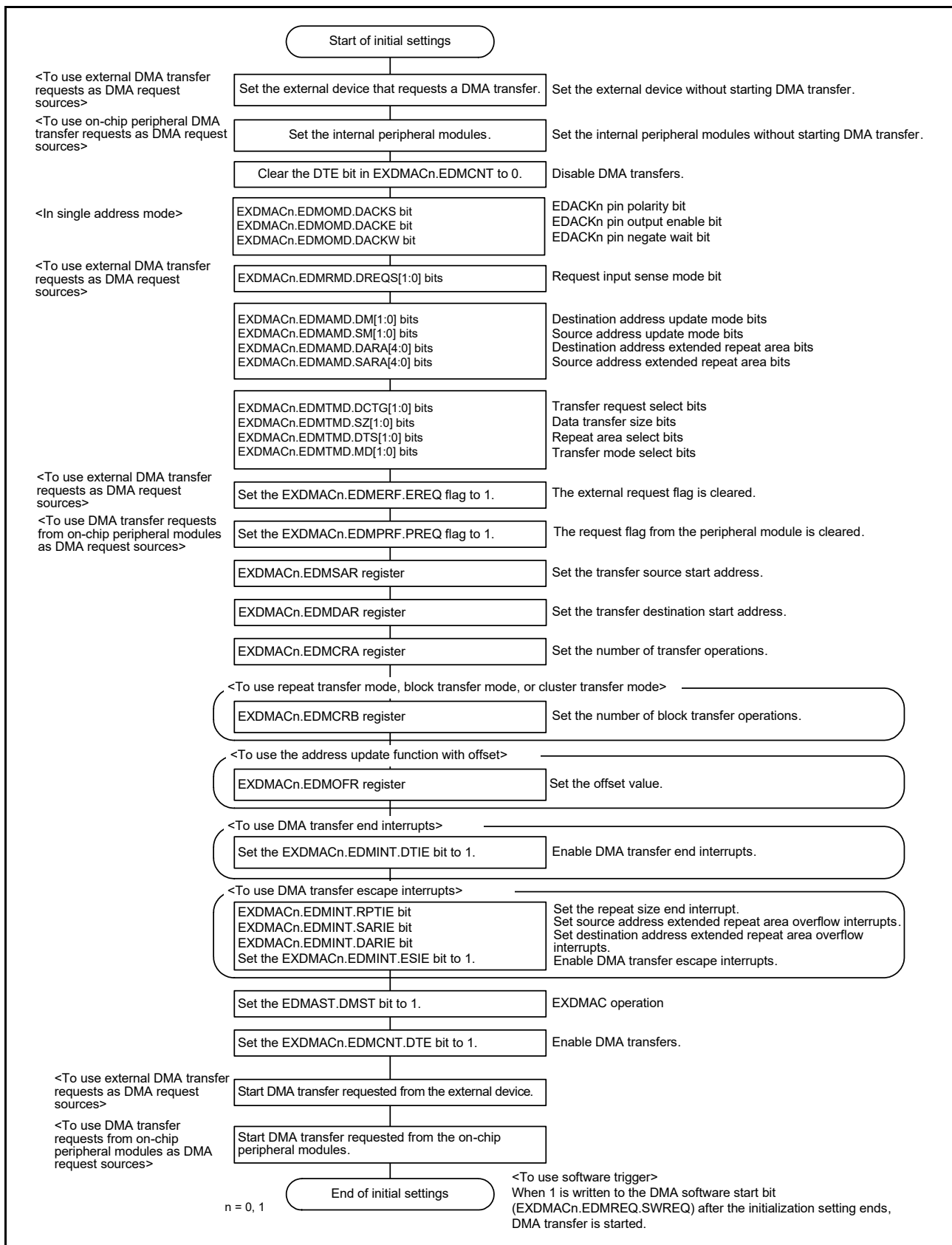


Figure 19.24 Register Setting Procedure

19.5.3 Starting DMA Transfer

Setting the DTE bit in EDMCNT of EXDMACn to 1 (DMA transfer enabled) and setting the DMST bit in EDMAST to 1 (EXDMAC start) enable DMA transfer of channel n (n = 0, 1).

When DMA transfer requests are generated, channel arbitration is performed where a DMA transfer request of higher-priority channel is accepted and DMA transfer of the channel starts. When a DMA transfer request is accepted and DMA transfer starts, the ACT flag in EDMSTS of EXDMACn is set to 1 (DMA transfer is in progress).

19.5.4 Registers during DMA Transfer

The EXDMAC registers are updated by a DMA transfer. The value to be updated differs according to the other settings and the transfer state. The registers to be updated are EDMSAR, EDMDAR, EDMCRA, EDMCRB, EDMCNT, and EDMSTS of EXDMACn.

(1) EXDMA Source Address Register (EXDMACn.EDMSAR)

When data has been transferred in response to one transfer request, the contents of EDMSAR are updated to the address to be accessed by the next transfer request.

For details on register update operation in each transfer mode, refer to Table 19.4 to Table 19.7.

(2) EXDMA Destination Address Register (EXDMACn.EDMDAR)

When data has been transferred in response to one transfer request, the contents of EDMDAR are updated to the address to be accessed by the next transfer request.

For details on register update operation in each transfer mode, refer to Table 19.4 to Table 19.7.

(3) EXDMA Transfer Count Register (EXDMACn.EDMCRA)

When data has been transferred in response to one transfer request, the count value is updated. The update operation depends on the transfer mode selected.

For details on register update operation in each transfer mode, refer to Table 19.4 to Table 19.7.

(4) EXDMA Block Transfer Count Register (EXDMACn.EDMCRB)

When data has been transferred in response to one transfer request, the count value is updated. The update operation depends on the transfer mode selected.

For details on register update operation in each transfer mode, refer to Table 19.4 to Table 19.7.

(5) DMA Transfer Enable Bit (EXDMACn.EDMCNT.DTE)

The EXDMACn.EDMCNT.DTE bit provides a way to control enabling or prohibition of transfer by writing to the relevant registers.

The EXDMACn.EDMCNT.DTE bit is set to 0 when any of the following conditions is generated by the DMA transfer.

- When the specified total volume of data transfer is completed
- When DMA transfer is stopped by the repeat size end interrupt
- When DMA transfer is stopped by the extended repeat area overflow interrupt

Writing to the registers of an EXDMAC channel is prohibited if the corresponding EXDMACn.EDMCNT.DTE bit is 1 (except to the EXDMACn.EDMCNT register itself).

Change the settings of the registers as required after writing 0 to the EXDMACn.EDMCNT.DTE bit.

(6) EXDMA Active Flag (EXDMACn.EDMSTS.ACT)

The ACT flag in EDMSTS of EXDMACn indicates whether the EXDMACn is in the idle or active state.

This flag is set to 1 when the EXDMACn starts data transfer, and is set to 0 when data transfer in response to one transfer request is completed.

Even when DMA transfer is stopped by writing 0 to the DTE bit in EDMCNT of EXDMACn, this flag remains 1 until DMA transfer is completed.

(7) Transfer End Interrupt Flag (EXDMACn.EDMSTS.DTIF)

The DTIF flag in EDMSTS of EXDMACn is set to 1 after transfer of the total transfer size of data is completed.

When both this flag and the DTIE bit in EDMINT of EXDMACn are set to 1, a transfer end interrupt is requested.

This flag is set to 1 when the DMA transfer bus cycle is completed and the ACT flag in EDMSTS of EXDMACn is set to 0 indicating the DMA transfer end.

This flag is automatically set to 0 when the DTE bit in EDMCNT of EXDMACn is set to 1 during the interrupt handling.

(8) Transfer Escape End Interrupt Flag (EXDMACn.EDMSTS.ESIF)

The ESIF flag in EDMSTS of EXDMACn is set to 1 when a repeat size end interrupt or extended repeat area overflow interrupt is requested. When this flag and the ESIE bit in EDMINT of EXDMACn are set to 1, a transfer escape end interrupt is requested.

This flag is set to 1 when the bus cycle of the DMA transfer having caused the interrupt request is completed and the ACT flag in EDMSTS of EXDMACn is set to 0 indicating the DMA transfer end.

This flag is automatically set to 0 when the DTE bit in EDMCNT of EXDMACn is set to 1 during an interrupt handling. Before sending an interrupt request from the EXDMACn to the CPU or the DTC, the interrupt control register must be set.

For details, refer to section 15, Interrupt Controller (ICUD).

19.5.5 Channel Priority

When multiple DMA transfer requests are present, the EXDMAC determines the priority of channels that have DMA transfer requests. The channel priority is fixed as channel 0 > channel 1.

When a DMA transfer request is generated during data transfer, channel arbitration is started after the final data has been transferred, and DMA transfer of the higher-priority channel starts.

19.6 Ending DMA Transfer

The operation for ending DMA transfer depends on the transfer end conditions. When DMA transfer ends, the DTE bit in EDMCNT and the ACT flag in EDMSTS of EXDMACn are changed from 1 to 0, indicating that DMA transfer has ended.

19.6.1 Transfer End by Completion of Specified Total Number of Transfer Operations

(1) In Normal Transfer Mode (EXDMACn.EDMTMD.MD[1:0] = 00b)

When the value of the EXDMACn.EDMCRAL register changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DTE bit in EDMCNT of EXDMACn is set to 0 and the DTIF flag in EDMSTS of EXDMACn is set to 1 at the same time. If the DTIE bit in EDMINT of EXDMACn is 1 at this time, a transfer end interrupt request is issued to the CPU or the DTC.

(2) In Repeat Transfer Mode (EXDMACn.EDMTMD.MD[1:0] = 01b)

When the value of DMCRB of EXDMACn changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DTE bit in EDMCNT of EXDMACn is set to 0 and the DTIF flag in EDMSTS of EXDMACn is set to 1 at the same time. If the DTIE bit in EDMINT of EXDMACn is 1 at this time, a transfer end interrupt request is issued to the CPU or the DTC.

(3) In Block Transfer Mode (EXDMACn.EDMTMD.MD[1:0] = 10b)

When the value of EDMCRB of EXDMACn changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DTE bit in EDMCNT of EXDMACn is set to 0 and the DTIF flag in EDMSTS of EXDMACn is set to 1 at the same time. If the DTIE bit in EDMINT of EXDMACn is 1 at this time, a transfer end interrupt request is issued to the CPU or the DTC.

(4) In Cluster Transfer Mode (EXDMACn.EDMTMD.MD[1:0] = 11b)

When the value of EDMCRB of EXDMACn changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DTE bit in EDMCNT of EXDMACn is set to 0 and the DTIF flag in EDMSTS of EXDMACn is set to 1 at the same time. If the DTIE bit in EDMINT of EXDMACn is 1 at this time, a transfer end interrupt request is issued to the CPU or the DTC.

Before sending an interrupt request from the EXDMACn to the CPU or the DTC, the interrupt control register must be set. For details, refer to section 15, Interrupt Controller (ICUD).

19.6.2 Transfer End by Repeat Size End Interrupt

In repeat transfer mode, a repeat size end interrupt is requested when transfer of a 1-repeat size of data is completed while the RPTIE bit in EDMINT of EXDMACn is set to 1. When the interrupt is requested to complete DMA transfer, the DTE bit in EDMCNT of EXDMACn is set to 0 and the ESIF flag in EDMSTS of EXDMACn is set to 1. If the ESIE bit in EDMINT of EXDMACn is 1 at this time, a transfer end interrupt request is issued to the CPU or the DTC. Here, the transfer can be resumed by writing 1 to the DTE bit in EDMCNT of EXDMACn.

A repeat size end interrupt can be requested also in block transfer mode (or cluster transfer mode). In block transfer mode (or cluster transfer mode), the interrupt is requested in the same way as in repeat transfer mode when transfer of a 1-block size (or 1-cluster) data is completed.

Before sending an interrupt request from the EXDMACn to the CPU or the DTC, the interrupt control register must be set. For details, refer to **section 15, Interrupt Controller (ICUD)**.

19.6.3 Transfer End by Interrupt on Extended Repeat Area Overflow

When an overflow on the extended repeat area occurs while the extended repeat area is specified and the SARIE or DARIE bit in EDMINT of EXDMACn is set to 1, an interrupt by an extended repeat area overflow is requested. When the interrupt is requested, the DMA transfer is terminated, the DTE bit in EDMCNT of EXDMACn is set to 0, and the ESIF flag in EDMSTS of EXDMACn is set to 1. If the ESIE bit in EDMINT of EXDMACn is 1 at this time, a transfer end interrupt request is issued to the CPU or the DTC.

Even if an interrupt by an extended repeat area overflow is requested during a read cycle, the following write cycle is performed.

In block transfer mode (or cluster transfer mode), even if an interrupt by an extended repeat area overflow is requested during a 1-block (or 1-cluster) transfer, the remaining data in the block (or the cluster) is transferred; transfer is terminated after a block (or cluster) transfer.

Before sending an interrupt request from the EXDMACn to the CPU or the DTC, the interrupt control register must be set. For details, refer to **section 15, Interrupt Controller (ICUD)**.

19.7 Interrupts

The EXDMAC can output one interrupt request to the CPU or the DTC for each channel. **Table 19.10** lists the relation among the interrupt sources, the interrupt status flags, and the interrupt enable bits. **Figure 19.25** shows the schematic logic diagram of interrupt outputs. The procedures for suspending or resuming DMA transfer by the EXDMAC interrupt are shown in **Figure 19.26**.

Table 19.10 **Relation among Interrupt Sources, Interrupt Status Flags, and Interrupt Enable Bits**

Interrupt Sources		Interrupt Enable Bits	Interrupt Status Flags	Request Output Enable Bits
Transfer end		—	EXDMACn.EDMSTS.DTIF	EXDMACn.EDMINT.DTIE
Escape transfer end	Repeat size end	EXDMACn.EDMINT.RPTIE	EXDMACn.EDMSTS.ESIF	EXDMACn.EDMINT.ESIE
	Source address extended repeat area overflow	EXDMACn.EDMINT.SARIE		
	Destination address extended repeat area overflow	EXDMACn.EDMINT.DARIE		

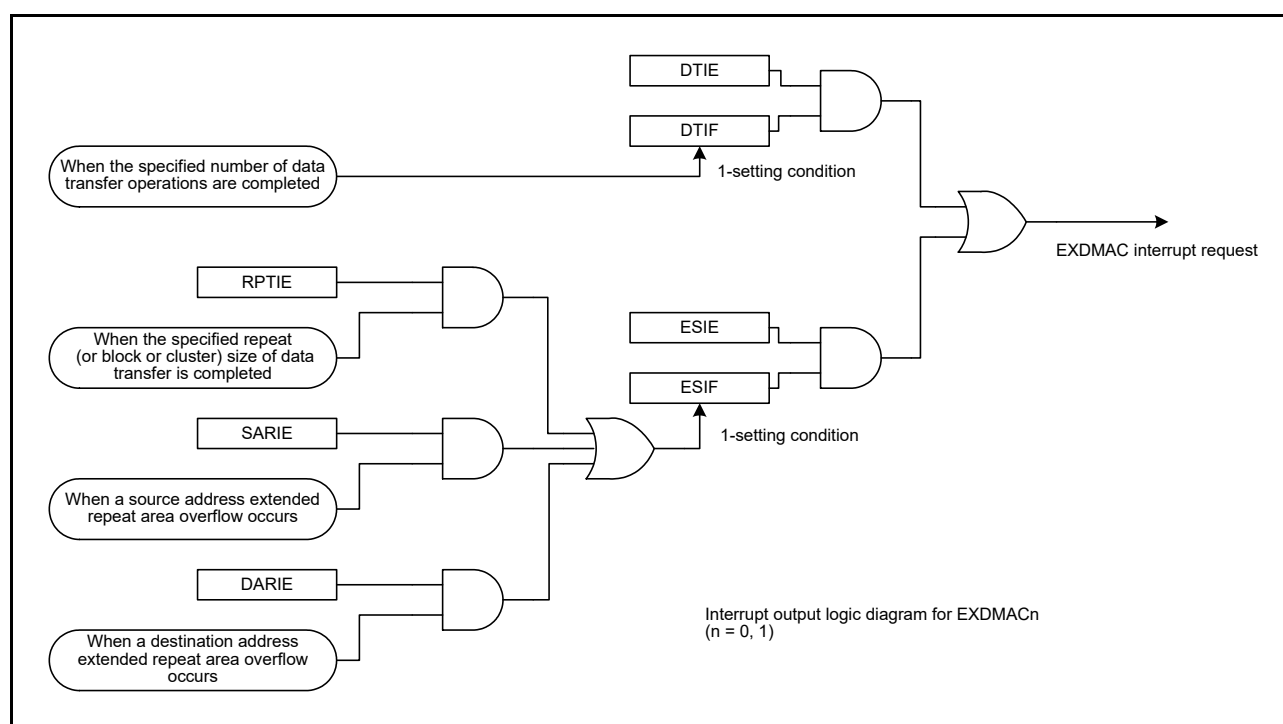


Figure 19.25 Schematic Logic Diagram of Interrupt Outputs

Specifically, the different procedures are used for canceling an interrupt to restart DMA transfer in the following two cases: (1) discontinuing or terminating DMA transfer and (2) continuing DMA transfer.

(1) When Discontinuing or Terminating DMA Transfer

Write 0 to the DTIF flag in EDMSTS of EXDMACn to clear a transfer end interrupt, and to the ESIF flag in EDMSTS of EXDMACn to clear a repeat size interrupt and an extended repeat area overflow interrupt. The EXDMACn remains in the stop state. When starting another DMA transfer after that, set the appropriate registers, and set the DTE bit in EDMCNT of EXDMACn to 1.

(2) When Continuing DMA Transfer

Write 1 to the DTE bit in EDMCNT of EXDMACn. The ESIF flag in EDMSTS of EXDMACn is automatically set to 0 (interrupt source cleared), and DMA transfer is resumed.

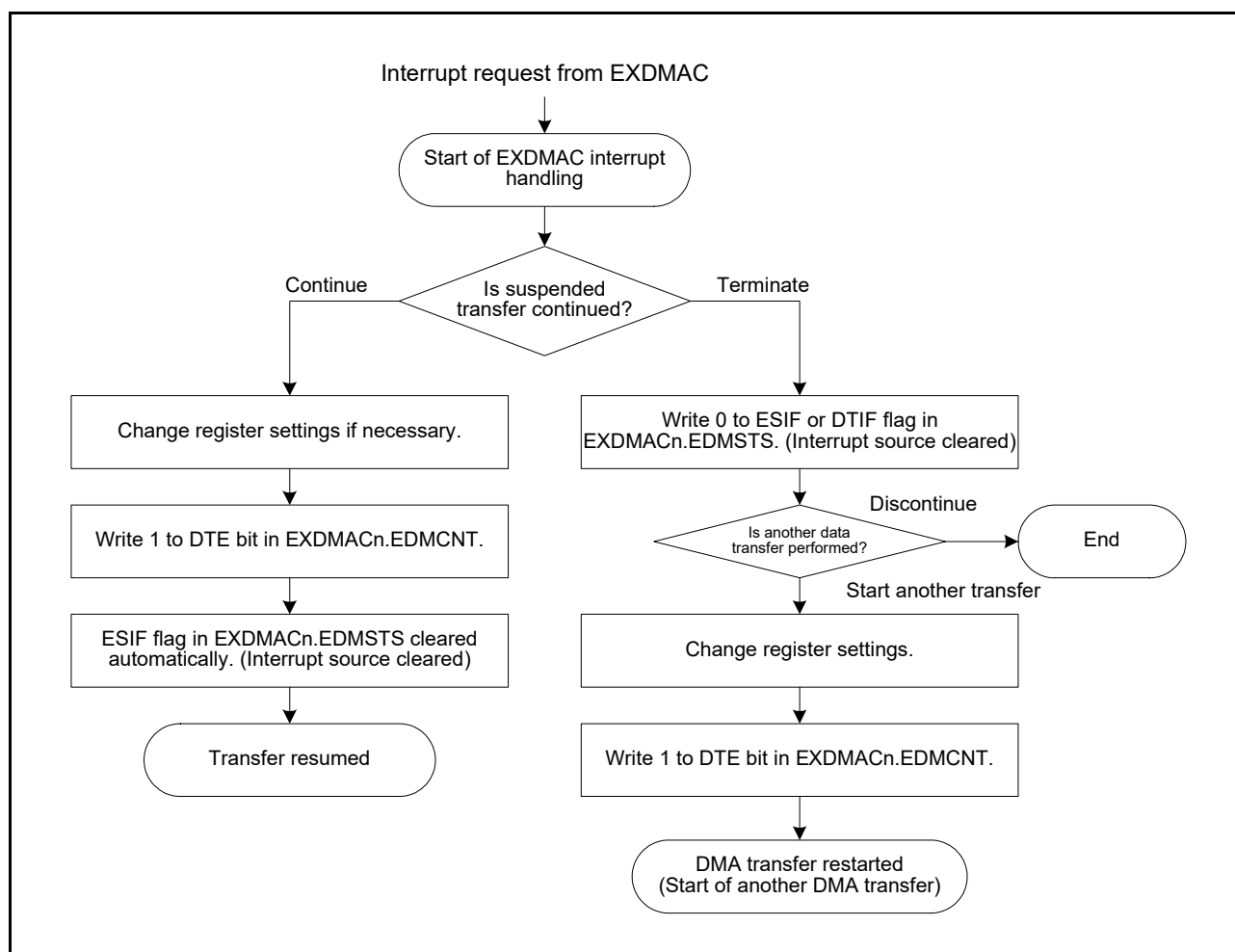


Figure 19.26 Procedures for Suspending or Resuming DMA Transfer by the EXDMAC Interrupt

19.8 Low-Power Consumption Function

To place the EXDMAC in the module-stop state, all-module clock stop mode, software-standby mode, or deep software-standby mode, set the EDMAST.DMST bit to 0 (EXDMAC stopped), and then perform the following processing.

(1) Module-Stop Function

Writing 1 to the MSTPCRA.MSTPA29 bit (transition to the module-stop state) enables the module-stop function of the EXDMAC. If DMA transfer is in progress at the time a 1 is written to the MSTPA29 bit, the transition to the module-stop state proceeds after DMAC transfer has ended.

Do not access the EXDMAC registers while the MSTPCRA.MSTPA29 bit is 1.

Writing 0 to the MSTPA29 bit releases the EXDMAC from the module-stop state.

(2) All-Module Clock Stop Mode

Follow the procedure in section 11.6.2.1, Transition to All-Module Clock Stop Mode, in section 11, Low Power Consumption. If DMA transfer is in progress at the time the WAIT instruction is executed, the EXDMAC can enter all-module clock stop mode after completion of the current DMA transfer.

After the EXDMAC returns from all-module clock stop mode, writing 0 to the MSTPA29 bit releases the EXDMAC from the module-stop state.

(3) Software Standby and Deep Software Standby Modes

Follow the procedure in section 11.6.3.1, Transition to Software Standby Mode or section 11.6.4.1, Transition to Deep Software Standby Mode, in section 11, Low Power Consumption.

If DMA transfer is in progress at the time the WAIT instruction is executed, the EXDMAC enters software standby mode or deep software stand by mode after completion of the current DMA transfer.

(4) Notes on Low-Power Consumption Function

For the timing of WAIT instruction execution and register settings, refer to section 11.7.6, Timing of WAIT Instructions, in section 11, Low Power Consumption.

To perform DMA transfer after returning from low-power consumption mode, set the EDMAST.DMST bit to 1 again.

19.9 EDACKn Operation in Single Address Mode

In single address mode, EDACKn is output to one of the external transfer-source or transfer-destination devices and the address is simultaneously output to the other transfer device for access.

When the external device receiving EDACKn transfers data to/from the CS area, the EDACKn negation timing can be adjusted by setting the DACKW bit in EDMOMD of EXDMACn. Specifically, the timing can be advanced by one BCLK cycle if the external device is a transfer destination and delayed by one BCLK cycle if the external device is a transfer source. When the external device receiving EDACKn transfers data to/from the SDRAM, the EDACKn negation timing cannot be adjusted by the DACKW bit EDMOMD of EXDMACn. If one of the parties for transfer by the EXDMAC (source or destination) is the SDRAM area, the EXDMACn.EDMOMD.DACKSEL bit can be set to 1 so that the EDACKn signal is only asserted for one-half of the SDCLK cycle in the latter half of the data-valid interval. The EXDMACn.EDMOMD.DACKSEL cannot be used to adjust the period for assertion of the EDACKn signal when a party for transfer is a CS area. For the CS area addresses and SDRAM area addresses, refer to **section 4, Address Space**.

The following sections show EDACKn operation examples in single address mode, in which data is transferred to/from the CS and SDRAM areas in normal and block transfer modes.

19.9.1 EDACKn Operation Example in Normal-Transfer (CS Area) Single Address Mode

Figure 19.27 shows the operation example in which data is transferred from the CS area to the device with EDACKn in normal transfer mode. Setting the DACKW bit in EDMOMD of EXDMACn to 1 allows EDACKn to be negated one BCLK cycle before the data read signal is negated.

Figure 19.28 shows the operation example in which data is transferred from the device with EDACKn to the CS area in normal transfer mode. Setting the DACKW bit to 1 allows EDACKn to be negated one BCLK cycle after the data write signal is negated.

For the data read signal, data write signal, and CS area access timing setting registers, refer to **section 16, Buses**.

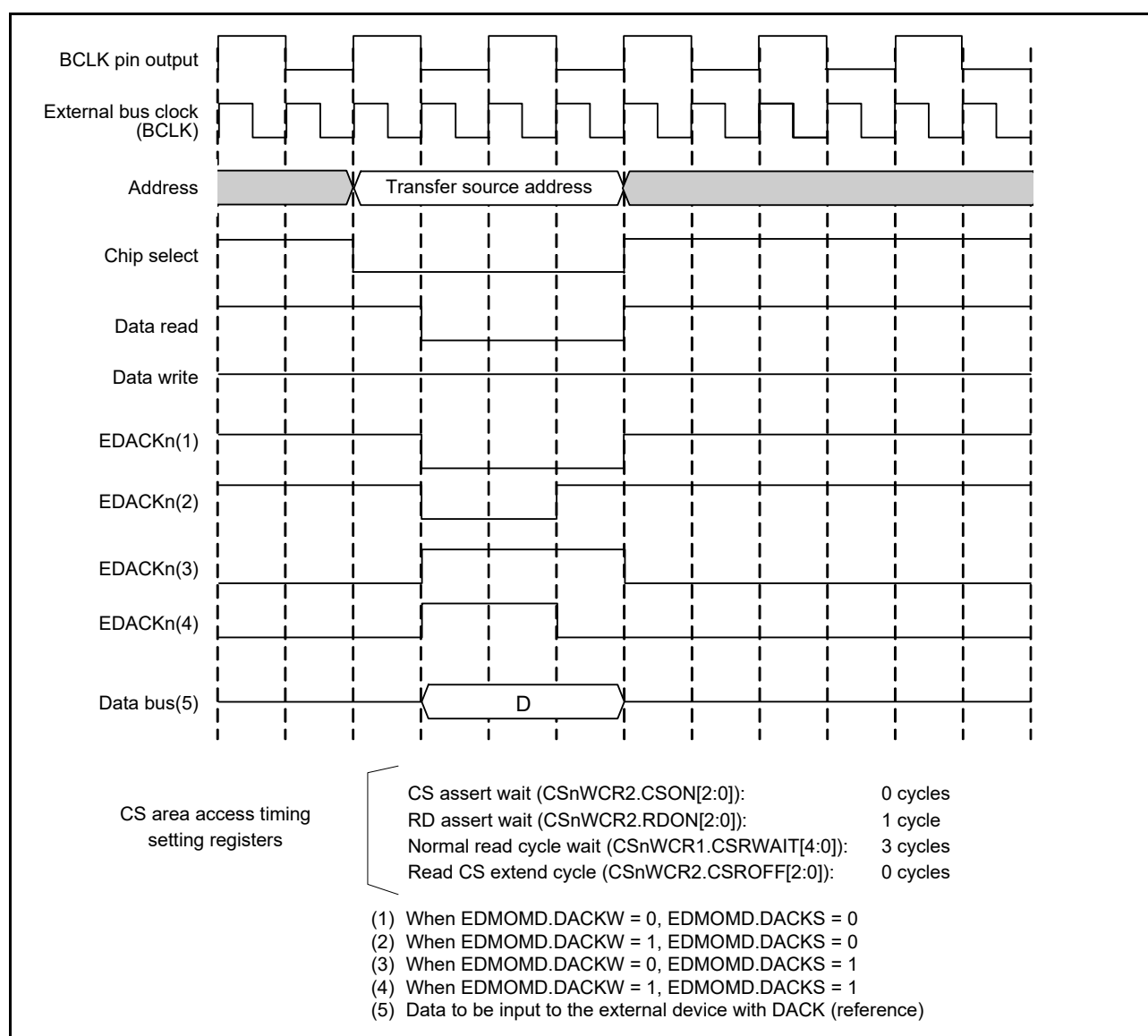


Figure 19.27 Operation Example in Normal-Transfer (CS Area Read) Single Address Mode

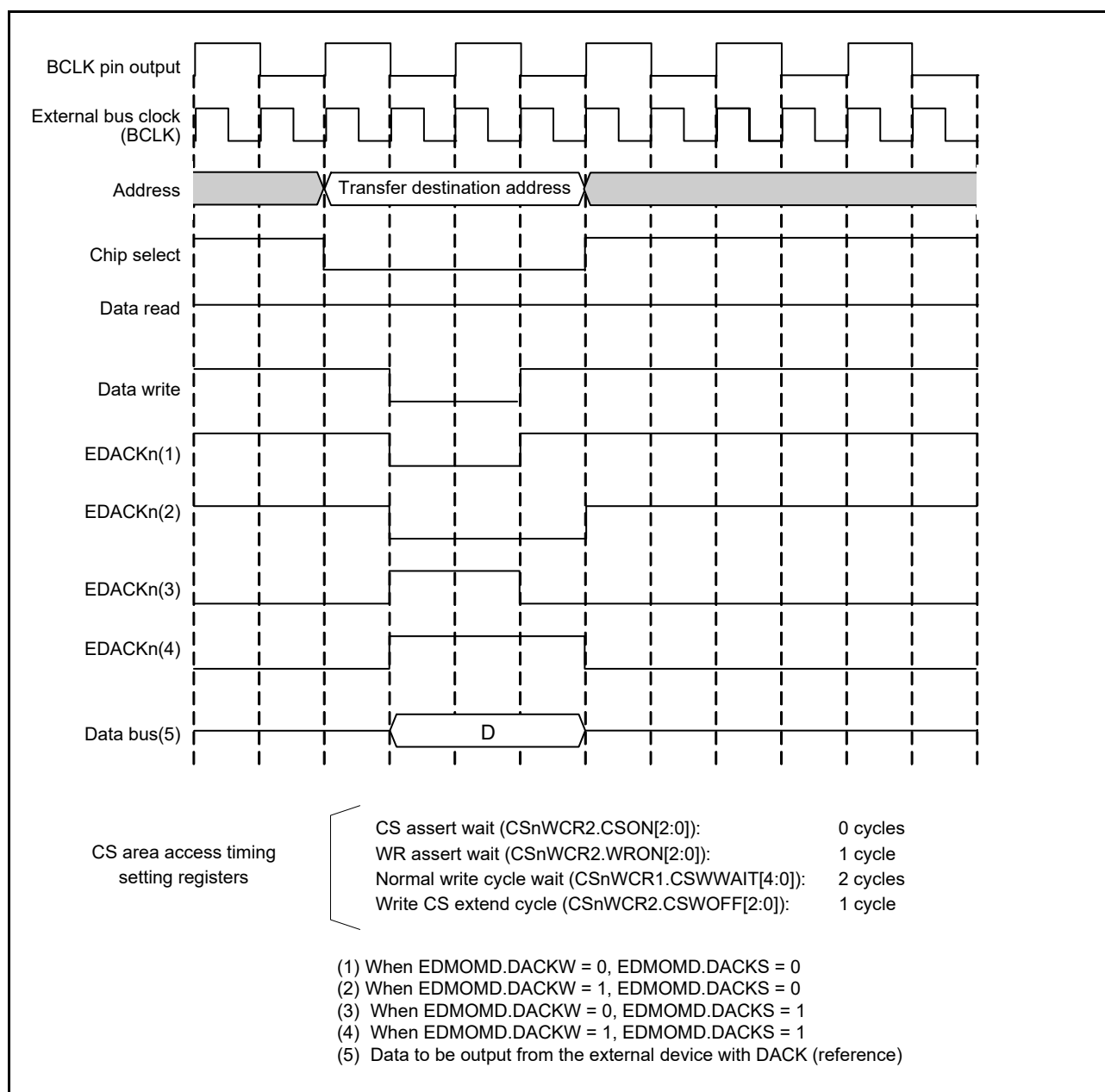


Figure 19.28 Operation Example in Normal-Transfer (CS Area Write) Single Address Mode

19.9.2 EDACKn Operation Example in Normal-Transfer (SDRAM Area) Single Address Mode

Figure 19.29 shows the operation example in which data is transferred from SDRAM to the device with EDACKn in normal transfer mode.

EDACKn is asserted while SDRAM is outputting data.

Figure 19.30 shows the operation example in which data is transferred from the device with EDACKn to SDRAM in normal transfer mode.

EDACKn is asserted while SDRAM is writing data.

For the SDRAM commands and SDRAM access timing setting registers, refer to section 16, Buses.

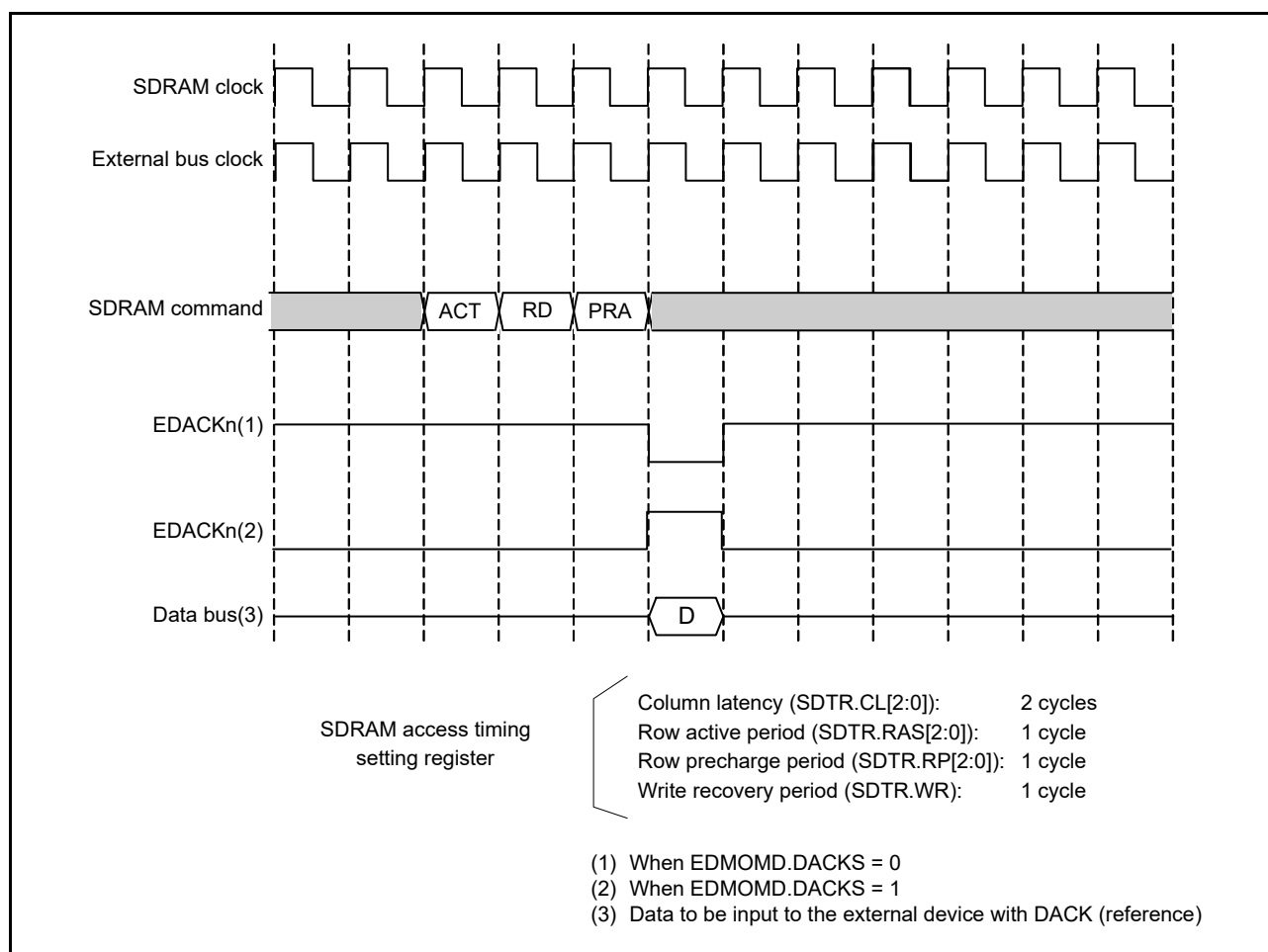


Figure 19.29 Operation Example in Normal-Transfer (SDRAM Area Read) Single Address Mode

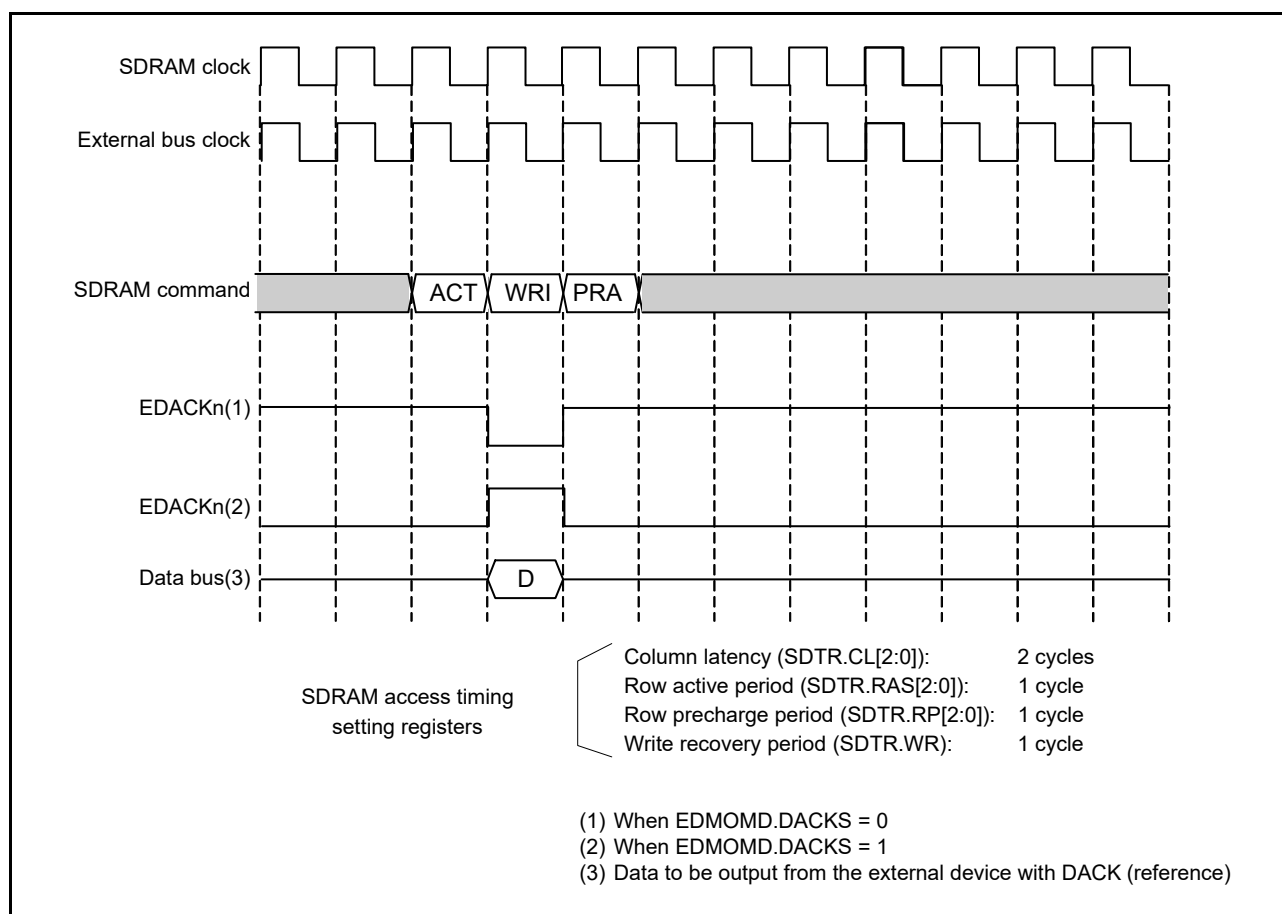


Figure 19.30 Operation Example in Normal-Transfer (SDRAM Area write) Single Address Mode

19.9.3 EDACKn Operation Example in Block-Transfer (CS Area) Single Address Mode

Figure 19.31 shows the operation example in which data is transferred from the CS area to the device with EDACKn in block transfer mode (block size = two). Setting the DACKW bit in EDMOMD of EXDMACn to 1 allows EDACKn to be negated one BCLK cycle before the data read signal is negated.

Figure 19.32 shows the operation example in which data is transferred from the device with EDACKn to the CS area in block transfer mode (block size = two). Setting the DACKW bit to 1 allows EDACKn to be negated one BCLK cycle after the data write signal is negated.

For the data read signal, data write signal, and CS area access timing setting registers, refer to section 16, Buses.

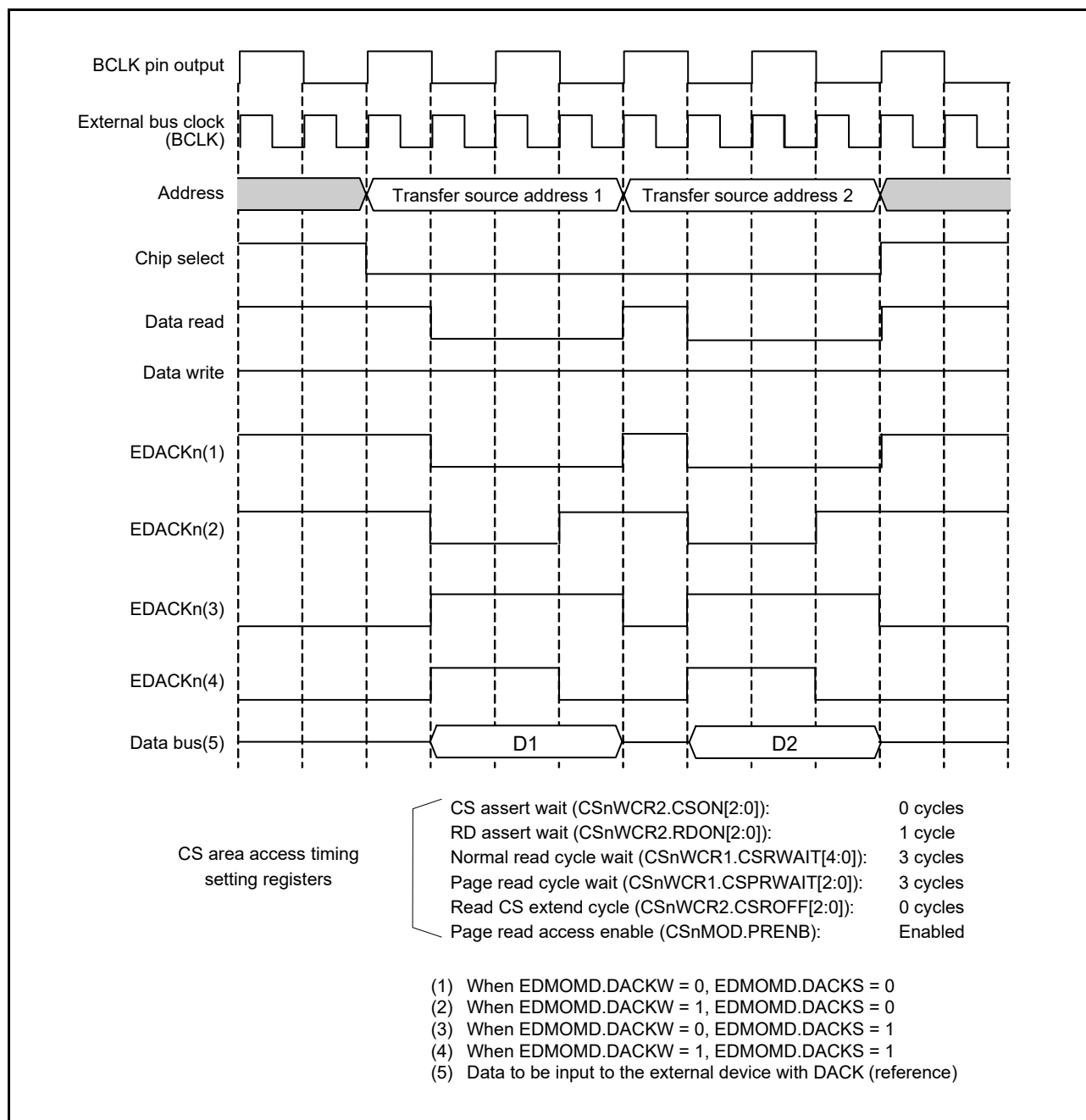


Figure 19.31 Operation Example in Block-Transfer (CS Area Read) Single Address Mode

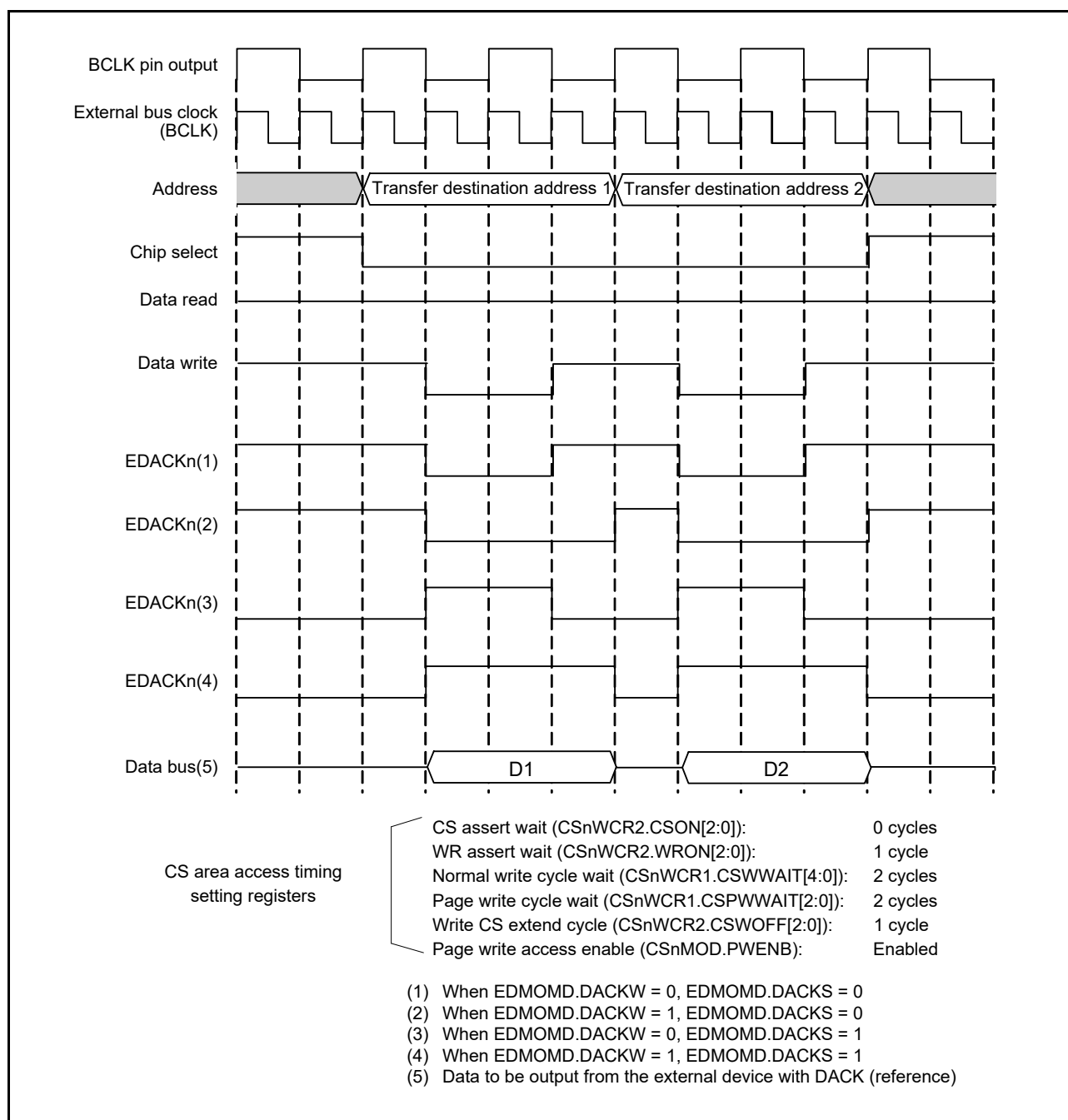


Figure 19.32 Operation Example in Block-Transfer (CS Area Write) Single Address Mode

19.9.4 EDACKn Operation Example in Block-Transfer (SDRAM Area) Single Address Mode

Figure 19.33 shows the operation example in which data is transferred from SDRAM to the device with EDACKn in block transfer mode (block size = four) when the SDRAM continuous access enable bit is enabled (SDAMOD.BE = 1) and EXDMACn.EDMOMD.DACKSEL set to 0.

EDACKn is asserted while SDRAM is outputting data.

Figure 19.34 shows the operation example in which data is transferred from the device with EDACKn to SDRAM in block transfer mode (block size = four) when the SDRAM continuous access enable bit is enabled (SDAMOD.BE = 1) and EXDMACn.EDMOMD.DACKSEL set to 0.

EDACKn is asserted while SDRAM is writing data.

Figure 19.35 shows the operation example in which data is transferred from SDRAM to the device with EDACKn in block transfer mode (block size = four) when the SDRAM continuous access enable bit is enabled (SDAMOD.BE = 1) and EXDMACn.EDMOMD.DACKSEL set to 1.

EDACKn is asserted while SDRAM is outputting data.

Figure 19.36 shows the operation example in which data is transferred from the device with EDACKn to SDRAM in block transfer mode (block size = four) when the SDRAM continuous access enable bit is enabled (SDAMOD.BE = 1) and EXDMACn.EDMOMD.DACKSEL set to 1.

EDACKn is asserted during SDRAM is writing data.

Figure 19.37 shows the operation example in which data is transferred from SDRAM to the device with EDACKn in block transfer mode (block size = two) when the SDRAM continuous access enable bit is disabled (SDAMOD.BE = 0) and EXDMACn.EDMOMD.DACKSEL set to 0.

Figure 19.38 shows the operation example in which data is transferred from the device with EDACKn to SDRAM in block transfer mode (block size = two) when the SDRAM continuous access enable bit is disabled (SDAMOD.BE = 0) and EXDMACn.EDMOMD.DACKSEL set to 0.

EDACKn is asserted while SDRAM is writing data.

For the SDRAM commands and SDRAM access timing setting registers, refer to section 16, Buses.

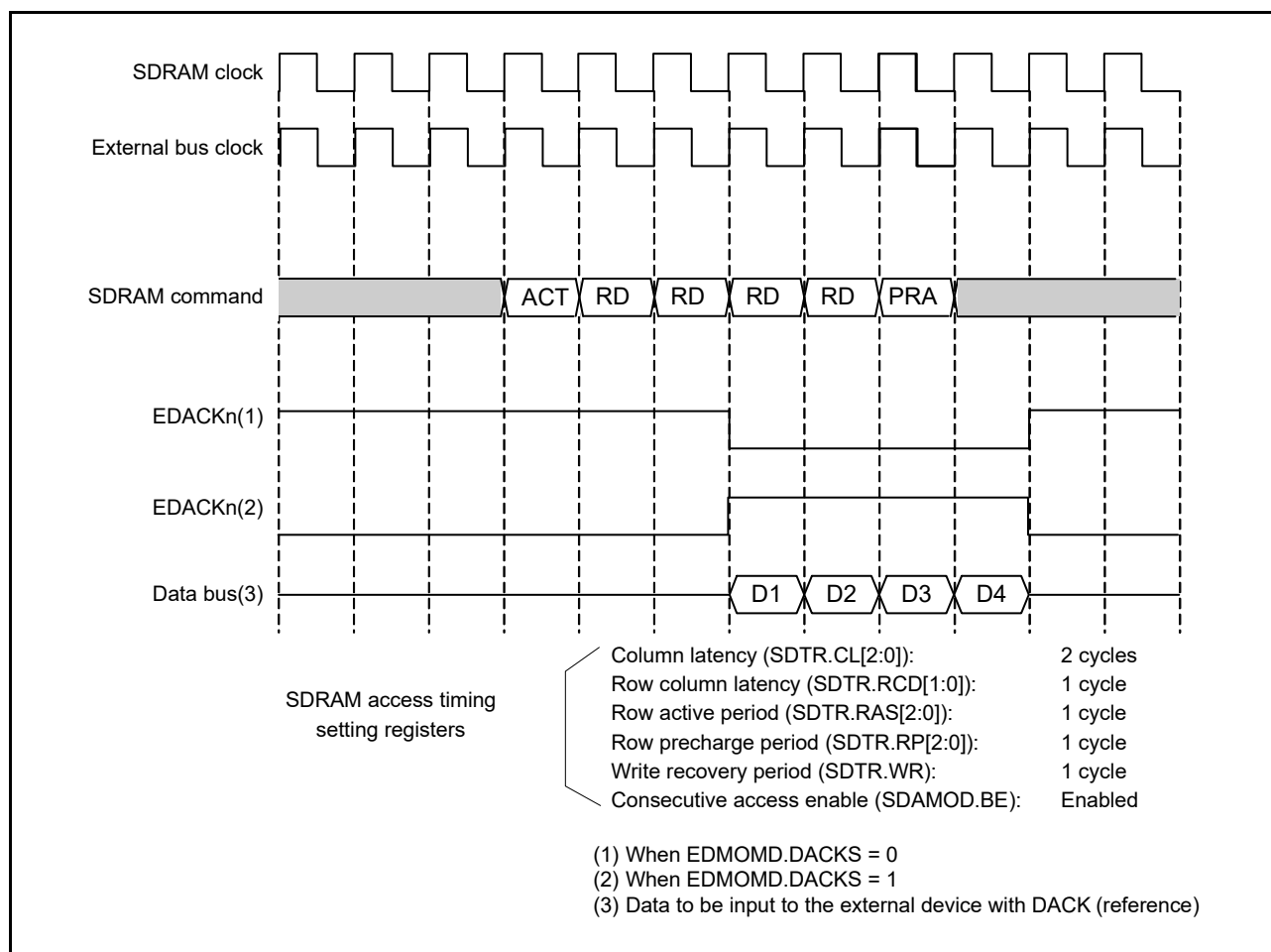


Figure 19.33 Operation Example in Block-Transfer
 (SDRAM Area Read: Consecutive Access Enabled, EXDMACn.EDMOMD.DACKSEL = 0) Single Address Mode

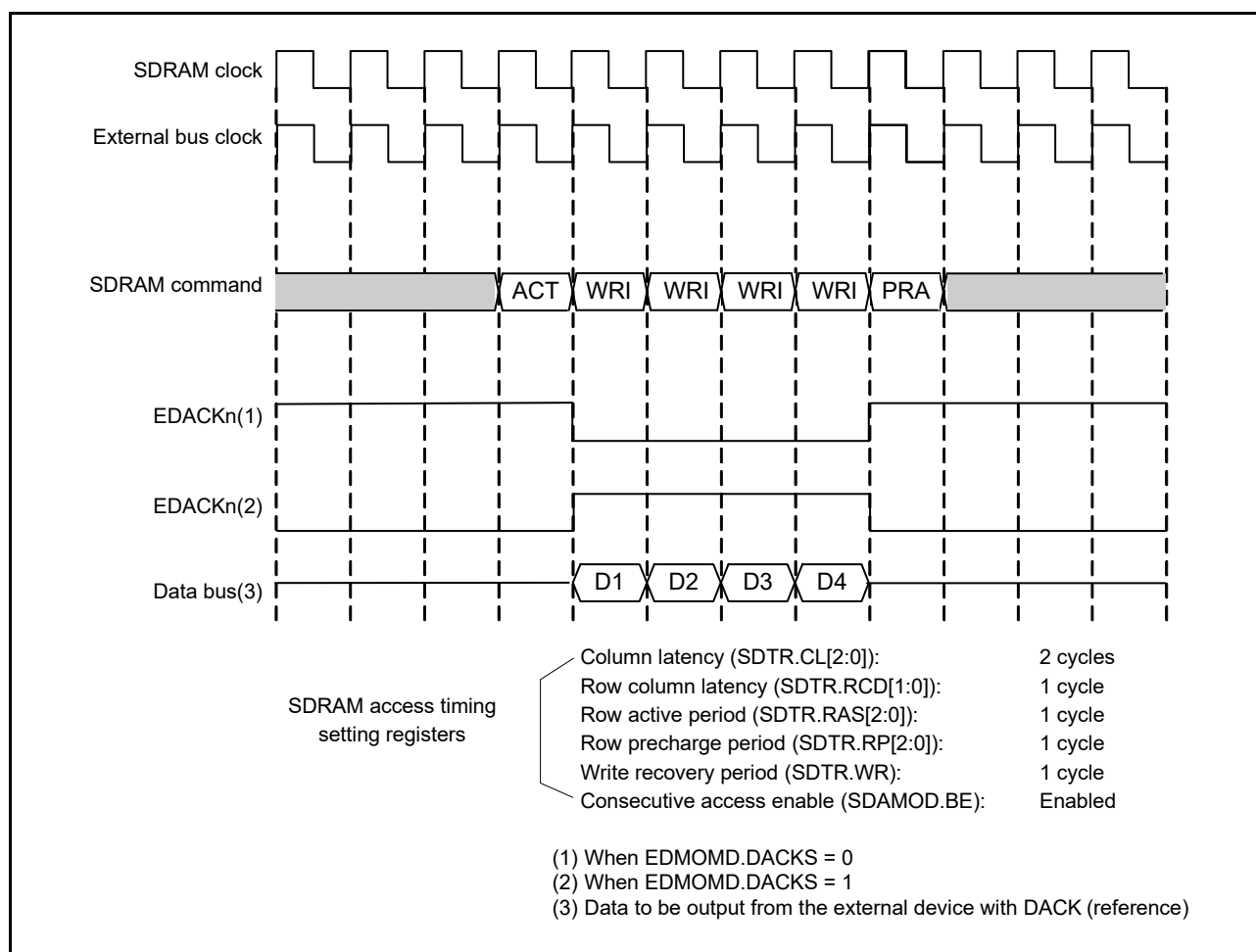


Figure 19.34 Operation Example in Block-Transfer
(SDRAM Area Write: Consecutive Access Enabled, EXDMACn.EDMOMD.DACKSEL = 0) Single Address Mode

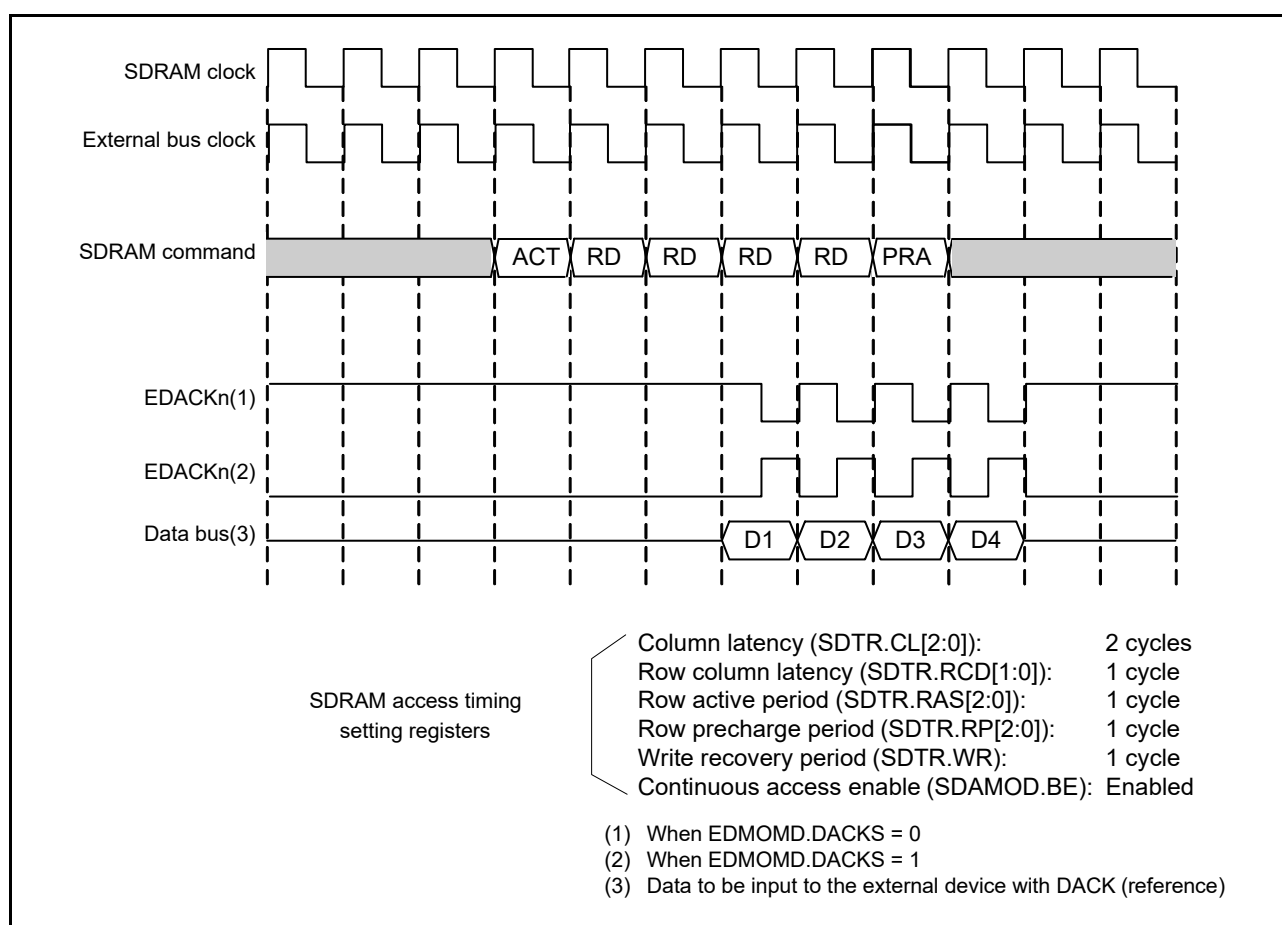


Figure 19.35 Operation Example in Block-Transfer
 (SDRAM Area Read: Continuous Access Enabled, EXDMACn.EDMOMD.DACKSEL = 1) Single Address Mode

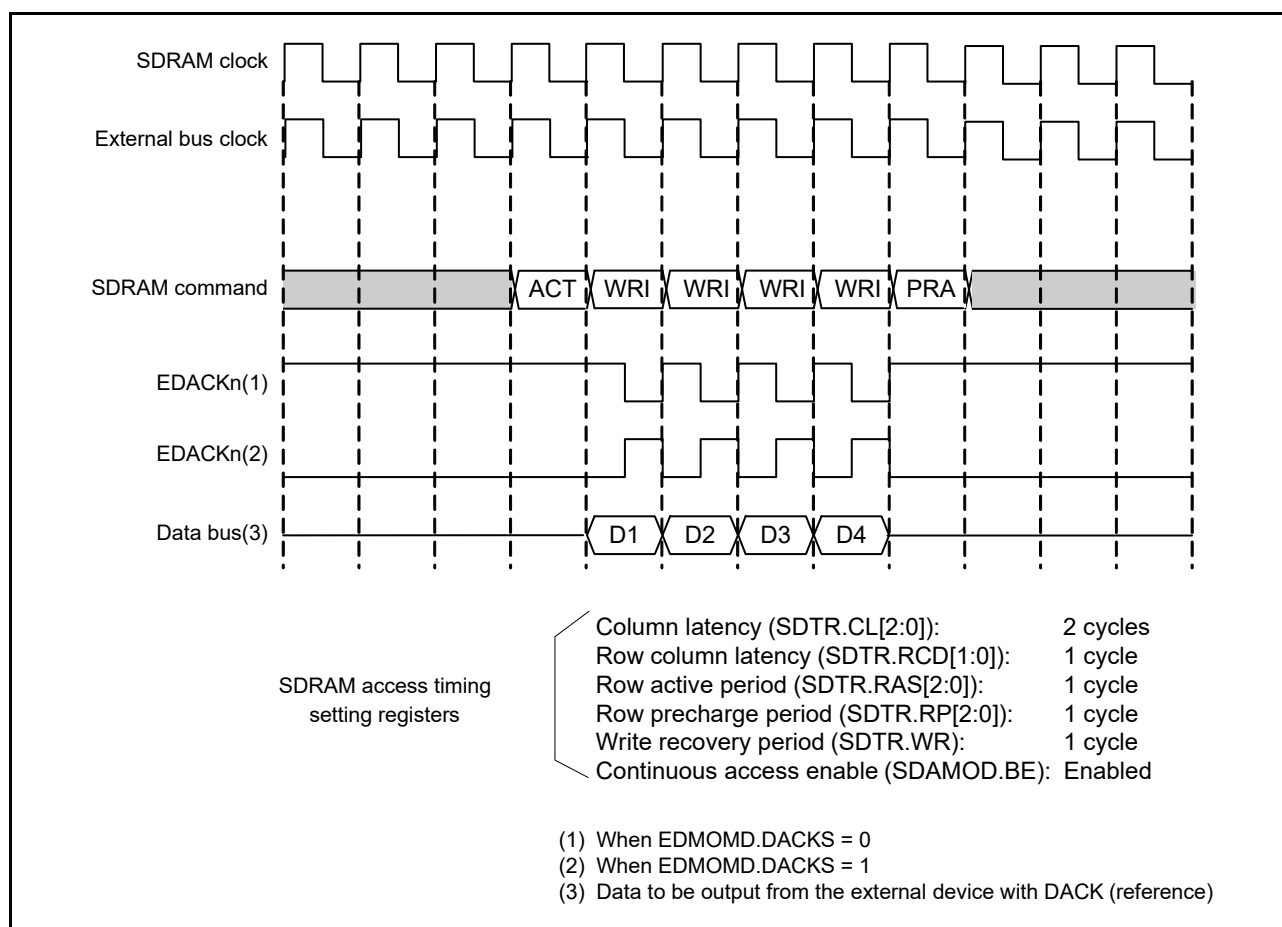


Figure 19.36 Operation Example in Block-Transfer
(SDRAM Area Write: Continuous Access Enabled, EXDMACn.EDMOMD.DACKSEL = 1) Single Address Mode

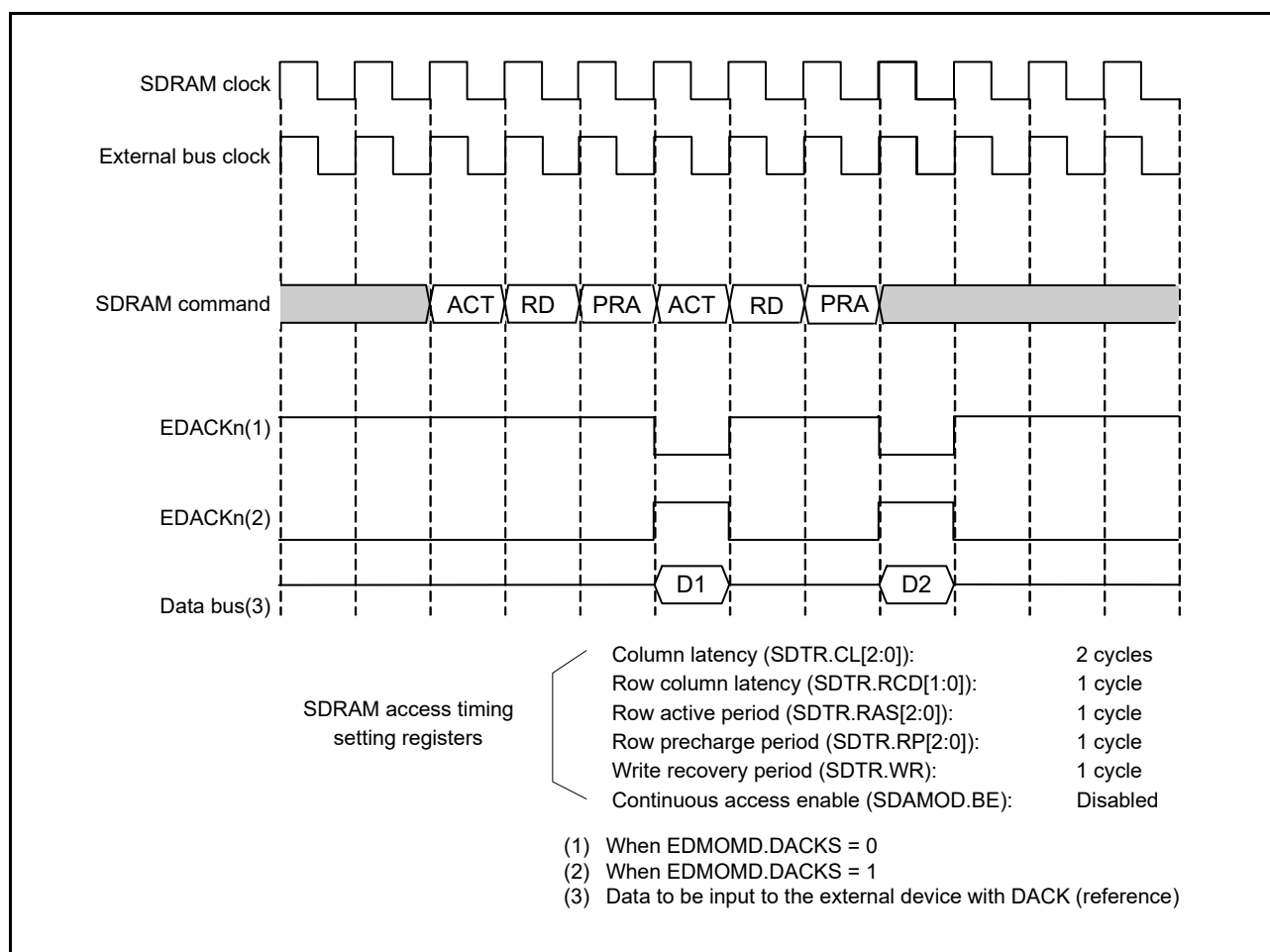
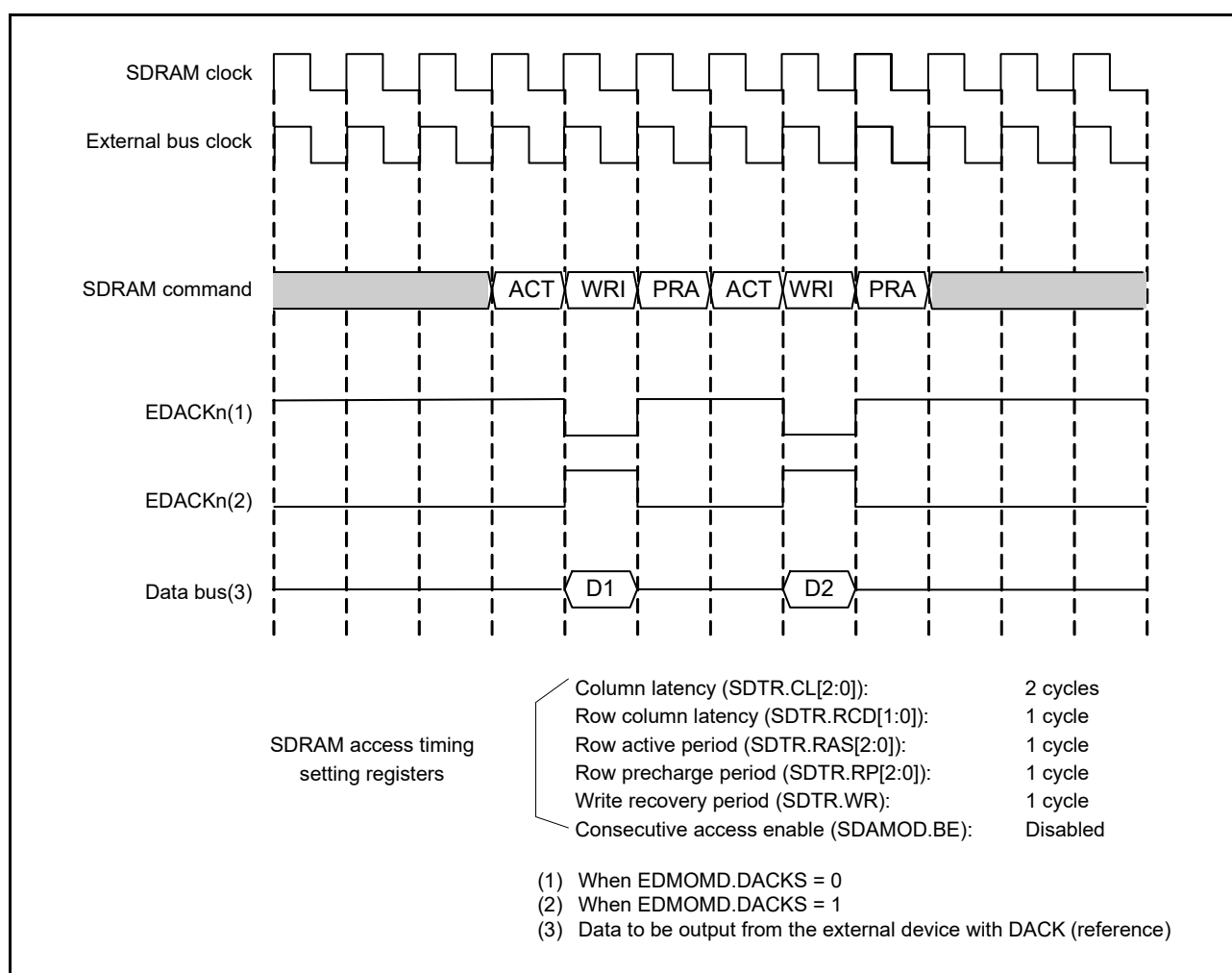


Figure 19.37 Operation Example in Block-Transfer
(SDRAM Area Read: Continuous Access Disabled, EXDMACn.EDMOMD.DACKSEL = 0) Single Address Mode



**Figure 19.38 Operation Example in Block-Transfer
 (SDRAM Area Write: Consecutive Access Disabled) Single Address Mode**

19.10 Usage Notes

19.10.1 Cluster Buffers

The EXDMAC provides eight 32-bit cluster buffers (CLSBR0 to CLSBR7), in which data is stored in the different manner depending on the transfer size setting (SZ[1:0] bits in EDMTMD of EXDMACn).

Figure 19.39 shows how data is stored in cluster buffers.

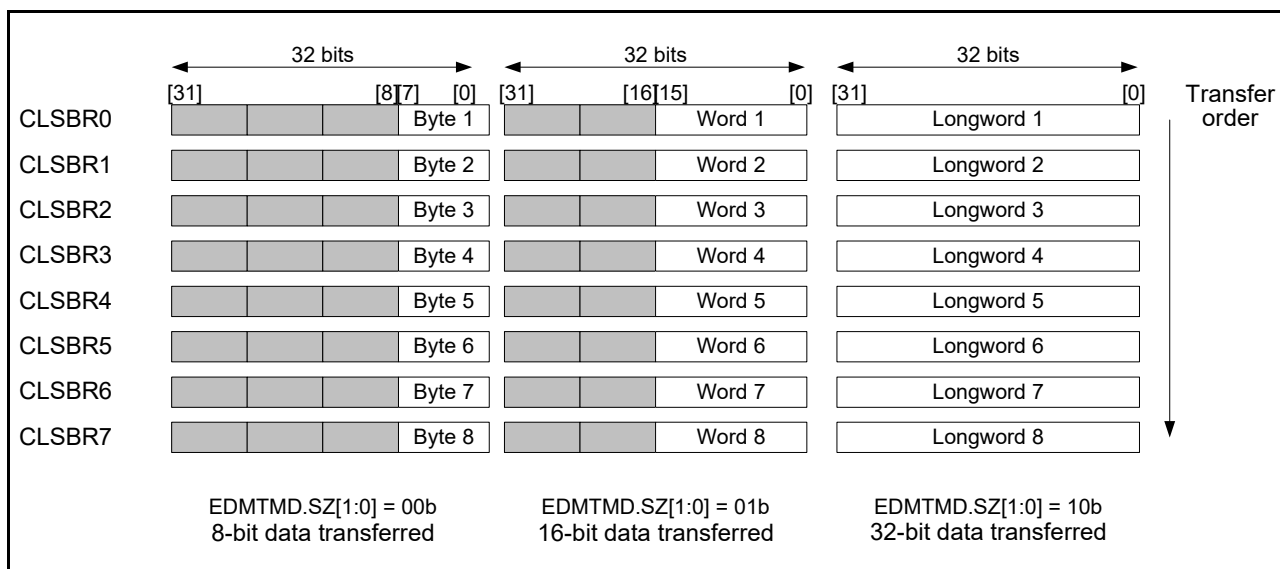


Figure 19.39 Data Storage in Cluster Buffers

19.10.2 Access to the Registers during DMA Transfer

Do not write to the EDMSAR, EDMDAR, EDMCRA, EDMCRB, EDMTMD, EDMOMD, EDMINT, EDMAMD, EDMOFR, and EDMRMD registers of EXDMACn while the EDMSTS.ACT flag of the same channel is set to 1 (DMA operating state) or the EDMCNT.DTE bit of the same channel is set to 1 (DMA transfer enabled).

19.10.3 DMA Transfer to Reserved Areas

DMA transfer to the reserved areas is prohibited. If such an access is made, transfer results are not guaranteed. For details on the reserved areas, refer to section 4, Address Space.